Reference-Dependent Utility over Prices and the Welfare Cost of Inflation*

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Abstract

Survey evidence documents anger and perceived unfairness among consumers facing rising prices. I study a mechanism through which these reactions can generate welfare losses by embedding reference-dependent demand in a New Keynesian model. When posted prices exceed reference prices formed from past purchases, the same physical consumption bundle delivers less utility. This generates a first-order welfare cost of inflation absent from standard models. I calibrate the parameters governing these losses using an online survey of about 1,500 U.S. respondents, with hypothetical shopping scenarios that vary reference prices while holding posted prices fixed. Under the baseline calibration, welfare losses with reference dependence are about four times those in the standard New Keynesian model for cost-push shocks scaled to generate 1% peak quarterly inflation in both models. To quantify this cost during the recent inflation surge, I build a heterogeneous-agent model with reference dependence in frequently purchased goods and household expenditure shares on these goods that vary with income. I fit this model to U.S. macroeconomic dynamics during 2021–2023 and estimate a total welfare loss of 5.8% of annual steady-state consumption for the average household, about one-fifth higher than without reference dependence.

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1 Introduction

Conventional models imply modest welfare costs of inflation, yet the public expresses a deep aversion to rising prices. U.S. consumer sentiment fell to a historic low in 2022 during the recent inflation surge, with rising prices at the center of public frustration. Federal Reserve Chair Jerome Powell linked this dissatisfaction in part to consumers "paying much more for the basics of life" than they had two or three years earlier. Paying more than one is accustomed to can leave consumers feeling overcharged and less satisfied with the transaction (Thaler, 1985). This reaction offers a possible explanation for the anger at higher prices documented in consumer surveys.[1] The preferences underlying standard macroeconomic models omit this psychological cost, so conventional welfare calculations may understate the losses households experience during inflationary episodes.

To quantify this cost, I integrate reference-dependent demand into a New Keynesian model. In this framework, consumers assess how good a deal each variety offers by comparing its posted price with a reference price formed from its past prices. Holding the posted price fixed, a lower reference price makes the purchase a worse deal, reducing both demand and utility. I show that reference-dependent demand generates a novel *first-order* cost of inflation that standard models fail to capture. Two parameters govern the magnitude of this cost: the reference price elasticity, which is the elasticity of demand with respect to the reference price *holding the posted price fixed*; and the speed at which reference prices update. I estimate these parameters with a survey of U.S. consumers, in which hypothetical shopping scenarios generate exogenous variation in reference prices while leaving posted prices unchanged. To quantify this cost during the recent inflation surge, I incorporate reference-dependent demand into a rich heterogeneous-agent New Keynesian model with heterogeneous consumption baskets, sticky wages, and a realistic wealth distribution. Fitting the model to U.S. data for 2021–2023, I find that the episode cost the average household 5.8% of its 2021 steady-state consumption, with reference dependence accounting for 16% of this loss.

At the core of this framework is a CES consumption aggregator in which the gap between each variety's posted and reference prices enters as a taste shifter. Reference prices are weighted moving averages of each variety's past prices and coincide with posted prices at the zero-inflation steady state. I show that a second-order approximation of household utility around an efficient steady state contains a new term that is linear in the gap between posted and reference prices. Intuitively, in the symmetric equilibrium, inflation pushes posted prices above their references across all varieties, making every purchase feel like a worse deal. The household therefore derives less utility from a given consumption bundle.

The welfare loss is greater when demand is more sensitive to reference prices and when reference prices update more slowly. A larger reference-price elasticity implies a greater utility

[1] See Shiller (1997) and Stantcheva (2024).

loss for a given gap between posted and reference prices. Slower updating of reference prices leads to more persistent gaps between posted and reference prices, increasing the cumulative welfare loss. In its general form, the demand system allows a kink at the reference price: demand can respond more strongly to a price increase above the reference than to an equal-sized decrease below it. The reference-price elasticity can therefore differ between the loss side, where posted prices exceed their references, and the gain side, where they fall below. Because inflation generates an increase in prices relative to their references, the loss-side elasticity governs the resulting welfare costs.

I estimate the reference-price elasticities and the speed at which reference prices update with a survey of about 1,500 U.S. respondents conducted on Prolific in April 2026. To isolate the reference-price elasticity, the ideal experiment varies the reference price while holding the posted price, income and other prices fixed. I implement this through novel hypothetical questions where respondents consider various shopping scenarios. First, each respondent names a grocery product they buy often, reports the price they typically expect to pay for it, which I take as their initial reference price, and states how much they would buy at that price over the next month. At baseline, the posted price equals their initial reference price. Next month's posted price remains fixed at the respondent's initial reference price. In the hypothetical scenario, however, the product usually sells for less, making next month's price a temporary increase. In this scenario, respondents report the price they would typically expect to pay: their reference price under the scenario, and how much they would buy next month, holding income and all other prices fixed. The gap between the initial reference price and the scenario's usual price is randomized across respondents, generating exogenous variation in reference prices while the posted price remains unchanged. Respondents also answer a similar gain-side scenario, in which the item usually sells for more. Since stated quantities are discrete, I estimate the elasticities by Poisson pseudo-maximum likelihood. The same posted price elicits 0.9% less consumption when the reference price is 1% below it, but only 0.3% more when the reference price is 1% above it. I obtain a similar loss-side reference-price elasticity when estimating the same demand system using observational NielsenIQ scanner data for 22 grocery categories.

The speed of reference-price updating is identified through another hypothetical question, in which respondents encounter a 20% price increase after several shopping trips at their initial reference price. A randomly chosen half of respondents report the price that would feel normal (neither too high nor too low) on subsequent trips, while the other half report the price they would typically expect to pay. Using the normal-price responses for the baseline, I invert the reference-price law of motion to infer persistence per shopping trip, then convert it to quarterly persistence using each respondent's shopping frequency. The mean quarterly persistence is 0.53: just over half of a gap between posted and reference prices remains after a quarter.

I use these estimates to quantify the welfare cost of a cost-push shock in the baseline New Keynesian model with reference-dependent demand. The estimated kink in demand at the

reference price makes the model non-standard in two ways: the firm's optimality conditions are one-sided inequalities rather than a single first-order condition, and the steady state, where prices equal their references, sits exactly at the kink. I therefore solve for the exact nonlinear perfect-foresight path after the shock, in which the equilibrium determines whether firms price above, below, or at their references in each period. Scaling the shock in each model to generate a one-percentage-point peak increase in quarterly inflation, I find that the consumption-equivalent welfare loss is 4.3 times that in the standard New Keynesian model. A decomposition shows that the new first-order term accounts for around 80% of the welfare losses. This amplification is larger for smaller shocks, because the new term is first order while the conventional costs are second order, and for more persistent shocks, which lead to more persistent gaps between prices and reference prices.

I then quantify these welfare costs during the 2021–2023 U.S. inflation surge using a heterogeneous-agent New Keynesian model. Two considerations motivate this extension. First, reference-dependent demand has been documented most extensively for frequently purchased goods, whose shares in household expenditure vary across income groups. Guided by this evidence, I define the reference-dependent sector to include food, energy, and other frequently purchased nondurables, with expenditure shares across income groups disciplined using data from the Consumer Expenditure Survey. Second, I incorporate sticky wages and heterogeneous nominal asset holdings to quantify the importance of reference-dependent demand relative to these other mechanisms emphasized in the recent literature. I find that reference dependence contributes a welfare loss equivalent to 0.91% of 2021 steady-state consumption for the average household, with this loss about 8% higher in the most exposed income group than in the least exposed group.

These results bear on the debate over how much weight monetary policy should place on food and energy inflation. Reference-dependent demand has been documented in both food and energy markets, including grocery products and gasoline. The conventional case for looking through temporary movements in these prices is that responding aggressively can increase volatility in core inflation and real activity. My results show that, in addition to disproportionately affecting lower-income households through their consumption baskets, food and energy inflation carries a first-order welfare cost through reference-dependent demand. These findings align with survey evidence on households' concerns during the inflation surge.[2] This cost must be weighed against the costs of stabilizing these prices.

[2] In a 2022 Pew survey, rising prices for food and consumer goods and for gasoline and energy ranked as the leading economic concerns, with three-quarters of U.S. adults reporting that they were very concerned about each (Pew Research Center, 2022). In the University of Michigan's Surveys of Consumers, respondents volunteered concerns about high food and gasoline prices during interviews about their finances and economic outlook. Over 2021–2023, respondents who mentioned these prices reported lower consumer sentiment than those who did not (Hsu, 2024).

Related literature. This paper contributes to four strands of the literature. First, it builds on a long-standing literature that seeks to understand and quantify the welfare costs of inflation. Standard accounts emphasize the welfare costs of economizing on money holdings (Fischer, 1981; Lucas, 2000) and of inefficient relative-price dispersion (Woodford, 2003; Galí, 2015). More recent work examines the welfare costs of real-wage erosion and costly wage adjustment (Afrouzi et al., 2024; Guerreiro et al., 2024), as well as redistribution across households with different net nominal positions (Doepke and Schneider, 2006; Pallotti et al., 2024). I introduce reference-dependent demand, grounded in consumer psychology and marketing evidence, as a new source of inflation's welfare costs. I estimate the parameters governing this demand through a consumer survey and use them to calibrate the demand system in a New Keynesian model, allowing me to quantify its importance relative to other costs of inflation. This mechanism offers a possible explanation for the anger at rising prices documented by Shiller (1997) and Stantcheva (2024). Thaler's (1985) discussion of transaction disutility in terms of annoyance and feeling ripped off provides a psychological interpretation of this utility loss.

I also contribute to work that incorporates insights from behavioral economics into models of price setting and monetary transmission. Loss aversion has been used to explain regular prices and sales (Heidhues and Kőszegi, 2014), price stickiness (Angelis, 2025), and asymmetric monetary transmission (Santoro et al., 2014). Blinder et al. (1998) document firms' concerns about antagonizing customers through price increases, while Rotemberg (2005) and Eyster et al. (2021) embed these concerns in macroeconomic models to study price rigidity. Relative to these papers, I show how reference-dependent consumer demand generates endogenous price rigidity and estimate the parameters governing this mechanism using a new consumer survey. I solve the model in dynamic general equilibrium, preserving the demand kink and accounting for how firms' pricing decisions affect future reference prices and demand. I further contribute by integrating this behavioral mechanism into a rich heterogeneous-agent New Keynesian model to quantify its aggregate and distributional welfare effects.

This paper also relates to research on heterogeneous consumption baskets and the distributional effects of inflation. Empirical work documents higher inflation for lower-income households over longer horizons, reflecting differences in the goods they purchase (Jaravel, 2019; Argente and Lee, 2021). Olivi et al. (2026) study how heterogeneous consumption baskets shape monetary policy trade-offs, while Bobasu et al. (2025) examine how energy expenditure shares and household wealth shape the distributional effects of energy price shocks. Relative to this literature, I show that reference-dependent demand creates an additional channel through which consumption-basket heterogeneity affects welfare during inflationary episodes. I examine how losses attributable to reference dependence vary across income groups during the 2021–2023 U.S. inflation surge.

Finally, I contribute to the extensive empirical marketing literature on reference prices and loss aversion in consumer demand (Winer, 1986; Mayhew and Winer, 1992; Hardie et al., 1993;

Kalyanaram and Little, 1994; Briesch et al., 1997; Bell and Lattin, 2000; Mazumdar et al., 2005). These papers typically focus on a few product categories and document reference-dependent demand. However, evidence is mixed on whether demand responds more strongly to price increases above the reference than to equal-sized decreases below it, as loss aversion would predict. Relative to these studies, I identify reference-price elasticities using hypothetical shopping scenarios that generate exogenous variation in reference prices while holding posted prices fixed. The survey also explicitly instructs respondents to exclude stockpiling and postponing purchases, and I find evidence of loss aversion in their stated consumption choices. I also estimate the same demand system using NielsenIQ scanner data for 22 grocery categories, finding a reference-price elasticity on the loss side close to the survey estimate but a substantially larger elasticity on the gain side. These contrasting findings suggest that stockpiling during sales can obscure loss aversion in purchase data, helping reconcile the survey evidence with the mixed findings in existing work that uses scanner data.

The rest of the paper proceeds as follows. Section 2 presents the model and its main theoretical results. Section 3 uses survey evidence to estimate the parameters of the reference-dependent demand system. Section 4 examines macroeconomic dynamics and welfare costs in the baseline New Keynesian model with reference dependence. Section 5 quantifies aggregate and distributional welfare losses during the 2021–2023 U.S. inflation surge using the HANK model. Section 6 concludes.

2 A New Keynesian Model with Reference-Dependent Demand

In this section, I show that reference-dependent demand generates a first-order welfare cost of inflation in an otherwise standard New Keynesian model. The gap between posted and reference prices enters the CES consumption aggregator as a taste shifter, with reference prices formed from past posted prices. Holding a variety's posted price fixed, a lower reference price makes the purchase a worse deal, reducing both demand and utility. Firms face Rotemberg adjustment costs and internalize how current prices affect future reference prices and demand. I show that a second-order approximation of household utility around the zero-inflation steady state contains a new term linear in the gap between posted and reference prices. Intuitively, prices above references reduce the consumption utility delivered by a given physical bundle, so maintaining consumption utility requires more output and labor. The welfare loss is greater when demand is more sensitive to reference prices and when reference prices update more slowly.

2.1 Reference-Dependent Demand

There is a unit mass of differentiated goods indexed by $i \in [0, 1]$. Good i has a posted price $P_t(i)$ and a reference price $R_t(i)$, the benchmark the consumer uses to assess how good a deal

the variety offers.

Let $C_t(i)$ denote physical consumption of variety i. Following Sibly (2002) and Ahrens et al. (2017), I define the consumption composite C_t^U using a reference-dependent CES (RD–CES) aggregator with variety-specific taste shifters $s_t(i)$ that depend on the gaps between posted and reference prices:

$$
\begin{aligned}
C_t^U &= \left[\int_0^1 s_t(i)^{1/\varepsilon} C_t(i)^{(\varepsilon-1)/\varepsilon}\,\mathrm{d}i\right]^{\varepsilon/(\varepsilon-1)}, \qquad \varepsilon > 1, \\
s_t(i) &= \begin{cases} \left[\dfrac{P_t(i)}{R_t(i)}\right]^{-\Gamma_L}, & P_t(i) \geq R_t(i), \\ \left[\dfrac{P_t(i)}{R_t(i)}\right]^{-\Gamma_G}, & P_t(i) < R_t(i). \end{cases}
\end{aligned} \tag{2.1}
$$

The parameters Γ_L and Γ_G govern reference-dependent demand when posted prices are above and below reference prices, respectively. For $\Gamma_L > 0$, purchases at posted prices above their reference prices are perceived as worse deals. The aggregator captures this by assigning a lower consumption composite C_t^U, and hence less consumption utility, for a given physical consumption bundle $\{C_t(i)\}_i$. Appendix A.1 provides a transaction-utility microfoundation for the relative demands implied by this aggregator.[3]

Taking posted prices and reference prices as given, the household chooses quantities $\{C_t(i)\}_i$ to minimize the expenditure required to attain a given level of C_t^U. The resulting demand for each variety and the ideal price index are:

$$
\begin{aligned}
C_t(i) &= s_t(i)\left(\frac{P_t(i)}{P_t^{\mathrm{RD}}}\right)^{-\varepsilon} C_t^U, \\
P_t^{\mathrm{RD}} &= \left[\int_0^1 s_t(i) P_t(i)^{1-\varepsilon}\,\mathrm{d}i\right]^{1/(1-\varepsilon)}.
\end{aligned} \tag{2.2}
$$

The ideal price index P_t^{RD} measures the expenditure required to obtain one unit of the consumption composite C_t^U.

The parameters Γ_L and Γ_G are the reference-price elasticities of demand, holding the variety's posted price fixed. When a variety's posted price exceeds its reference, a 1% reduction in the reference price makes the purchase a worse deal and lowers demand by approximately Γ_L%. When the posted price lies below its reference, a 1% increase in the reference price makes the purchase a better deal and raises demand by approximately Γ_G%.

Reference dependence also changes how demand responds to the posted price. In the standard CES model, the own-price elasticity is ε. Here, a higher posted price also makes the

[3]This construction uses the link between discrete-choice models and CES demand (Anderson et al., 1989), incorporating transaction utility as in Thaler (1985).

purchase a worse deal relative to its reference price. Holding the reference price fixed, the absolute own-price elasticity is therefore $\varepsilon + \Gamma_L$ when the posted price exceeds its reference and $\varepsilon + \Gamma_G$ when it lies below. These elasticities are evaluated holding the aggregate price index P_t^{RD} and consumption composite C_t^U fixed.

Reference prices are formed from past posted prices:

$$R_{t+1}(i) = R_t(i)^{\rho} P_t(i)^{1-\rho} e^{\bar{\pi}}, \qquad \rho \in [0, 1). \tag{2.3}$$

Here $\bar{\pi}$ denotes trend inflation. The factor $e^{\bar{\pi}}$ makes posted and reference prices coincide in any steady state with constant inflation $\bar{\pi}$, so only departures from trend inflation open gaps between them. I set $\bar{\pi} = 0$ throughout. In logs, the reference price is therefore an exponentially weighted average of past posted prices. It equals last period's posted price when $\rho = 0$ and adjusts more slowly as ρ rises.

To understand the effect of reference-dependent demand on the ideal price index, let P_t denote the conventional CES price index. For the remainder of this section, I assume all varieties share a common posted price, $P_t(i) = P_t$, and reference price, $R_t(i) = R_t$.[4] With lowercase letters denoting natural logs, the common taste shifter when $P_t > R_t$ is

$$s_t = e^{-\Gamma_L (p_t - r_t)} = e^{-(\varepsilon - 1)\chi_t}, \qquad \chi_t \equiv \frac{\Gamma_L}{\varepsilon - 1}(p_t - r_t). \tag{2.4}$$

This yields the following result.

Lemma 1 (The ideal RD–CES price index)**.** *Suppose all varieties share a common posted price P_t and a common reference price R_t, with $P_t > R_t$. Then the ideal price index satisfies*

$$P_t^{\mathrm{RD}} = P_t e^{\chi_t}, \qquad p_t^{\mathrm{RD}} = p_t + \chi_t. \tag{2.5}$$

The reference wedge χ_t is the log difference between the ideal price index under reference-dependent demand and the conventional CES price index. When $\Gamma_L > 0$, posted prices above their references make purchases worse deals, so a given physical bundle delivers a lower consumption composite C_t^U and hence lower utility. Attaining the same consumption composite consequently requires greater expenditure, so $P_t^{\mathrm{RD}} > P_t$. Without reference dependence, the two indices coincide. For $P_t < R_t$, the same price-index relationship holds with $\chi_t = \Gamma_G(p_t - r_t)/(\varepsilon - 1)$.[5]

[4] This corresponds to a symmetric equilibrium of the Rotemberg model introduced below, with identical firms and common initial posted and reference prices.

[5] When posted and reference prices differ across varieties, the relationship in Lemma 1 holds to first order around a symmetric allocation with posted prices equal to reference prices:

$$p_t^{\mathrm{RD}} - p_t \approx \int_0^1 \chi_t(i)\, \mathrm{d}i, \qquad \chi_t(i) \equiv -\frac{\log s_t(i)}{\varepsilon - 1}.$$

2.2 The Intertemporal Household Problem

The intratemporal problem implies that expenditure on the consumption composite C_t^U is $P_t^{\mathrm{RD}} C_t^U$. The household chooses consumption, labor supply, and saving in an otherwise standard intertemporal problem, taking the ideal price index as given.

Given initial bond holdings B_0, the representative household maximizes expected lifetime utility subject to its budget constraint and a no-Ponzi condition:

$$\begin{aligned} \max_{\{C_t^U, N_t, B_{t+1}\}_{t=0}^{\infty}} \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(C_t^U)^{1-\sigma} - 1}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \right] \\ \text{s.t.} \quad & P_t^{\mathrm{RD}} C_t^U + Q_t B_{t+1} \leq B_t + W_t N_t + D_t^F + T_t, \qquad t \geq 0, \\ & \liminf_{T \to \infty} \left(\prod_{s=0}^{T} Q_s \right) B_{T+1} \geq 0. \end{aligned}$$

Here $\beta \in (0,1)$, $\sigma \geq 1$, and $\varphi \geq 0$, with the usual logarithmic limit at $\sigma = 1$. The variable Q_t is the price of a one-period nominal bond, W_t is the nominal wage, D_t^F denotes distributed firm profits, and T_t denotes lump-sum transfers net of taxes. Bonds are in zero net supply.

Let $I_t \equiv Q_t^{-1}$ denote the gross nominal interest rate. Consider a deterministic zero-inflation steady state, where posted and reference prices coincide and $\bar{I} = 1/\beta$. I normalize the initial steady-state price level to one, $\bar{P} = \bar{R} = 1$. For the analysis that follows, every lowercase variable denotes the log deviation of its level counterpart from this initial steady state, so p_t and r_t are also the natural logs of posted and reference prices, and $i_t \equiv \log(I_t/\bar{I})$. To first order, the Euler equation is[6]

$$c_t^U = \mathbb{E}_t c_{t+1}^U - \frac{1}{\sigma} \left(i_t - \mathbb{E}_t \pi_{t+1} \right) + \frac{1}{\sigma} \left(\mathbb{E}_t \chi_{t+1} - \chi_t \right), \tag{2.6}$$

where $\pi_{t+1} \equiv p_{t+1} - p_t$ is posted-price inflation.[7] Holding all else constant, an increase in the reference wedge χ_t, the scaled gap between posted and reference prices, lowers the consumption utility delivered by a given physical bundle today. With expected future conditions unchanged, the Euler equation implies a lower current consumption composite C_t^U.

Around the same steady state, the labor-supply condition is

$$w_t - p_t = \sigma c_t^U + \varphi n_t + \chi_t. \tag{2.7}$$

Relative to the standard model, the reference wedge reduces what wage income buys in units of the consumption composite. A higher χ_t therefore requires a higher posted real wage to

[6] The household conditions are smooth in χ_t, so I linearize in the wedge while retaining its piecewise-linear dependence on the posted–reference price gap.

[7] By Lemma 1, inflation in the ideal price index satisfies $\pi_{t+1}^{\mathrm{RD}} = \pi_{t+1} + \chi_{t+1} - \chi_t$. The household's real return to saving therefore depends on inflation in the price of the consumption composite.

support the same consumption composite and hours worked.

2.3 Firms and Inflation Dynamics

Firms set prices facing reference-dependent demand and Rotemberg adjustment costs. When $\Gamma_L > \Gamma_G$, demand is kinked at the reference price, making the firm's pricing problem nonsmooth. To build intuition about how reference-dependent demand affects firms' pricing decisions, I first consider the smooth case, $\Gamma_L = \Gamma_G \equiv \Gamma$. Appendix B extends the results to the kinked case, $\Gamma_L > \Gamma_G$.

Firm i is owned by the household and produces physical output $Y_t(i)$ of variety i. Let Y_t^U aggregate these quantities using the RD–CES aggregator in (2.1). Taking aggregate variables and initial conditions $R_0(i)$ and $P_{-1}(i)$ as given, firm i chooses its price sequence to maximize the present discounted value of profits:[8]

$$\begin{aligned}
\max_{\{P_t(i)\}_{t=0}^{\infty}} \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{\Lambda_t}{P_t^{\mathrm{RD}}} \left[P_t(i) Y_t(i) - (1-\tau) W_t N_t(i) - \frac{\psi_p}{2} \left(\frac{P_t(i)}{P_{t-1}(i)} - 1 \right)^2 P_t^{\mathrm{RD}} Y_t^U \right] \\
\text{s.t.} \quad & Y_t(i) = \left(\frac{P_t(i)}{R_t(i)} \right)^{-\Gamma} \left(\frac{P_t(i)}{P_t^{\mathrm{RD}}} \right)^{-\varepsilon} Y_t^U, \\
& Y_t(i) = A_t N_t(i)^{1-\alpha}; \qquad R_{t+1}(i) = R_t(i)^{\rho} P_t(i)^{1-\rho}.
\end{aligned}$$

Here $\Lambda_t \equiv (C_t^U)^{-\sigma}$ is marginal utility of the consumption composite, $\alpha \in [0,1)$ governs returns to labor, τ is an employment subsidy, and $\psi_p \geq 0$ governs price-adjustment costs, which are incurred in units of the consumption composite.[9] At a symmetric equilibrium, define nominal marginal cost relative to the posted price:

$$mc_t \equiv \frac{(1-\tau) W_t}{P_t \mathrm{MPL}_t}, \tag{2.8}$$

where MPL_t is the marginal product of labor.

The reference price is an additional state variable, so optimality also requires a costate condition. Let Θ_t denote the shadow value of the reference price: the additional firm value from a marginal proportional increase in that price, normalized by $\Lambda_t Y_t^U$. It satisfies

$$\Theta_t = \Gamma(1 - mc_t) + \beta\rho\, \mathbb{E}_t \left[\mathcal{M}_{t+1} \Theta_{t+1} \right], \tag{2.9}$$

where $\mathcal{M}_{t+1} \equiv \Lambda_{t+1} Y_{t+1}^U / (\Lambda_t Y_t^U)$. Holding the posted price fixed, a 1% higher reference price raises sales by approximately Γ%, with a profit margin $1 - mc_t$ on those additional sales. A fraction ρ of this increase carries into tomorrow's reference price; the second term captures its

[8] Posted prices satisfy $P_t(i)/P_t \in [1/K, K]$ for some finite $K > 1$. These bounds are slack under symmetry and, together with the regularity conditions in Appendix B, ensure that firm value is finite.

[9] Under symmetry, $P_t^{\mathrm{RD}} Y_t^U = P_t Y_t$, so these costs can equivalently be expressed in units of physical output Y_t.

discounted value to the firm.[10]

Let $\Pi_t \equiv P_t/P_{t-1}$ denote gross posted-price inflation. At an interior optimum, the firm's exact price-setting condition is

$$\psi_p\Big[(\Pi_t-1)\Pi_t-\beta\mathbb{E}_t\{\mathcal{M}_{t+1}(\Pi_{t+1}-1)\Pi_{t+1}\}\Big]=(\varepsilon+\Gamma)mc_t-(\varepsilon+\Gamma-1)+\beta(1-\rho)\mathbb{E}_t[\mathcal{M}_{t+1}\Theta_{t+1}]. \tag{2.10}$$

Relative to the standard Rotemberg model, reference dependence changes the optimal pricing condition in two ways. First, with reference prices predetermined, firms face a higher static demand elasticity, $\varepsilon+\Gamma$ rather than ε. Second, a fraction $1-\rho$ of today's proportional price increase carries into tomorrow's reference price, raising future demand at given posted prices: a benefit valued by Θ_{t+1}. Appendix A.2 gives the derivation.

Steady state and subsidy. At the deterministic zero-inflation steady state, posted and reference prices coincide and price-adjustment costs are zero. Evaluating (2.9) and (2.10) at this steady state yields the effective demand elasticity and gross markup:

$$\varepsilon_{\text{eff}}=\varepsilon+\Gamma\frac{1-\beta}{1-\beta\rho},\qquad \bar{\mu}=\frac{\varepsilon_{\text{eff}}}{\varepsilon_{\text{eff}}-1}. \tag{2.11}$$

A permanent 1% increase in a firm's price reduces its discounted total demand by approximately $\varepsilon_{\text{eff}}\%$, holding aggregate conditions at steady state. The demand loss initially reflects the elasticity $\varepsilon+\Gamma$, then diminishes as the reference price catches up.[11] I offset the steady-state monopoly distortion with the employment subsidy $1-\tau=1/\bar{\mu}$.

In a symmetric equilibrium, all firms produce the same quantity, so aggregate physical output is $Y_t=Y_t(i)$ for all i. To derive the Phillips curve below, I specialize to log utility, $\sigma=1$, and define $\widetilde{y}_t\equiv y_t-a_t$ as the log deviation of physical output from its technology benchmark.[12] Without reference dependence, $\widetilde{y}_t$ coincides with both the efficient output gap and the natural output gap under the employment subsidy. With one-period reference prices, $\rho=0$, the pricing conditions yield the following Phillips curve.

Proposition 1 (The RD–Rotemberg Phillips curve)**.** *Suppose* $\Gamma_L=\Gamma_G\equiv\Gamma$*,* $\sigma=1$*,* $\rho=0$*, and* $\psi_p>0$*, with the subsidy* $1-\tau=1/\bar{\mu}$*. Around the zero-inflation steady state, inflation*

[10] The regularity conditions in Appendix A.2 rule out non-fundamental solutions to the costate recursion and ensure that the firm's transversality conditions hold.

[11] The demand elasticity j periods after the permanent price increase is $\varepsilon+\Gamma\rho^j$, so

$$\varepsilon_{\text{eff}}=(1-\beta)\sum_{j=0}^{\infty}\beta^j(\varepsilon+\Gamma\rho^j)=\varepsilon+\Gamma\frac{1-\beta}{1-\beta\rho}.$$

[12] With log utility, the log deviation of marginal cost is proportional to $\widetilde{y}_t$ to first order; Appendix A.2 derives this relationship.

satisfies, to first order,

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa_{\text{RD}} \left(\widetilde{y}_t - \beta \gamma \, \mathbb{E}_t \widetilde{y}_{t+1} \right), \tag{2.12}$$

where

$$\kappa_{\text{RD}} \equiv \frac{(\varepsilon + \Gamma)(1 + \varphi)}{\psi_p \bar{\mu}(1 - \alpha)}, \qquad \gamma \equiv \frac{\Gamma}{\varepsilon + \Gamma}. \tag{2.13}$$

The higher static demand elasticity makes inflation more responsive to current marginal cost. A persistent cost increase also compresses future profit margins, reducing the value of a higher reference price tomorrow and weakening the incentive to raise today's price, as captured by the negative expected future output-gap term in equation (2.12). Solving the Phillips curve forward for an AR(1) marginal-cost process shows that reference dependence increases pass-through to inflation relative to the standard Rotemberg model, with a smaller percentage increase for more persistent shocks.[13] When $\Gamma = 0$, the standard Rotemberg Phillips curve is recovered.

With $\Gamma_L > \Gamma_G$, one-sided conditions replace the smooth pricing condition (2.10) at the reference price. Under the conditions described in Appendix B.2, the kink generates an inaction band near the zero-inflation steady state: the firm optimally sets $P_t = R_t$ over a range of marginal costs.[14] Within this band, a price increase loses too much demand, while a price cut attracts too few additional sales to offset the lower margin. A sufficiently large increase in marginal costs narrows profit margins, making it profitable to raise the price above the reference despite the larger demand elasticity. The optimal price therefore satisfies the nonlinear pricing condition (2.10), with Γ_L replacing Γ in the current-demand terms. Conversely, a sufficiently large fall in marginal costs makes additional sales more profitable, inducing the firm to cut its price below the reference despite the weaker demand response. The corresponding pricing condition uses Γ_G in the current-demand terms. Appendix B.3 provides the formal pricing conditions and their implications for inflation dynamics.

2.4 Monetary Policy and Equilibrium

Under symmetry, aggregate production and goods-market clearing imply

$$Y_t \equiv \int_0^1 Y_t(i) \, \mathrm{d}i = A_t N_t^{1-\alpha} = e^{\chi_t} Y_t^U, \tag{2.14}$$

$$Y_t^U = C_t^U + \frac{\psi_p}{2} (\Pi_t - 1)^2 Y_t^U. \tag{2.15}$$

[13] With $\widehat{mc}_t$ following an AR(1) process with persistence ρ_m, solving (A.13) forward gives

$$\pi_t = \frac{\varepsilon + \Gamma(1 - \beta\rho_m)}{\psi_p \bar{\mu}(1 - \beta\rho_m)} \widehat{mc}_t.$$

For the same ψ_p, the proportional increase over the standard model, approximately $\Gamma(1 - \beta\rho_m)/\varepsilon$, falls as ρ_m rises.

[14] For a related mechanism of price inaction, see Angelis (2025).

Monetary policy. The central bank responds to posted-price inflation and to the log deviation of physical output from its deterministic steady state:

$$i_t = \phi_\pi \pi_t + \phi_y \log\left(\frac{Y_t}{\bar{Y}}\right) + \nu_t, \qquad \phi_\pi > 0, \quad \phi_y \geq 0. \tag{2.16}$$

Here ν_t is an exogenous monetary-policy shock.

Equilibrium. Given exogenous sequences $\{A_t, \nu_t\}_{t=0}^{\infty}$ and initial conditions $B_0 = 0$, R_0, and P_{-1}, a symmetric equilibrium is a path along which the household's choices solve its intratemporal and intertemporal problems in Sections 2.1 and 2.2, each firm's price sequence maximizes the present discounted value of its profits, reference prices follow (2.3), the nominal interest rate follows (2.16), the government finances the employment subsidy with lump-sum taxes, $T_t = -\tau W_t N_t$, and the bond, labor, and composite-good markets clear, the last through the resource constraint (2.15). With $\Gamma_L = \Gamma_G$, firm optimality implies (2.9) and (2.10). With $\Gamma_L > \Gamma_G$, these conditions hold with the active branch's elasticity away from the reference price, and one-sided conditions replace the pricing equation at the reference price (Appendix B). Appendix A.4 states the definition formally, including the price-aggregation and symmetry conditions.

2.5 Welfare Costs of Inflation under Reference-Dependent Demand

The economy is initially at its deterministic zero-inflation steady state. At time 0, an unanticipated aggregate shock occurs. Its subsequent path is known, and no further shocks arrive, so the economy follows a perfect-foresight transition. I use a standard second-order approximation of household utility around the initial zero-inflation steady state to evaluate the welfare consequences of this transition (Galí, 2015; Woodford, 2003). I measure the discounted lifetime utility loss relative to the no-shock continuation of the initial steady state.

Proposition 2 (Welfare following an unanticipated shock)**.** *Suppose* $\Gamma_L = \Gamma_G \equiv \Gamma$*,* $\sigma = 1$*, and* $\rho = 0$*, with the employment subsidy* $1 - \tau = 1/\bar{\mu}$*. Following an unanticipated shock at time* 0*, the representative household's discounted lifetime utility loss has the second-order expansion*

$$\mathcal{L}_0 = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left[\frac{1+\varphi}{1-\alpha} \widetilde{y}_t^2 + \frac{2\Gamma}{\varepsilon - 1} \pi_t + \psi_p \pi_t^2 \right] + \text{t.i.p.} + O(3). \tag{2.17}$$

Appendix A.3 provides the derivation. As in the standard model, the quadratic output-gap term reflects the consumption–labor trade-off, while the quadratic inflation term captures resources consumed by Rotemberg adjustment costs. Reference-dependent demand adds a term linear in the reference wedge $\chi_t = \Gamma(p_t - r_t)/(\varepsilon - 1)$, which is proportional to the gap between posted and reference prices. With one-period references, this gap equals current inflation.

The aggregation identity (2.14), $Y_t^U = e^{-\chi_t} Y_t$, shows how this wedge enters welfare. When posted prices exceed references, purchases are perceived as worse deals, and a given quantity of physical output delivers fewer units of composite output. Together with goods-market clearing (2.15), this implies a lower consumption composite C_t^U, and hence lower utility, for given physical output and inflation. For a given positive log gap between posted and reference prices, a higher reference-price elasticity Γ implies a larger reduction in the consumption composite and hence a larger first-order welfare loss. Compensating for this loss would require more output and labor. This is the general-equilibrium welfare counterpart of the price-index wedge in Lemma 1. This first-order welfare effect arises even when the employment subsidy eliminates the steady-state monopoly distortion.[15]

With gradual reference-price updating ($\rho > 0$), the utility loss from a price increase can persist even after inflation has subsided. Starting from the zero-inflation steady state, the discounted first-order welfare loss from reference dependence is

$$\sum_{t=0}^{\infty} \beta^t \chi_t = \frac{\Gamma}{(\varepsilon - 1)(1 - \beta\rho)} \sum_{t=0}^{\infty} \beta^t \pi_t. \tag{2.18}$$

Slower updating keeps reference prices below posted prices for longer following a price increase, prolonging the associated utility loss. Holding the inflation path fixed, this amplifies the first-order welfare loss by $1/(1 - \beta\rho)$ relative to one-period references.

Unequal reference-price elasticities preserve this first-order channel: Γ_L governs losses above references, while Γ_G governs gains below them. When $\Gamma_L > \Gamma_G$, equal and opposite log price–reference gaps generate larger losses than gains at first order. The approximation exploits the smoothness of household utility in composite consumption and labor, with the reference wedge entering through the aggregation identity linking physical and composite output. This identity holds on both sides of the kink, so the welfare evaluation along the nonlinear perfect-foresight transition retains the kink. Appendix B.4 extends the approximation to unequal elasticities and general CRRA utility.

3 Survey Design: Estimating Reference-Price Elasticities and Persistence

In this section, I estimate reference-price elasticities and the speed at which reference prices update using hypothetical shopping scenarios in a Prolific survey. Observational purchase data do not record reference prices, and constructing them from past prices can confound their

[15] This differs from Benigno and Woodford (2005), where linear terms in the initial utility expansion arise from steady-state distortions.

effects with shifts in purchases across time.[16] In contrast, the survey directly elicits respondents' reference prices and intended purchases. I then present scenarios that hold the posted price fixed but frame it as a temporary deviation from what the product usually costs. Holding the rest of the choice environment fixed, changes in elicited reference prices and purchases relative to baseline identify the reference-price elasticities.

To infer how quickly reference prices update, I use a separate scenario in which I re-elicit respondents' reference prices after they encounter a hypothetical price increase on a single shopping trip.

3.1 Survey Design, Sample, and Reference-Price Descriptives

The survey was administered on Prolific in late April 2026, recruiting 1,501 U.S. respondents from Prolific's standard participant pool.[17] I exclude 69 respondents who fail attention checks or report implausible prices or quantities (see Appendix D.2), leaving 1,432 respondents for the descriptive analysis.

Table 1 compares the sample with the U.S. population aged 18–65 in the American Community Survey (ACS). The sample is younger, has a higher share of women, and has a much higher share of unemployed respondents (15.0% versus 3.6%). To address these differences, I also report weighted estimates of the reference-price elasticities and persistence using weights calibrated to the ACS population's joint age-sex distribution and employment margin (see Appendix D.6).

Respondents begin the survey by naming a product they buy frequently, the package size and the store where they usually buy their chosen product, anchoring the scenarios to a familiar shopping environment. They report the price they typically expect to pay for the product at that store, which I take as their initial reference price (median \$3.99). They also report how much they would buy over the next month at that price (median three units; Appendix Figure D.1 shows both distributions). Appendix Figures C.1–C.3 reproduce these screens.

Ninety percent of respondents shop for their chosen product at least once every two weeks, suggesting that they have a clear sense of the price they typically expect to pay (Figure 1, panel a). Reference prices are also salient when shopping: 67% report having their typically expected price in mind most of the time or always when shopping for the item (panel b). 83% of respondents judge a posted price equal to their reported expected price to be "about right," supporting its interpretation as the benchmark consumers use to evaluate deals (Appendix Table D.4).[18] Eighty-one percent of respondents cite their own past shopping experiences as

[16] For example, consumers may stock up during a sale and buy less afterward (Hendel and Nevo, 2006). Once the sale ends, the posted price exceeds a reference price constructed by averaging past prices, including the discount, so lower purchases due to earlier stockpiling may be misattributed to reference dependence.

[17] Every question screen is reproduced in Appendix C.

[18] Price expectations need not coincide with the internal reference prices consumers use to evaluate deals (Thomas and Menon, 2007).

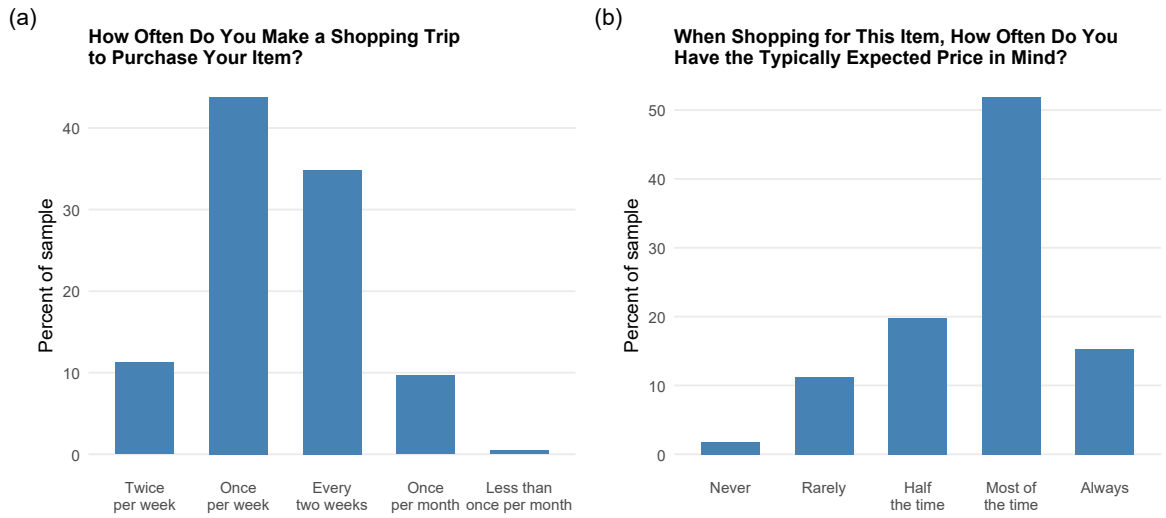
Table 1: Sample Composition and ACS Benchmarks

	Full sample	ACS 2024
Age (years), mean (sd)	38.7 (12.3)	41.0 (13.8)
Sex		
Female	784 (52.5%)	50.3%
Male	698 (46.7%)	49.7%
Prefer not to say	12 (0.8%)	–
Employment status		
Full-time	774 (56.0%)	61.6%
Unemployed, seeking work	207 (15.0%)	3.6%
Part-time	191 (13.8%)	13.8%
Not in paid work	139 (10.1%)	21.0%
Other	50 (3.6%)	–
Starting a new job within a month	20 (1.4%)	–
Student		
Yes	214 (15.3%)	11.5%

Notes: The sample column covers all respondents in the Prolific sample ($N = 1{,}501$). For categorical variables, cells report counts and their shares of responses to the item; for numeric variables, the mean with the standard deviation in parentheses. The ACS column reports the same statistic for U.S. residents aged 18–65 in the 2024 one-year ACS; a dash indicates survey categories that do not have a direct ACS counterpart. The ACS counts an individual as unemployed only if they had no job and had actively looked for work in the previous four weeks, whereas the survey records respondents' self-reported employment status. Appendix D.6 describes the reweighting to ACS benchmarks.

a factor shaping the price they typically expect to pay (Figure 2). This supports modeling reference prices as backward-looking, as in Section 2.[19]

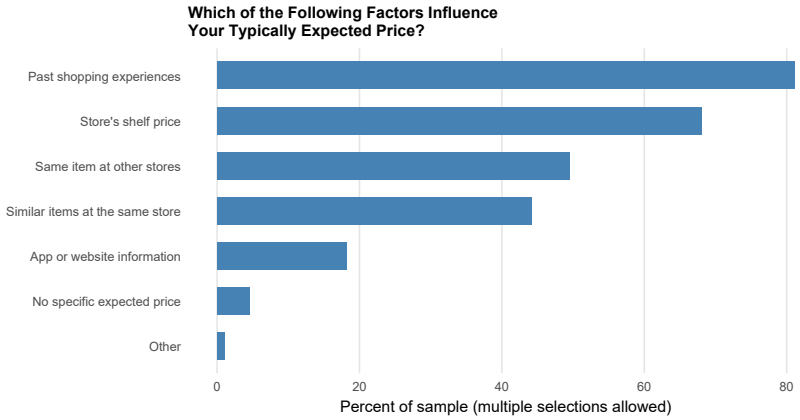
Figure 1: Purchase Frequency and Reference-Price Salience



Notes: Panel (a): how often respondents make a shopping trip for their chosen item. Panel (b): how often respondents report having the typically expected price in mind when shopping for the item. Sample: $N = 1{,}432$ after baseline exclusions.

[19] An extensive marketing literature constructs internal reference prices from past prices and estimates their effects on demand. For reviews, see Kalyanaram and Winer (1995); Mazumdar et al. (2005).

Figure 2: How Consumers Form Their Reference Prices



Notes: Share of respondents selecting each factor as an influence on the price they typically expect to pay; multiple selections allowed. Sample: $N = 1{,}432$ after baseline exclusions.

3.2 Identifying the Reference-Price Elasticities

Choice environment and identification. To fix ideas, suppose respondent i has M_i purchase occasions during the month. On each occasion m, they choose between the product they identified in the survey, f, and the available alternatives. I assume that this choice set is a finite approximation to the continuum of varieties in Section 2. Appendix A.1 shows that, under additional assumptions, this environment delivers the aggregate relative demands implied by equation (2.2). Respondent i's utility from each option on occasion m is:

$$u_{ifm} = \delta_{if} - \varepsilon \log P_i + \Gamma_b \log\left(\frac{R_i}{P_i}\right) + \xi_{ifm}, \qquad u_{ijm} = \delta_{ij} + \xi_{ijm}, \quad j \neq f,$$
$$\Gamma_b = \begin{cases} \Gamma_L, & P_i \geq R_i, \\ \Gamma_G, & P_i < R_i. \end{cases} \tag{3.1}$$

Here, P_i is the posted price respondent i faces for product f next month and R_i is their reference price for that product. The terms δ_{if} and δ_{ij} capture respondent-specific components of mean utility. The term $\Gamma_b \log(R_i/P_i)$ captures transaction utility (Thaler, 1985), reflecting the respondent's evaluation of the deal relative to their reference price. When P_i exceeds R_i, the purchase is perceived as a loss and carries a utility cost. The taste shocks ξ_{ijm} are i.i.d. standard Type I extreme-value draws across alternatives and purchase occasions, yielding logit choice probabilities. Expected monthly purchases of f equal M_i times its choice probability.

Let R_{0i} denote respondent i's baseline reference price. At baseline, $P_i = R_{0i}$, so transaction utility is zero. Increasing P_i while holding $R_i = R_{0i}$ combines the conventional price effect, governed by ε, with the transaction-utility effect, governed by Γ_L. By contrast, a scenario with a lower reference price, $R_i < R_{0i}$, holding $P_i = R_{0i}$ and the rest of the choice environment

fixed, affects choice *only* through the perceived deal. The following proposition formalizes this argument.

Proposition 3 (Identification of reference-price elasticities)**.** *Under the choice model above, suppose the posted price is fixed at the baseline reference price, $P_i = R_{0i}$, and the mean-utility components $\{\delta_{ij}\}_j$ and number of purchase occasions M_i remain unchanged.*

Let q_{0i} and q_{ib} denote purchases of f at reference prices R_{0i} and R_{ib}, respectively, where $b \in \{L, G\}$ and $R_{iL} < R_{0i} < R_{iG}$. Let ω_i denote the baseline choice probability of f. For each b,

$$\lim_{\omega_i \to 0} \frac{\mathbb{E}[q_{ib}]}{\mathbb{E}[q_{0i}]} = \left(\frac{R_{ib}}{R_{0i}}\right)^{\Gamma_b}. \tag{3.2}$$

Appendix D.1 provides the derivation. With the posted price and the rest of the choice environment fixed, quantity responses to exogenous reference-price shifts below R_{0i} identify Γ_L, while responses to shifts above R_{0i} identify Γ_G.[20]

Hypothetical shopping scenarios. To implement this design, I present each respondent with two hypothetical shopping scenarios involving the product, package size, and store they have already identified. Both scenarios keep next month's posted price equal to the respondent's baseline reference price. They differ in what the product usually costs, framing the same posted price as a temporary increase in the loss scenario and a temporary discount in the gain scenario.

Consider a respondent whose baseline reference price is \$5. In the loss scenario, the respondent is told that the product usually costs \$4 at that store, will temporarily increase to \$5 next month, and will return to \$4 afterward. They first report how many units they would buy next month at \$5. The next screen restates this price path and asks: "What price per unit would you typically expect to pay for this item after next month?" I take this response as the respondent's reference price in the loss scenario. Figure 3 illustrates the scenario.

In the gain scenario, the same respondent is told that the product usually costs \$6, will be on sale for \$5 next month, and will return to \$6 afterward. They again report how many units they would buy next month at \$5, followed by the price they would typically expect to pay after next month. Thus, the scenarios are designed to make the same \$5 purchase feel like a worse deal when the product usually costs \$4 and a better deal when it usually costs \$6. The aim is to shift demand by changing the perceived attractiveness of the deal while holding the posted price fixed.

Across respondents, the percentage gap between the baseline reference price and the scenario's usual price is randomly assigned to be 12%, 20%, or 28%. In the example above, a usual price of \$4.40 instead of \$4 makes next month's \$5 price a smaller temporary increase, so the purchase may feel like a better deal and induce a smaller reduction in intended purchases.

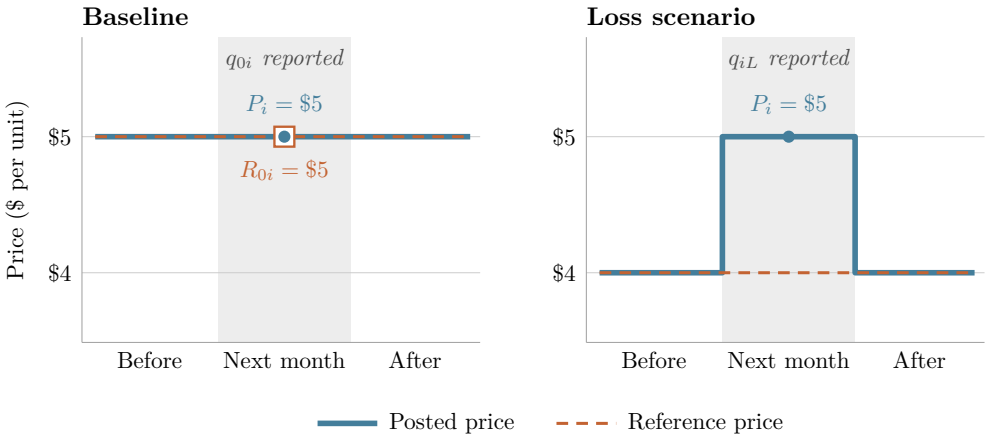
[20] The small-ω_i approximation treats the choice set as a counterpart to the continuum of varieties in Section 2. For $\omega_i > 0$, the approximation in Proposition 3 yields a lower bound on Γ_b.

Conversely, a usual price of \$3.60 makes the same \$5 price a larger temporary increase, potentially making the purchase feel like a worse deal and inducing a larger reduction in intended purchases. Consistent with this prediction, larger assigned gaps in the loss scenario are associated with larger average reductions in intended purchases (Appendix Table D.3, Panel A). Randomizing this gap therefore provides the exogenous variation used to identify Γ_L and Γ_G, while holding the posted price and the rest of the choice environment fixed.

For each respondent, the same assigned percentage places the usual price below baseline in the loss scenario and above baseline in the gain scenario. The order of the loss and gain scenarios is also randomized. The estimates use respondents' reported reference prices, which need not equal the usual prices stated in the scenarios. The question screens are reproduced in Appendix Figures C.4–C.7.

Each quantity question instructs respondents to hold their income and the prices of other goods fixed and to report purchases for next month's use only. Respondents are explicitly told to ignore buying extra for later use or delaying purchases until a later month; reporting zero purchases is allowed. These instructions are intended to align reported purchases with next month's consumption, consistent with the model.

Figure 3: Illustration of the Loss Scenario

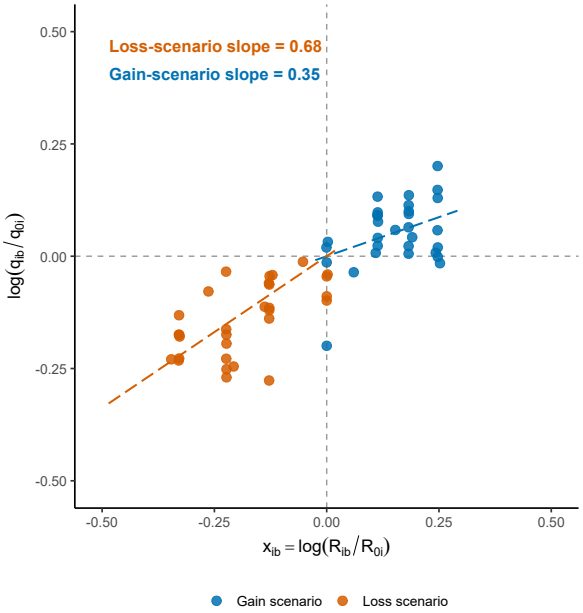


Notes: The dashed line illustrates a respondent whose reported reference price equals the usual price.

Figure 4 plots purchases against reference prices in both scenarios, with each variable expressed as a log change from the respondent's baseline. With the posted price held fixed, a lower reference price discourages purchases more strongly than a higher reference price encourages them. Appendix Table D.3 reports the raw purchase responses by assigned price gap.

Estimation. Let $x_{ib} = \log(R_{ib}/R_{0i})$ denote the change in respondent i's elicited reference price relative to baseline in scenario b. Under the small-share approximation in Proposition 3,

Figure 4: Demand Responses to Reference-Price Shifts



Notes: Binned means (30 equal-frequency bins per scenario) of the log quantity response $\log(q_{ib}/q_{0i})$ against the elicited reference-price gap $x_{ib} = \log(R_{ib}/R_{0i})$, where R_{ib} is the reference price elicited in scenario b and R_{0i} the baseline reference price, which equals the posted price. The dashed lines represent least-squares fits to the individual responses in each scenario, with the intercepts fixed at zero. The sample applies the baseline exclusions and reference-price filter and excludes observations with zero reported purchases (loss $N = 1{,}182$; gain $N = 1{,}212$).

expected purchases satisfy:

$$\mathbb{E}[q_{ib} \mid q_{0i}, x_{ib}] = q_{0i} \exp(\Gamma_b x_{ib}) \,. \tag{3.3}$$

I estimate this restriction by Poisson pseudo-maximum likelihood, separately for the loss and gain scenarios. This estimator is a natural choice for quantity responses recorded as counts.[21] The restriction further imposes that when the respondent's reference price is unchanged, expected purchases should not differ from baseline.

Imperfect understanding of the hypothetical scenarios or lapses in attention may introduce measurement error into elicited reference prices, potentially biasing the elasticity estimates. The extent of this error may vary with the order in which the scenarios are presented: in the second scenario, familiarity with the scenario may improve comprehension, while survey fatigue may reduce attention. I therefore instrument x_{ib} with the randomly assigned log ratio of the scenario's usual price to the baseline reference price, denoted by z_{ib}. The exclusion restriction holds if z_{ib} affects purchases only through the respondent's reference price.

I implement this instrumental-variable strategy with a control function (Wooldridge, 2015).

[21] The Poisson estimator is consistent when the conditional mean is correctly specified; quantities need not follow a Poisson distribution (Gourieroux et al., 1984; Santos Silva and Tenreyro, 2006).

Let $O_{ib} = 1$ indicate that scenario b is presented first. The first stage is:

$$x_{ib} = \pi_0 + \pi_1 z_{ib} + \pi_O O_{ib} + \pi_{zO} O_{ib} z_{ib} + v_{ib}. \tag{3.4}$$

This specification allows the first-stage intercept and slope to vary by presentation order. Let $\hat{v}_{ib}$ denote the first-stage residual, capturing variation in x_{ib} not explained by the assigned gap z_{ib} or presentation order.

In the second stage, I include $\hat{v}_{ib}$ and its interaction with presentation order in the Poisson regression:

$$\mathbb{E}[q_{ib} \mid q_{0i}, x_{ib}, \hat{v}_{ib}, O_{ib}] = q_{0i} \exp(\Gamma_b x_{ib} + \delta \hat{v}_{ib} + \delta_O O_{ib} \hat{v}_{ib}) \,. \tag{3.5}$$

Under the control-function assumptions, controlling for $\hat{v}_{ib}$ allows Γ_b to be identified from variation in x_{ib} induced by the randomly assigned price gap z_{ib}.

I apply an additional filter that restricts elicited scenario reference prices to the ranges described in Appendix D.2.[22] The filter leaves 1,238 loss-scenario and 1,223 gain-scenario observations. I compute standard errors for the control-function estimates using a respondent-level bootstrap with 1,000 replications (Appendix D.3).

Main estimates. Panels A and B of Table 2 report estimates of Γ_L and Γ_G, respectively. Column (1) reports Poisson pseudo-maximum-likelihood estimates without the control function. Column (2) presents control-function estimates without correcting for the order in which the scenarios are presented. Column (3), the preferred specification, allows the first-stage intercept and slope and the second-stage residual coefficient to vary by presentation order. Column (4) re-estimates this specification using weights calibrated to match ACS demographic benchmarks.

The preferred specification in column (3) yields $\hat{\Gamma}_L = 0.875$ and $\hat{\Gamma}_G = 0.326$. At baseline, the posted price equals the reference price. Holding the posted price fixed, the estimates imply that a 1% reduction in the reference price lowers expected purchases by approximately 0.9%, while a 1% increase raises them by approximately 0.3%. ACS reweighting changes the loss-side estimate little, from 0.875 to 0.933. Appendix Table D.5 reports a joint specification on the balanced sample, showing that Γ_L exceeds Γ_G at the 1% significance level. The estimates imply a loss-to-gain elasticity ratio of approximately 2.7, indicating substantially greater sensitivity to unfavorable than favorable reference-price gaps. This asymmetry is broadly consistent with experimental evidence on loss aversion (Tversky and Kahneman, 1992).

Robustness. Appendix Tables D.6 and D.7 report elasticity estimates across several specifications and sample restrictions. Estimates of Γ_L, which governs the first-order welfare loss from reference dependence during inflationary episodes (as discussed in Section 2), remain

[22] In the loss scenario, for example, reference prices substantially above the baseline reference price suggest a misunderstanding of the stated return to a lower usual price after next month.

Table 2: One-Sided Reference-Price Elasticities

	PPML (1)	CF (2)	CF + Order (3)	CF + Order (ACS wt.) (4)
Panel A. Loss side (Γ_L)				
$\hat{\Gamma}_L$	0.788***	0.873***	0.875***	0.933***
	(0.084)	(0.095)	(0.097)	(0.131)
Respondents	1,238	1,238	1,238	1,187
First-stage F		751.7	375.6	338.1
Panel B. Gain side (Γ_G)				
$\hat{\Gamma}_G$	0.347***	0.324***	0.326***	0.310**
	(0.105)	(0.108)	(0.109)	(0.141)
Respondents	1,223	1,223	1,223	1,174
First-stage F		523.6	262.0	264.3

Notes: Poisson pseudo-maximum likelihood of equation (3.3), with $\log q_0$ as offset and no intercept, after the baseline exclusions and the reference-price screen described in Appendix D.2. Column (1) treats the elicited gap as exogenous and reports respondent-clustered standard errors. Columns (2)–(4) add the control function of equation (3.5), with the randomized gap as instrument; column (2) omits the presentation-order terms, and column (4) re-estimates column (3) with weights calibrated to the 2024 ACS population ages 18–65 (Appendix D.6), on the respondents inside the calibration universe. Columns (2)–(4) report respondent-level bootstrap standard errors (1,000 replications). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

broadly stable, ranging from 0.69 to 1.03 across the checks in column (3). I restrict the sample to respondents whose baseline price feels "about right," since this assessment is consistent with zero transaction utility at baseline, as assumed in the identification argument. This restriction leaves the loss-side estimate almost unchanged ($\hat{\Gamma}_L = 0.864$). Allowing for an intercept and order-specific level controls relaxes the restriction that purchases equal baseline purchases when the reference price is unchanged, yielding $\hat{\Gamma}_L = 0.789$. Restricting the sample to perishable products, for which stockpiling is less feasible, yields $\hat{\Gamma}_L = 0.766$.[23] Log-linear 2SLS, which changes the functional form and excludes zero purchases, yields $\hat{\Gamma}_L = 0.732$. Allowing the elasticities to vary by presentation order checks for sensitivity to learning or fatigue; neither difference is statistically significant at the 5% level in the control-function specifications. Estimates of Γ_G remain positive in most specifications, although their magnitude and statistical significance are more sensitive to the specification and sample restrictions.

Appendix E describes an alternative empirical design that estimates reference-price elasticities using observational data from NielsenIQ Retail Measurement Services (RMS). The analysis covers 22 product categories selected to align with respondents' product choices in the survey. The average estimate of Γ_L across the categories reported in Appendix Table E.1 is 0.900, while the expenditure-weighted average reported in Appendix Table E.2 is 0.761. Both are similar in magnitude to the survey estimate of 0.875. The expenditure-weighted estimate of Γ_G is 1.117 (Appendix Table E.2), substantially above the survey estimate of 0.326. This difference is

[23] Products are classified as perishable based on respondents' own assessments.

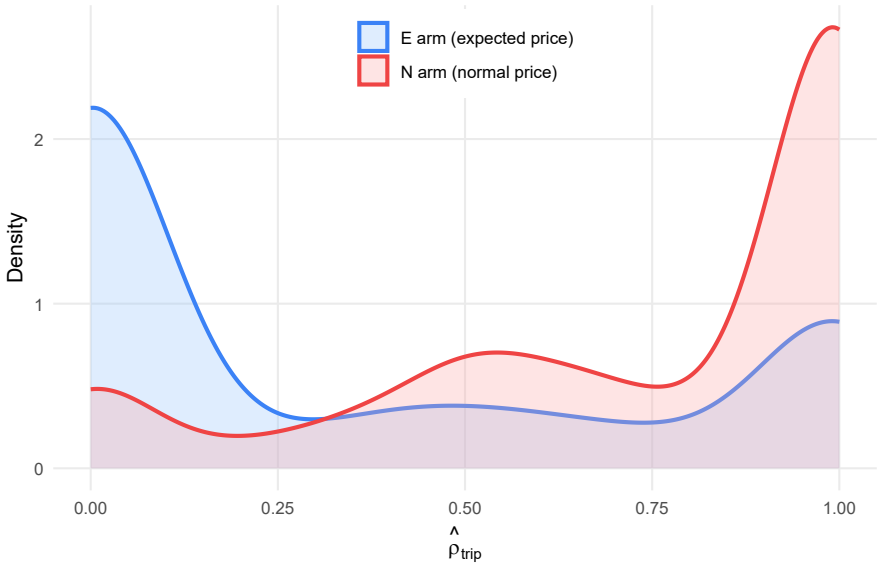
consistent with stockpiling during temporary discounts: scanner purchases can reflect inventory accumulation, whereas the survey asks about purchases for next month's use.[24]

3.3 Reference-Price Persistence

To estimate the speed at which reference prices update, respondents consider a scenario in which the price of their product has been equal to their baseline reference price on recent shopping trips and rises by 20% on the next trip. A randomly selected half report the price they would typically expect to pay on subsequent shopping trips (Appendix Figure C.10), while the other half report the price that would feel normal, defined as neither too high nor too low (Appendix Figure C.11). This distinction matters because respondents may come to expect a higher price while still feeling that it is too high for the product.

Inverting the reference-price law of motion in equation (2.3) yields a trip-level estimate of ρ for each respondent. Figure 5 shows the distribution of these estimates separately for the two questions. Respondents who report expected prices are more concentrated near zero, indicating full adjustment after one shopping trip, whereas those reporting normal prices are more concentrated near one, indicating little adjustment. Mean trip-level persistence is 0.33 for expected prices and 0.74 for normal prices.

Figure 5: Distribution of Trip-Level Reference-Price Persistence



Notes: Kernel densities of respondent-level trip persistence by randomized question, restricted to $\hat{\rho}_{\text{trip},i} \in [0, 1]$. A value of 0 means the reference moves fully to the new posted price after one trip; 1 means it does not move. The expected-price question (E arm) has $N = 651$; the normal-price question (N arm) has $N = 629$.

I convert these estimates to the quarterly level using each respondent's reported shopping

[24]The five largest category-level estimates of Γ_G are for ice cream, butter and margarine, bacon, soup and ramen, and frozen pizza. The average of these five estimates is approximately 2.00 (Appendix Table E.1).

frequency (Appendix Figure C.8) and then average across respondents. The resulting quarterly estimates are 0.53 for the normal-price question and 0.22 for the expected-price question. Table 3 reports the quarterly estimates by shopping frequency, together with the ACS-weighted overall estimates.

Table 3: Quarterly Reference-Price Persistence

	Expected-price question		Normal-price question	
Shopping frequency	N	Mean $\hat{\rho}_{\text{qtr}}$	N	Mean $\hat{\rho}_{\text{qtr}}$
Twice per week	73	0.192	63	0.608
Once per week	281	0.215	289	0.509
Every two weeks	235	0.222	208	0.534
Once per month	57	0.237	68	0.540
Overall	646	0.217	628	0.531
Overall (ACS-weighted)	617	0.211	607	0.544

Notes: Quarterly persistence is calculated using each respondent's reported shopping frequency and then averaged across respondents. Sample restrictions and frequency mappings appear in Appendix D.5; ACS weighting is described in Appendix D.6.

The normal-price estimate serves as the baseline for the quantitative analysis. The estimate implies that approximately 47% of the log price gap closes within one quarter. This choice is supported by responses to a follow-up question asking the expected-price group to assess how the higher price would feel after a month. Among respondents who judged their baseline price to be "about right" and whose price expectations had adjusted almost fully to the new price after one shopping trip, 76% say that the higher price would still feel too high after a month (Appendix Table D.4, Panel B). This suggests that respondents can update their price expectations while still viewing the purchase as a bad deal.

Taken together, the survey estimates indicate that purchases respond more strongly to unfavorable than favorable reference-price gaps and that reference prices adjust gradually following a price increase. The preferred estimates are $\Gamma_L = 0.87$, $\Gamma_G = 0.33$, and quarterly persistence $\rho = 0.53$. In the model of Section 2, the loss-side elasticity governs the strength of the first-order welfare effect, while persistence determines how long reference prices lag behind posted prices. The next section uses these estimates to quantify the welfare costs of inflationary episodes and compare them with those implied by the standard model.

4 Quantitative analysis

I use the survey estimates to quantify how reference dependence changes the welfare costs of inflation. I evaluate these losses relative to those in the standard New Keynesian model, scaling cost-push shocks to generate the same peak quarterly inflation rate. Under the baseline calibration, the consumption-equivalent loss is 4.3 times as large, with the first-order cost accounting for approximately 79% of the total. I then examine how these results vary with

reference-price persistence and the size and persistence of the shock.

4.1 Environment

The model is the same as in Section 2, with the parameters governing reference dependence taken from the survey estimates in Section 3 (hereafter RD-RANK). I compare the welfare costs of inflation in this economy with those in the standard representative-agent New Keynesian model (RANK), obtained by setting both reference-price elasticities to zero. I also allow for interest-rate smoothing in the monetary-policy rule. Appendix F contains all the equilibrium conditions.

Starting at the deterministic zero-inflation steady state, I consider an unanticipated proportional increase in firms' marginal costs. Firms face real marginal cost $z_t \equiv \upsilon_t mc_t$, where mc_t is defined in Section 2 and the cost-push factor υ_t follows

$$\log \upsilon_t = \rho_u \log \upsilon_{t-1} + \varepsilon_t^u, \qquad \rho_u = 0.90, \tag{4.1}$$

with $\upsilon_t = 1$ in steady state. The innovation occurs once, and its subsequent effects are fully anticipated.

The survey estimates identify a larger reference-price elasticity for losses than for gains, $\Gamma_L > \Gamma_G$, creating a kink in demand at the firm's reference price. To characterize the firm's optimal pricing decisions in this setting, let $\mathcal{D}_t^+$ and $\mathcal{D}_t^-$ denote the right and left derivatives of its dynamic pricing objective with respect to its current log posted price.[25] The pricing conditions in symmetric equilibrium are:

$$\begin{aligned} P_t > R_t: &\quad \mathcal{D}_t^+ = 0, \\ P_t < R_t: &\quad \mathcal{D}_t^- = 0, \\ P_t = R_t: &\quad \mathcal{D}_t^+ \leq 0 \leq \mathcal{D}_t^-. \end{aligned} \tag{4.2}$$

When $P_t \neq R_t$, the pricing and costate conditions take the form of equations (2.9)–(2.10), with Γ replaced by Γ_L when $P_t > R_t$ and by Γ_G when $P_t < R_t$.[26] When marginal cost exceeds an upper threshold $\overline{z}_t$, the firm has an incentive to raise its price from the reference; when it falls below a lower threshold $\underline{z}_t$, the firm has an incentive to cut it. Appendix B.2 derives these thresholds conditional on inherited states and continuation values.

When $P_t = R_t$, the one-sided inequalities hold for marginal costs $z_t \in [\underline{z}_t, \overline{z}_t]$. This inaction region generates endogenous nominal rigidity at a reference price that evolves with the firm's past pricing decisions. The reference-price costate in this regime measures the marginal value of a higher reference price when the posted price tracks it. Appendix F gives the corresponding

[25] The derivatives are normalized by $\Lambda_t Y_t^U$; their expressions at $P_t = R_t$ are given in Appendix B.2.

[26] In both conditions, mc_t is replaced by z_t.

Table 4: Baseline quarterly calibration

Parameter	Symbol	Value	Source or target
Discount factor	β	0.99	Standard quarterly value
CRRA coefficient	σ	2	Conventional curvature
Decreasing-returns parameter	α	1/3	Standard value
Inverse Frisch elasticity	φ	1	Standard value
CES demand elasticity	ε	4	Steady-state markup of one third
Loss-side reference elasticity	Γ_L	0.87	Survey estimate, Section 3
Gain-side reference elasticity	Γ_G	0.33	Survey estimate, Section 3
Reference persistence	ρ	0.53	Normal-price survey arm
Cost-push persistence	ρ_u	0.90	Persistent inflation episode
Interest-rate smoothing	ρ_i	0.80	Baseline policy rule
Inflation-rule coefficient	ϕ_π	1.50	Baseline policy rule
Output-rule coefficient	ϕ_y	0.125	Baseline policy rule
Calvo non-reset probability	θ	2/3	Three-quarter price duration
Rotemberg adjustment coefficient	ψ_p	52.94	Duration-preserving mapping
Perfect-foresight horizon	T	200	Numerical convergence
Target peak quarterly inflation	Peak π_t	1.00%	Matched-experiment normalization

Notes: The cost-push innovation is chosen separately in the RANK and RD-RANK models to deliver a 1% peak quarterly inflation rate. The welfare tables also report $\rho = 0.22$, the mean of the expected-price arm, and $\rho = 0.88$, the median of the normal-price arm (Appendix D.5). The duration-preserving mapping gives $\psi_p = 52.94$ at $\varepsilon = 4$ and $\theta = 2/3$.

recursion.

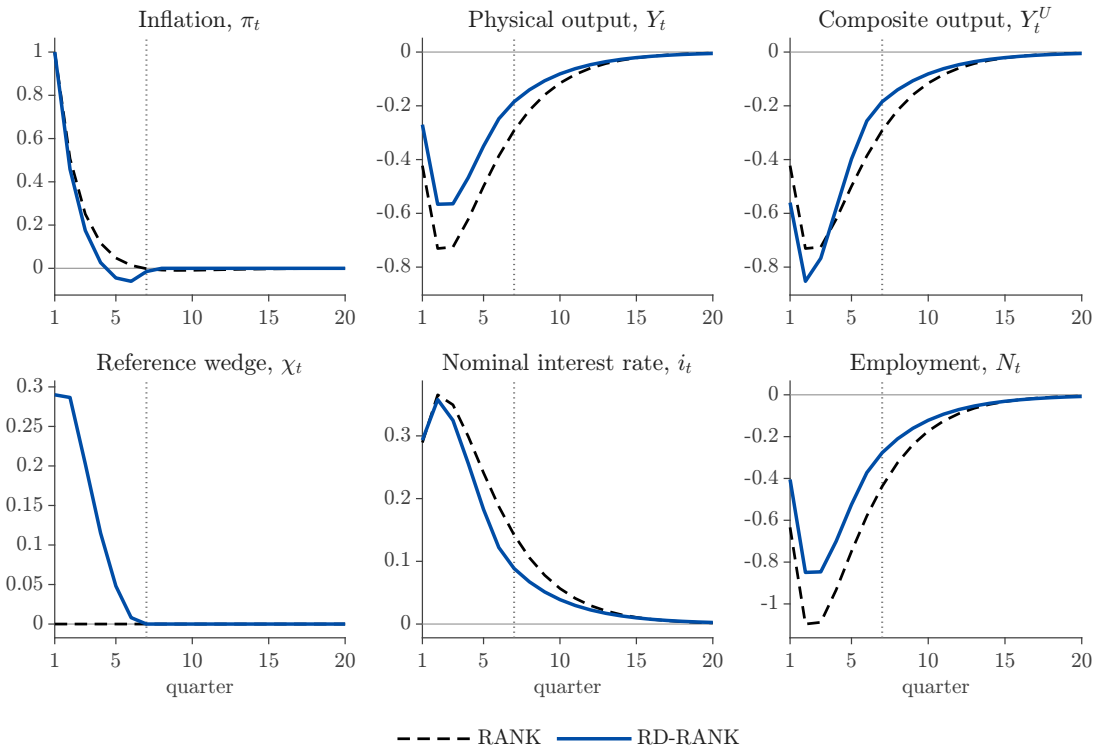
To capture endogenous switches between pricing regimes, I solve the nonlinear model under perfect foresight over 200 quarters, imposing the steady state as the terminal condition. The algorithm solves the nonlinear equilibrium conditions conditional on a conjectured sequence of pricing regimes. It then updates the regime sequence until the resulting equilibrium path satisfies the conditions defining each regime. Appendix F describes the solution algorithm and reports the numerical checks.

4.2 Calibration and macroeconomic dynamics

I scale the initial cost-push innovation ε_0^u separately in each model to produce a 1% peak in quarterly inflation. Table 4 reports the calibrated parameters. The survey estimates imply $\Gamma_L = 0.87$ and $\Gamma_G = 0.33$. I set reference-price persistence to $\rho = 0.53$, the mean quarterly persistence in the normal-price arm of the survey (Section 3.3). Household utility has curvature $\sigma = 2$, and the CES demand elasticity is $\varepsilon = 4$. I set the Rotemberg coefficient to $\psi_p = 52.94$ so that the model without reference dependence matches the Phillips-curve slope of a Calvo model with an average price duration of three quarters. I use the same coefficient in both models. In RD-RANK, I set the employment subsidy to the midpoint of the interval supporting the zero-inflation steady state derived in Proposition B.1.

Columns 4 and 5 of Panel A in Table 5 show that RD-RANK requires a smaller cost-push

Figure 6: Matched-peak cost-push shock with lower persistence: $\rho_u = 0.70$



Notes: Responses to a cost-push shock with persistence $\rho_u = 0.70$, scaled separately in each model to produce a 1% peak in quarterly inflation (innovations ε_0^u of 10.83 log points in RANK and 10.30 in RD-RANK). Reference-price persistence is $\rho = 0.53$; all other parameters are those of Table 4. Quarter 1 is the quarter of the shock, $t = 0$. Physical output Y_t, composite output Y_t^U, employment N_t, and the gross nominal interest rate are expressed as log deviations from steady state multiplied by 100; inflation is $100\,\pi_t$ and the reference wedge is $100\,\chi_t$. The two output panels share a vertical scale. The vertical dotted line marks quarter 7, when the RD-RANK model returns to $P_t = R_t$; inflation is exactly zero from quarter 8 onward.

innovation to generate the same inflation peak, consistent with the higher pass-through discussed in Section 2. Figure 6 shows the impulse responses for $\rho_u = 0.70$. Appendix Figure F.1 shows the baseline case, $\rho_u = 0.90$. The smaller innovation reduces cost pressure, while forward-looking firms anticipate eventually setting $P_t = R_t$ as the shock dissipates. Rotemberg adjustment costs encourage firms to spread this return over several periods. These forces contribute to faster disinflation in RD-RANK and, through the monetary-policy rule, a faster subsequent decline in the nominal interest rate. Inflation in RD-RANK becomes negative in quarter 5 and reaches a trough of approximately -0.060% in quarter 6.[27] Prices equal their references in quarter 7, and inflation is zero from quarter 8 onward.

Physical output and employment contract less in RD-RANK, while composite output initially falls more. A positive reference wedge χ_t reduces the consumption utility delivered by

[27]The small inflation undershoot in RANK reflects interest-rate smoothing.

a given physical bundle.[28] With $\sigma > 1$, the intertemporal elasticity of substitution is below one, making households reluctant to accept a temporary decline in consumption utility. They therefore have an incentive to purchase more physical goods to partly offset this loss. This response limits the contraction in physical output and employment but does not fully restore consumption utility. At the quarter-2 trough, physical output falls by 0.57% and composite output by 0.85%. In RANK, where the two output measures coincide, both fall by 0.73%. The reference wedge disappears when prices equal their references in quarter 7. The associated reduction in consumption utility is central to the welfare losses examined in the next subsection.

4.3 Welfare losses

I measure welfare losses from the episode relative to the initial steady state. To this end, I calculate discounted household utility from composite consumption C_t^U and labor N_t along each nonlinear transition. Holding labor at its steady-state level, I define the consumption-equivalent loss λ as the permanent proportional reduction in steady-state consumption that generates the same lifetime utility loss (see Lucas, 2003):

$$\sum_{t=0}^{\infty} \beta^t \left[\frac{(\bar{C}^U)^{1-\sigma}}{1-\sigma} - \frac{\left[(1-\lambda)\bar{C}^U\right]^{1-\sigma}}{1-\sigma} \right] = \Delta W. \tag{4.3}$$

Here, $\Delta W \equiv \sum_{t=0}^{T-1} \beta^t (\bar{u} - u_t)$ is the discounted utility loss along the transition, where u_t denotes period household utility and $\bar{u}$ its steady-state value. I report consumption-equivalent losses in percentage terms as 100λ.[29]

Panel A of Table 5 reports welfare losses for a cost-push shock with persistence $\rho_u = 0.90$, scaled separately in each model to generate a 1% peak in quarterly inflation. The consumption-equivalent loss is 0.0087% in RANK, compared with 0.0266% in RD-RANK when $\rho = 0.22$, the expected-price arm, and 0.0376% when $\rho = 0.53$. Reference dependence therefore raises the welfare cost by factors of 3.05 and 4.31, respectively. Using the median of the normal-price arm instead, $\rho = 0.88$, raises the RD-RANK loss to 0.1018%, 11.7 times the RANK loss, with the reference wedge accounting for 92.1% of the second-order loss. Slower reference-price updating makes the utility loss from a given price increase more persistent (Appendix Figure F.2).

To interpret these losses, I use the extension of Section 2's welfare approximation to the case $\Gamma_L > \Gamma_G$ and general CRRA utility derived in Appendix B.4. For the cost-push experiment,

[28] By the aggregation identity (2.14), $\chi_t = \log(Y_t/Y_t^U)$. When $P_t > R_t$, it equals $\Gamma_L \log(P_t/R_t)/(\varepsilon - 1)$, linking the gap between physical and composite output to the gap between posted and reference prices.

[29] A permanent loss of λ has the same present value as a one-time loss of $\lambda/(1-\beta)$ quarters of steady-state consumption, or 25λ years at $\beta = 0.99$. The baseline RD-RANK loss of 0.0376% thus corresponds to about 0.94% of one year's consumption, the unit used in Section 5.

the normalized second-order loss is[30]

$$\mathcal{L}^{(2)} = \sum_{t=0}^{T-1} \beta^t \left[\frac{\sigma - 1}{2} (y_t^U)^2 + \frac{1+\varphi}{2(1-\alpha)} \widetilde{y}_t^2 + \chi_t + \frac{\psi_p}{2} \pi_t^2 \right], \tag{4.4}$$

where $\widetilde{y}_t = y_t - a_t$ (lowercase quantities denote log deviations from steady state). Relative to the log-utility case in Section 2, general CRRA utility introduces the additional term $\frac{\sigma-1}{2}(y_t^U)^2$. This term captures the additional welfare cost of the decline in composite consumption when $\sigma > 1$: marginal utility rises more sharply as consumption falls, making further reductions increasingly costly. The last two columns of Panel B in Table 5 show that the approximation closely reproduces the exact consumption-equivalent losses, with relative errors below 0.5% in RD-RANK and 1.4% in RANK. Column 6 of Panel B in Table 5 shows that the reference wedge χ_t accounts for 69.4% and 78.5% of the approximated RD-RANK loss at $\rho = 0.22$ and $\rho = 0.53$, respectively. The increase in welfare losses is therefore primarily accounted for by the first-order contribution of the reference wedge, χ_t.

Panel A of Table 5 shows how welfare losses vary with cost-push persistence. For a 1% peak in quarterly inflation, reducing cost-push persistence from $\rho_u = 0.90$ to 0.70 lowers the RD-RANK loss to 0.0130%, or 3.24 times the RANK loss, compared with 4.31 in the baseline. More persistent shocks keep posted prices above their references for longer, increasing the cumulative welfare loss. With $\rho_u = 0.90$, halving the inflation peak to 0.5% lowers losses to 0.0022% in RANK and 0.0158% in RD-RANK, raising their ratio to 7.10. As shocks become larger and peak inflation rises, the quadratic terms in the welfare decomposition become more important relative to χ_t.[31]

[30] Utility losses are normalized by steady-state marginal utility times consumption, $(\bar{C}^U)^{1-\sigma}$.

[31] Under the baseline parameters, a 4% peak in quarterly inflation gives the first-order reference contribution a share of approximately 51% of the second-order RD-RANK loss, roughly matching the three quadratic components combined.

Table 5: Welfare losses from a matched cost-push shock and their decomposition

Panel A. Exact consumption-equivalent losses

Peak (% per quarter)	ρ_u	ρ	Innovation ε_0^u (log points)		Loss, $100 \times \lambda$ (% of $\bar{C}$, permanent)		Difference (pp)	Ratio (RD/RANK)
			RANK	RD-RANK	RANK	RD-RANK		
1.0	0.90	0.22	8.36	7.95	0.0087	0.0266	0.0179	3.052
1.0	0.90	0.53	8.36	7.89	0.0087	0.0376	0.0289	4.310
1.0	0.90	0.88	8.36	8.15	0.0087	0.1018	0.0931	11.673
1.0	0.70	0.53	10.83	10.30	0.0040	0.0130	0.0090	3.244
0.5	0.90	0.53	4.23	3.98	0.0022	0.0158	0.0136	7.100

Panel B. Component shares of the second-order loss

Model	Peak (%)	ρ_u	Share of second-order loss (%)				Approximate	Exact
			$\frac{\sigma-1}{2}(y_t^U)^2$	$\frac{1+\varphi}{2(1-\alpha)}\widetilde{y}_t^2$	$\boldsymbol{\chi}_t$	$\frac{\psi_p}{2}\pi_t^2$	$100 \times \lambda^{(2)}$	$100 \times \lambda$
RANK	1.0	0.90	6.2	18.6	**0.0**	75.2	0.0086	0.0087
RD-RANK, $\rho = 0.22$	1.0	0.90	2.2	5.0	**69.4**	23.3	0.0265	0.0266
RD-RANK, $\rho = 0.53$	1.0	0.90	1.8	3.3	**78.5**	16.5	0.0375	0.0376
RD-RANK, $\rho = 0.88$	1.0	0.90	1.0	0.9	**92.1**	6.0	0.1017	0.1018
RANK	1.0	0.70	2.7	8.2	**0.0**	89.1	0.0040	0.0040
RD-RANK, $\rho = 0.53$	1.0	0.70	0.9	1.3	**72.4**	25.4	0.0130	0.0130
RANK	0.5	0.90	6.4	19.2	**0.0**	74.3	0.0022	0.0022
RD-RANK, $\rho = 0.53$	0.5	0.90	1.0	1.9	**87.3**	9.8	0.0158	0.0158

Notes: Panel A reports exact discounted CRRA losses at $\sigma = 2$. The innovation ε_0^u is the impact value of the cost-push innovation in equation (4.1), in log points, so 8.36 raises marginal cost by 8.7% on impact; it is chosen separately within each row to match the peak quarterly inflation rate in the first column. Losses are consumption equivalents, $100 \times \lambda$ from equation (4.3): the permanent percentage reduction in steady-state consumption $\bar{C}$ that produces the same lifetime utility loss as the episode, holding labor at its deterministic steady state. Composite and physical consumption coincide in the steady state, where the reference wedge is zero, so $\bar{C}$ is the same object in both models. The difference is in percentage points and the ratio is RD-RANK over RANK. RANK is independent of reference persistence ρ, but not of cost-push persistence ρ_u or the size of the episode. The rows at $\rho = 0.88$ use the median quarterly persistence in the normal-price arm with trip-level persistence capped at 0.995 (Appendix D.5). Panel B reports each component's share of the loss in equation (4.4), evaluated on the nonlinear path of each experiment; shares sum to 100 up to rounding. The last two columns report approximate and exact consumption-equivalent losses. The second-order consumption-equivalent conversion is described in Appendix F; its correction is smaller than 10^{-4} percentage point and is excluded from the component shares.

5 Welfare costs during the 2021–2023 inflation episode

Section 4 examines the welfare implications of reference dependence following a cost-push shock in a representative-agent economy. To quantify its importance during the 2021–2023 U.S. inflation surge, I extend the framework to a two-sector heterogeneous-agent New Keynesian model. Two considerations motivate this extension. First, reference-dependent demand has been extensively documented for frequently purchased goods, and exposure to this sector varies across the income distribution. Second, recent work emphasizes sticky wages and heterogeneity in net nominal positions as important determinants of welfare during the episode, motivating their inclusion in the model.[32] I fit the model to U.S. data to assess whether reference dependence is a quantitatively important driver of welfare losses during this episode.

5.1 Model

Time is discrete and quarterly. The economy is populated by a unit continuum of households $i \in [0, 1]$ that face idiosyncratic earnings risk and differ in wealth and discount factors. Households save in a one-period nominal bond subject to a borrowing constraint. They consume goods from two sectors, combined using a CES aggregator. Expenditure shares across the two sectors vary with household earnings. Sector R features reference-dependent demand as developed in Section 2; sector S has standard CES demand. In both sectors, monopolistically competitive firms produce using labor and face Rotemberg price adjustment costs. Nominal wages are sticky and set by unions, which allocate hours uniformly across households. The government issues nominal debt and levies labor-income taxes to finance spending on goods and services, transfers, and subsidies. Monetary policy follows a Taylor rule.

5.1.1 Households

Household i enters quarter t with state $(\beta_{it}, z_{it}, b_{i,t-1})$, where $b_{i,t-1}$ denotes inherited bond holdings, z_{it} its earnings state, and β_{it} its discount factor. The earnings state z_{it} follows a finite-state Markov chain with transition matrix Π_z and determines labor efficiency $e_{z_{it}}$, normalized to have a stationary mean of one. Independently of the earnings process, $\beta_{i,t+1} = \beta_{it}$ with probability $1 - p_\beta$. With probability p_β, the household draws a new discount factor, equal to β_H with probability $\pi_\beta(\beta_H)$ and $\beta_L < \beta_H$ otherwise (Auclert et al., 2025). All households supply N_t hours, determined by union labor demand. Appendix G.1 gives further details of the household problem.

[32] See Afrouzi et al. (2024) on real-wage erosion and costly wage adjustment, and Pallotti et al. (2024) on redistribution across households with different net nominal positions.

Preferences and consumption baskets. Household i's period utility is

$$u(c_{it}) - v(N_t) = \frac{c_{it}^{1-\sigma}}{1-\sigma} - \varphi_N \frac{N_t^{1+\varphi}}{1+\varphi},$$

where σ is relative risk aversion, φ is the inverse Frisch elasticity, and φ_N scales labor disutility. Composite consumption is

$$c_{it} = \left[\omega_R(z_{it})^{1/\nu}(c_{it}^R)^{(\nu-1)/\nu} + (1-\omega_R(z_{it}))^{1/\nu}(c_{it}^S)^{(\nu-1)/\nu}\right]^{\nu/(\nu-1)}. \tag{5.1}$$

Here ν is the elasticity of substitution across sectors and $\omega_R(z_{it})$ is household i's weight on sector R. The quantities c_{it}^R and c_{it}^S are consumption in utility units. In sector S, these coincide with physical quantities. In sector R, the reference-dependent aggregator of Section 2 implies that $e^{\chi_{R,t}}$ physical units are required to deliver one unit of c_{it}^R, where $\chi_{R,t}$ is the reference wedge for sector-R goods.

Let $P_{R,t}$ and $P_{S,t}$ denote posted sectoral prices, $\tilde{q}_t = P_{R,t}/P_{S,t}$ their relative price, and $\Pi_{j,t} = P_{j,t}/P_{j,t-1}$ gross inflation in sector j. The effective relative price of sector-R consumption is $q_t = e^{\chi_{R,t}}\tilde{q}_t$. Expressing expenditure in units of the sector-S good, cost minimization gives household i's consumption price index

$$P_{z_{it}}^C(q_t) = \left[\omega_R(z_{it})q_t^{1-\nu} + 1 - \omega_R(z_{it})\right]^{1/(1-\nu)}. \tag{5.2}$$

Thus $P_{z_{it}}^C(q_t)c_{it}$ is the expenditure required to attain composite consumption c_{it}. At the normalized steady state, $q_{\text{ss}} = 1$, the weight $\omega_R(z_{it})$ equals the household's sector-R expenditure share and satisfies $\partial \log P_{z_{it}}^C(q)/\partial \log q\Big|_{q=1} = \omega_R(z_{it})$. Away from steady state, expenditure shares also respond to relative prices.

Intertemporal household problem. Taking aggregate paths as given, household i chooses consumption c_{it} and end-of-quarter bond holdings b_{it} to maximize utility subject to its budget constraint. In the recursive problem, I suppress household subscripts and write $(\beta, z, b) = (\beta_{it}, z_{it}, b_{i,t-1})$, $c = c_{it}$, and $b' = b_{it}$. Primes on the exogenous states denote their next-quarter values. The common value function satisfies

$$V_t(\beta, z, b) = \max_{c, b' \geq 0} \left\{ u(c) - v(N_t) + \frac{\beta}{\zeta_t}\mathbb{E}\left[V_{t+1}(\beta', z', b') \mid \beta, z\right] \right\}, \tag{5.3}$$

subject to household i's budget and borrowing constraints, expressed in units of the sector-S good,

$$P_{z_{it}}^C(q_t)c_{it} + b_{it} = R_t^b b_{i,t-1} + (1-\tau_L)\mathcal{T}^w w_t e_{z_{it}} N_t + \xi_{z_{it}}\mathcal{D}_t + Tr_t^{\text{hh}}, \qquad b_{it} \geq 0, \tag{5.4}$$

where $w_t = W_t/P_{S,t}$ is the real wage, τ_L the labor-income tax rate, $\mathcal{T}^w$ the gross wage subsidy, $\mathcal{D}_t$ aggregate dividends, $\xi_{z_{it}}$ household i's dividend incidence weight, and Tr_t^{hh} the lump-sum transfer common to all households. The realized return on nominal bonds is

$$R_t^b = \frac{1 + i_{t-1}}{\Pi_{S,t}}, \tag{5.5}$$

where i_{t-1} is the nominal interest rate fixed in the preceding quarter. The variables $b_{i,t-1}$ and b_{it} denote inherited and new real bond holdings in units of the sector-S good. Aggregate demand shocks enter through the discount-factor shifter ζ_t, so household i's effective discount factor is β_{it}/ζ_t (Auclert et al., 2021).

Sector-R inflation affects the cost of consumption through the effective relative price q_t, which incorporates both relative prices and the reference wedge. Sector-S inflation also affects q_t and enters the realized real return on nominal bonds. For a given nominal interest rate, higher sector-S inflation reduces the real bond payoff in units of the sector-S good.

5.1.2 Firms and unions

Firms and price setting. In each sector $j \in \{R, S\}$, a continuum of monopolistically competitive firms produces differentiated goods using labor with constant returns and productivity $Z_t Z_{j,t}$, where Z_t is aggregate productivity and $Z_{j,t}$ is sectoral productivity. The within-sector CES elasticity is ϵ_j. Firms choose prices to maximize the present value of profits, subject to demand and quadratic price-adjustment costs with coefficient ψ_j. Constant production subsidies offset markup distortions in steady state.

In sector S, optimal pricing in a symmetric equilibrium gives the Rotemberg Phillips curve

$$(\Pi_{S,t} - 1)\Pi_{S,t} = \frac{\epsilon_S}{\psi_S}\left(\upsilon_{S,t} mc_{S,t} - \frac{\epsilon_S - 1}{\epsilon_S}\right) + \delta_t \mathbb{E}_t\left[\frac{Y_{S,t+1}}{Y_{S,t}}(\Pi_{S,t+1} - 1)\Pi_{S,t+1}^2\right], \tag{5.6}$$

where $Y_{S,t}$ is sectoral output, $mc_{S,t}$ is real marginal cost inclusive of the production subsidy, and $\upsilon_{S,t}$ is a cost-push shock. Firms discount nominal profits at $\delta_t = (1 + i_t)^{-1}$ (Auclert et al., 2021).

Sector-R demand and reference updating follow (2.2) and (2.3) in Section 2. Price setting satisfies the smooth branch conditions away from the reference price and the one-sided optimality conditions at the kink, derived in Appendix B. Appendix G.2 collects the equations governing firms' production, pricing, and dividends.

Unions and wage setting. A continuum of unions sets nominal wages for differentiated labor services. Each union represents all households and chooses its wage to maximize their average utility, taking household consumption-saving decisions as given (Auclert et al., 2021). Hours are allocated uniformly across households. Wage adjustment costs are quadratic in log

wage changes and are borne in utility units. In a symmetric equilibrium, gross wage inflation $\Pi_{w,t} = W_t/W_{t-1}$ satisfies

$$\log \Pi_{w,t} = \kappa_w \left[\varphi_N N_t^{1+\varphi} - \frac{(1-\tau_L)\mathcal{T}^w}{\mu_w \upsilon_{w,t}} w_t N_t \int_0^1 e_{z_{it}} \frac{c_{it}^{-\sigma}}{P_{z_{it}}^C(q_t)} \, \mathrm{d}i \right] + \beta_w \mathbb{E}_t[\log \Pi_{w,t+1}] \,, \tag{5.7}$$

where κ_w is the Phillips-curve slope, μ_w is the wage markup, $\upsilon_{w,t}$ a wage markup shock, and β_w the union discount factor, the stationary population mean of the households' discount factors (Appendix G.3). The factor $1/P_{z_{it}}^C(q_t)$ captures how much composite consumption an additional unit of income (measured in sector-S goods) can purchase. The wage subsidy offsets the steady-state union markup. Appendix G.3 provides further details on wage setting.

5.1.3 Policy and equilibrium

The central bank follows a Taylor rule with interest-rate smoothing,

$$1 + i_t = (1 + i_{t-1})^{\rho_i} \left[(1 + i_{\mathrm{ss}}) \prod_{j \in \{R,S\}} \Pi_{j,t}^{\phi_\pi \vartheta_j^\Pi} \left(\frac{Y_{j,t}^{\mathrm{phys}}}{\bar{Y}_j} \right)^{\phi_y \vartheta_j^Y} \right]^{1-\rho_i} e^{m_t}, \qquad m_t = \rho_m m_{t-1} + \varepsilon_t^i. \tag{5.8}$$

The coefficients $\phi_\pi \vartheta_j^\Pi$ and $\phi_y \vartheta_j^Y$ govern the responses to sectoral inflation and deviations of physical sectoral output from steady state, respectively. The parameter ρ_i governs interest-rate smoothing, and m_t is a monetary policy shock with persistence ρ_m and innovation ε_t^i.

Fiscal policy. The government issues one-period nominal bonds, purchases sector-S goods, and finances transfers to households and subsidies to firms and unions. Its budget constraint, expressed in units of the sector-S good, is

$$B_t = R_t^b B_{t-1} + G_t + Subs_t + Tr_t^{\mathrm{hh}} - \tau_L \mathcal{T}^w w_t N_t, \tag{5.9}$$

where B_t denotes real government debt, G_t government purchases, and $Subs_t$ the subsidy bill. The labor-income tax rate τ_L is fixed. Net household transfers Tr_t^{hh} comprise exogenous transfers less a lump-sum tax that responds to deviations of debt from steady state after a delay. This fiscal adjustment is inactive during 2021–2023: debt absorbs budget imbalances, and inflation-induced debt-revaluation gains are not automatically rebated to households during the episode. Appendix G.3 specifies the subsidy bill and fiscal adjustment rule.

Market clearing. Goods and labor markets clear, and aggregate household bond holdings equal government debt. Appendix G.4 defines equilibrium and collects the equilibrium conditions.

5.2 Calibration and estimation

The curvature of consumption utility (σ), reference-dependence parameters ($\Gamma_G, \Gamma_L, \rho_R$), sector-$R$ demand elasticity (ϵ_R), and Taylor-rule parameters (ρ_i, ϕ_π, ϕ_y) follow the calibration in Table 4. In the monetary-policy rule, sectoral inflation weights (ϑ_j^Π) equal fixed CPI expenditure shares, while output weights (ϑ_j^Y) equal steady-state GDP shares.

I calibrate steady-state sectoral expenditure weights $\omega_R(z)$ using the 2020 Consumer Expenditure Survey. The baseline reference-dependent basket contains food, energy, and frequently purchased nondurables; a broader definition adds rent and recurring services. Sector S contains the remaining consumption categories. Appendix I.1 describes both constructions, and Table I.3 reports the weights across earnings groups. I set the elasticity of substitution between the two sectors to $\nu = 0.40$ (Bobasu et al., 2025).

I calibrate the dividend-incidence rule to the distribution of dividend income across earnings groups in the 2019 Survey of Consumer Finances. Figure I.1 in Appendix I.2 compares the fitted rule with the data.

I use the monthly regular-price change frequencies reported by Nakamura and Steinsson (2008) to construct expenditure-weighted medians of 10.6% in sector R and 8.8% in sector S. Matching the corresponding Calvo Phillips-curve slopes gives $\psi_R = 34.6$ and $\psi_S = 89.7$. Wage-adjustment costs imply a wage Phillips-curve slope of $\kappa_w \simeq 0.085$, corresponding through a Calvo slope comparison to wage adjustment approximately once a year.

I set the quarterly persistence and unconditional standard deviation of idiosyncratic log earnings to 0.966 and 0.50, respectively (Auclert et al., 2021). I calibrate the household discount-factor distribution to match an aggregate quarterly marginal propensity to consume of 0.25 and government debt equal to 1.40 times annual GDP. I estimate the persistence of the cost-push, wage-markup, discount-factor and monetary-policy disturbances so that the model's expectations of CPI inflation, the short-term interest rate, real GDP growth and employment growth match those of professional forecasters (Appendix H.3). Table I.1 in Appendix I reports all parameter values and their sources or calibration targets.

Estimation. I recover eight aggregate innovations in each quarter of 2021Q1–2023Q4 to match two sectoral CPI inflation rates,[33] nominal wage inflation measured by the BLS Employment Cost Index, the effective federal funds rate, real GDP, nonfarm business hours, and federal social benefits and the federal primary deficit, both as shares of GDP. Output and hours are detrended against their 2015–2019 log-linear trends, with an additional level adjustment for hours. The policy rate and fiscal ratios are measured relative to their 2015–2019 means; price inflation is measured relative to 2% annually and wage inflation relative to this benchmark plus

[33] To align the price measures with the CEX basket definitions, I map consumption categories to CPI item-level price indices and aggregate them using fixed December 2020 CPI weights. Sectoral inflation is computed from the resulting indices.

average real wage growth over 2015–2019. I recover the innovations separately for economies with and without reference dependence, matching the same eight observed series in each case. Appendix Table H.1 links these observables to the innovations they jointly identify. Appendix Figures H.2 and H.3 report the observed and fitted series and the recovered shock innovations, respectively. Appendix H.3 details the data transformations and estimation.

I assume perfect foresight with respect to aggregate variables (Auclert et al., 2021). At each date t, agents observe a vector of unanticipated aggregate innovations. They know the persistence of these shocks and anticipate no further innovations. Taking the vector of inherited state variables and the cross-sectional distribution of household states as initial conditions, I compute the nonlinear equilibrium transition. I solve the kinked pricing problem in sector R using a complementarity formulation of the one-sided optimality conditions used in Section 4; Appendix H.4 describes the numerical implementation. Only the date-t allocation is realized. The resulting aggregate state and household distribution carry forward to date $t+1$, when new unanticipated innovations arrive and the transition is recomputed. Appendix H.2 details the sequential solution.

5.3 Welfare

I measure the welfare cost of the 2021–2023 episode by comparing households' allocations with those in an economy that remains in steady state. The calculation includes utility during the episode and the continuation value of households' positions at its end. In the steady-state economy, all aggregate inputs, including the discount-factor shifter $\zeta = 1$, remain at their steady-state values.

Let $t = 0$ denote 2021Q1 and $T = 12$ denote 2024Q1. For household i, lifetime welfare in scenario $x \in \{\text{ss}, tr\}$ is

$$V_i^x = \mathbb{E}_{i0}^x\left[\sum_{t=0}^{T-1} D_{it}\left\{u(c_{it}^x) - v(N_t^x) - \frac{\psi_w}{2}\left(\log \Pi_{w,t}^x\right)^2\right\} + D_{iT}\mathcal{V}_T^x(s_{iT}^x)\right],$$

where $s_{iT}^x = (\beta_{iT}, z_{iT}, b_{i,T-1}^x)$ is the household's state entering 2024, $D_{i0} = 1$, and $D_{it} = \prod_{s=0}^{t-1}\beta_{is}$. Expectations integrate over households' idiosyncratic earnings and discount-factor transitions, taking the aggregate path in each scenario as given. The continuation value $\mathcal{V}_T^x$ includes all subsequent consumption utility, labor disutility, and wage adjustment costs.

During the episode, households' allocations reflect decisions made under each quarter's information. Their continuation value follows the equilibrium path anticipated at the end of 2023, assuming no further aggregate innovations. Future prices, wages, bond returns, and fiscal policy determine consumption and saving along this path, including the consequences of eventual debt repayment. In the steady-state economy, continuation utility is evaluated using the steady-state value function. Both scenarios are evaluated with the aggregate discount shifter

fixed at $\zeta = 1$, while retaining households' idiosyncratic discount-factor process. This measures how much consumption households would be willing to surrender to avoid the episode when its consequences are valued using steady-state preferences. Holding the preference specification fixed provides a common basis for comparing the two allocations.

I express the welfare difference as a fraction of annual steady-state consumption. Household i's consumption-equivalent loss, CE_i, is the proportional reduction in its steady-state consumption during the four quarters of 2021 that makes it indifferent to the episode. The compensating allocation scales steady-state consumption uniformly across idiosyncratic states for $t = 0, \ldots, 3$, holding consumption from $t = 4$ onward and the entire paths of hours and wage adjustment costs at their steady-state allocations. With CRRA utility,

$$V_i^{\mathrm{ss}} + \left[(1 - \mathrm{CE}_i)^{1-\sigma} - 1\right] \underline{V}_i^{c,\mathrm{ss}} = V_i^{tr}, \qquad \underline{V}_i^{c,\mathrm{ss}} = \mathbb{E}_{i0}^{\mathrm{ss}} \sum_{t=0}^{3} D_{it} u(c_{it}^{\mathrm{ss}}). \tag{5.10}$$

Thus, the consumption adjustment is confined to one year, while the welfare difference includes the episode and its full continuation. I report the population mean,

$$100 \int_0^1 \mathrm{CE}_i \, \mathrm{d}i,$$

with initial household states distributed according to Ψ_{ss}. The contribution of reference dependence is the difference in average losses between the two economies fitted to the observed episode. Appendix H.6 provides the computational details.

5.3.1 Welfare costs and the contribution of reference dependence

The average household loses the equivalent of 5.80% of its 2021 steady-state consumption in the baseline economy, compared with 4.89% without reference dependence (Table 6). Reference dependence therefore adds 0.91 percentage points, accounting for approximately 16% of the total welfare cost. Under the broader basket definition, which includes rent and recurring services, the corresponding losses are 6.40% and 5.05%, raising the contribution of reference dependence to 1.35 percentage points, or 21%.

Reference dependence raises welfare costs because a given physical consumption bundle delivers less utility when posted prices exceed reference prices. Differences in consumption baskets determine households' exposure to this effect, with the broader basket placing greater weight on reference-dependent goods among lower- and middle-earning households. The additional welfare cost generally declines across the first six initial earnings groups before rising in the highest group (Figure 7).

Under the baseline welfare measure, which includes the continuation anticipated at the end of 2023, the highest initial earnings group gains relative to steady state. During the episode, inflation erodes the purchasing power of wage income and nominal bond payoffs,

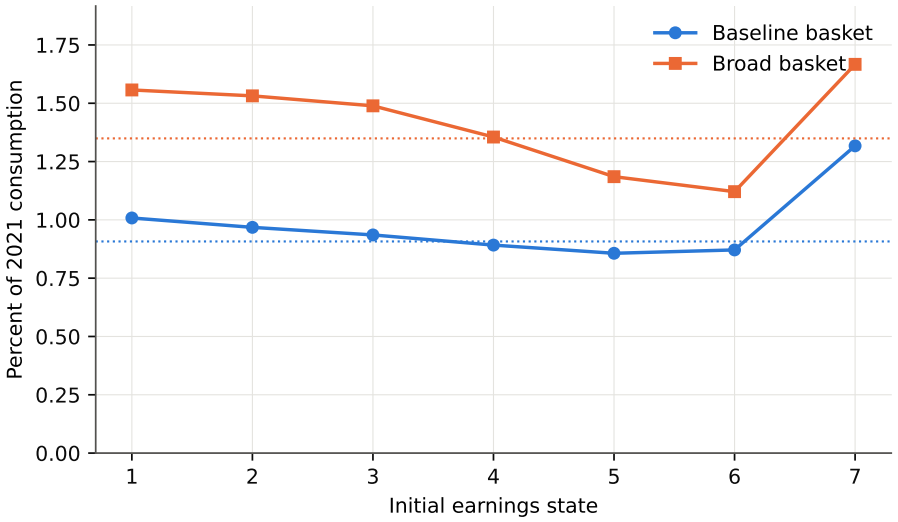
Table 6: The welfare cost of the episode and the contribution of reference dependence

	Baseline basket	Broad basket
Cost of the episode	5.80	6.40
Cost without reference dependence	4.89	5.05
Contribution of reference dependence	0.91	1.35
Share of the cost	16%	21%
d_R peak (2022Q2)	0.075	0.054
ε^{uR} peak, baseline $\rightarrow$ counterfactual	$0.12 \rightarrow 0.12$	$0.08 \rightarrow 0.08$
Loss quarters, of twelve	11	12

Notes: Results are shown for the two basket definitions. The alternative economy sets $(\Gamma_G, \Gamma_L, \rho_R) = (0, 0, 0)$, so sector R has a standard Phillips curve, and re-estimates all innovations, quarter by quarter under the same sequential information, against the same eight observables. Both economies therefore fit the same data. Welfare costs are population means of household consumption equivalents: the constant percentage of consumption in each quarter of 2021 that a steady-state household would give up to be indifferent to the episode, rounded independently. Values follow the fitted history through 2023Q4 and the model's continuation from 2024Q1 (Appendix H.6). d_R is the fitted reference-gap peak; ε^{uR} is the sector-R cost-push peak in surprise units, shown for the baseline and counterfactual.

while reference dependence further raises the effective cost of consumption. These forces contribute to consumption losses during 2021–2023, which are more than offset by anticipated consumption gains afterward. Higher expected dividend income and real bond returns support asset accumulation and sustained consumption gains. Reference dependence reduces this overall gain.

Figure 7: The welfare cost of reference dependence by earnings group



Notes: Difference in mean household consumption-equivalent losses between the fitted economies with and without reference dependence, by initial earnings state (1 is the lowest), in percent of 2021 steady-state consumption. Losses include wage adjustment costs and follow the fitted history through 2023Q4 and the model's continuation from 2024Q1. Dotted lines denote population means, which equal the contributions of reference dependence in Table 6.

Reference dependence raises welfare losses under every continuation convention, although the magnitude of the total loss depends on how post-2023 outcomes are valued. Counting only consumption utility, labor disutility, and wage adjustment costs during 2021–2023, with no terminal value assigned to households' positions at the end of 2023, yields losses of 4.37 and 4.43% of 2021 steady-state consumption under the baseline and broader baskets, with reference dependence accounting for approximately 14% and 17%, compared with 16% and 21% when continuation utility is evaluated along the equilibrium path anticipated at the end of 2023. Valuing households' positions at the end of 2023 using steady-state value functions, net of future debt repayment, gives 3.78 and 3.95%, with the contribution of reference dependence unchanged at 0.61 and 0.76 percentage points. The full forecast is larger because the estimated persistence carries the episode's cost-push, wage-markup and monetary disturbances into the continuation (Appendix H.6).

I also assess sensitivity to the treatment of preference shocks in welfare evaluation. First, I evaluate both scenarios using the realized aggregate discount shifters rather than their pre-episode value, retaining steady-state consumption and saving rules in the benchmark. Second, I allow households in the steady-state benchmark to re-optimize consumption and saving in response to aggregate discount-factor shocks, holding all other aggregate inputs at steady state. This allows the benchmark allocation to adjust to the preferences used to evaluate it. Including the post-2023 equilibrium adjustment anticipated at the end of 2023, both exercises leave overall welfare losses and the contribution of reference dependence close to the baseline estimates. Appendix Tables H.3 and H.4 report these results and further robustness checks.

6 Conclusion

Reference dependence in consumer demand creates a first-order welfare cost of inflation: when posted prices exceed slowly adjusting reference prices, a given physical bundle delivers less consumption utility. Using survey estimates to discipline the model, I find that this mechanism substantially increases inflation's welfare costs in both a representative-agent economy and a richer heterogeneous-agent economy with sticky wages and differences in households' wealth and consumption baskets.

The findings of this paper suggest several directions for future research. First, optimal monetary policy could be studied in an economy where sectors differ both in price rigidity and in the presence of reference-dependent demand. This would establish whether reference dependence warrants greater stabilization of inflation in relatively flexible-price sectors, given the consequences for inflation elsewhere and real activity.

In this paper, I assume that reference prices are formed solely from past posted prices. A second extension would allow reference prices to incorporate expected future prices, so that they can respond before posted prices change. This would allow us to quantify differences in

the welfare costs of anticipated and unanticipated inflation. Finally, solving the kinked model under aggregate uncertainty would clarify how the asymmetry between gains and losses affects the welfare costs of business-cycle fluctuations and the potential gains from stabilization.

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A Proofs and Derivations

A.1 Transaction Utility and RD–CES Relative Demand

This appendix provides a transaction-utility foundation for the relative demand system in Section 2.1. Its scope is deliberately narrow: it derives relative demand across inside varieties. The outside option determines total category demand but does not affect the composition of inside demand.

Consider consumer h on purchase occasion m. The utility from choosing inside variety $j \in \{1, \ldots, J\}$ is

$$u_{hjm} = \bar{v}_j - \varepsilon \log P_j - (\varepsilon - 1)\chi_{hj} + \xi_{hjm}, \tag{A.1}$$

where

$$\chi_{hj} = \begin{cases} \dfrac{\Gamma_L}{\varepsilon - 1} \log(P_j / R_j^h), & P_j \geq R_j^h, \\ \dfrac{\Gamma_G}{\varepsilon - 1} \log(P_j / R_j^h), & P_j < R_j^h. \end{cases} \tag{A.2}$$

The outside option delivers $u_{h0m} = v_0^h + \xi_{h0m}$. The term $-(\varepsilon - 1)\chi_{hj}$ is transaction utility: a price below the reference raises choice utility, while a price above it lowers choice utility.

Assumption A.1 (Taste shocks)**.** *The shocks ξ_{hjm}, including the outside option $j = 0$, are independent and identically distributed standard Type I extreme-value random variables across consumers, alternatives, and purchase occasions.*

Under Assumption A.1, the inside-good choice probability is

$$\Pr_h(j) = \frac{e^{\bar{v}_j} e^{-(\varepsilon-1)\chi_{hj}} P_j^{-\varepsilon}}{e^{v_0^h} + \sum_{k=1}^{J} e^{\bar{v}_k} e^{-(\varepsilon-1)\chi_{hk}} P_k^{-\varepsilon}}. \tag{A.3}$$

Assumption A.2 (Aggregation)**.** *Consumers share the reference-price elasticities (Γ_L, Γ_G) and a common market reference R_j for each variety. Deterministic inside-good quality is symmetric, $\bar{v}_j = \bar{v}$.*

Define

$$s_j \equiv e^{-(\varepsilon-1)\chi_j}, \qquad \Delta \equiv \sum_{k=1}^{J} s_k P_k^{-\varepsilon}. \tag{A.4}$$

If each inside purchase delivers one unit and consumer h has M_h purchase occasions, expected aggregate purchases satisfy

$$C_j = s_j P_j^{-\varepsilon} \Omega(\Delta), \qquad \Omega(\Delta) \equiv e^{\bar{v}} \int \frac{M_h}{e^{v_0^h} + e^{\bar{v}} \Delta} \, \mathrm{d}h. \tag{A.5}$$

Heterogeneity in shopping intensity and outside-option values enters only through the common market-size term $\Omega(\Delta)$.

Proposition A.1 (Transaction-utility foundation)**.** *Under Assumptions A.1 and A.2, relative inside-good demand satisfies*

$$\frac{C_j}{C_k} = \frac{s_j P_j^{-\varepsilon}}{s_k P_k^{-\varepsilon}}. \tag{A.6}$$

Proof. Equation (A.5) factors demand into the variety-specific term $s_j P_j^{-\varepsilon}$ and the common term $\Omega(\Delta)$. Taking a ratio cancels the common term and gives (A.6). □

Connection to the CES aggregate. The relative-demand system in Proposition A.1 is exactly the one generated by the mass-weighted finite-variety RD–CES aggregate

$$C^U = \left[\frac{1}{J}\sum_{j=1}^{J} s_j^{1/\varepsilon} C_j^{(\varepsilon-1)/\varepsilon}\right]^{\varepsilon/(\varepsilon-1)}, \tag{A.7}$$

when expenditure is measured as $J^{-1}\sum_j P_j C_j$. The standard finite-to-continuum limit yields (2.1) and the ideal price index in (2.2). Expenditure minimization under (A.7), using the same mass weights in expenditure and the aggregate, gives (A.6). Under Riemann-sum regularity, $J^{-1}\sum_j s_j P_j^{1-\varepsilon}$ converges to $\int_0^1 s_t(i) P_t(i)^{1-\varepsilon}\,\mathrm{d}i$.

Remark A.1 (Scope of the microfoundation). Proposition A.1 microfounds relative demand across inside varieties. Deriving the complete intertemporal representative-household problem from the purchase-occasion model would require additional assumptions about category expenditure and intertemporal aggregation.

A.2 The Smooth Phillips Curve

Maintain $\Gamma_L = \Gamma_G \equiv \Gamma$ and the firm's problem in Section 2.3. This appendix derives its nonlinear optimality conditions and obtains Proposition 1 by linearizing around the symmetric zero-inflation steady state.

Optimality conditions. Let $\mathcal{V}_t(R_t(i), P_{t-1}(i))$ be the present discounted value of the firm's profits, with each period's profits weighted by the household's marginal utility of income, $\Lambda_t / P_t^{\mathrm{RD}}$. Under symmetry, the shadow value of the reference price in Section 2.3 is $\Theta_t \equiv (\Lambda_t Y_t^U)^{-1} \partial \mathcal{V}_t / \partial \log R_t(i)$. Let $MC_t(i) \equiv (1-\tau) W_t / \mathrm{MPL}_t(i)$ denote nominal marginal cost. Demand has elasticities $-(\varepsilon+\Gamma)$ and Γ with respect to the posted and reference prices. The reference law maps their logs into tomorrow's log reference with weights $1-\rho$ and ρ. The reference-price envelope condition is

$$\frac{\partial \mathcal{V}_t}{\partial \log R_t(i)} = \frac{\Lambda_t}{P_t^{\mathrm{RD}}}\,\Gamma\,[P_t(i) - MC_t(i)]\,Y_t(i) + \beta\rho\,\mathbb{E}_t \frac{\partial \mathcal{V}_{t+1}}{\partial \log R_{t+1}(i)}. \tag{A.8}$$

The lagged-price envelope condition is $\partial \mathcal{V}_t / \partial \log P_{t-1}(i) = \Lambda_t \psi_p(\Pi_t(i) - 1)\Pi_t(i) Y_t^U$, where $\Pi_t(i) \equiv P_t(i)/P_{t-1}(i)$. At an interior price optimum, the first-order condition is

$$\begin{aligned} 0 = \frac{\Lambda_t}{P_t^{\mathrm{RD}}} \Big\{ P_t(i) Y_t(i) - (\varepsilon + \Gamma)\big[P_t(i) - MC_t(i)\big] Y_t(i) - \psi_p(\Pi_t(i) - 1)\Pi_t(i) P_t^{\mathrm{RD}} Y_t^U \Big\} \\ + \beta \mathbb{E}_t \Big[(1-\rho) \frac{\partial \mathcal{V}_{t+1}}{\partial \log R_{t+1}(i)} + \Lambda_{t+1} \psi_p(\Pi_{t+1}(i) - 1)\Pi_{t+1}(i) Y_{t+1}^U \Big]. \end{aligned} \tag{A.9}$$

The last term is the lagged-price envelope condition one period ahead. Under symmetry, $Y_t(i) = e^{\chi_t} Y_t^U$, $P_t^{\mathrm{RD}} = P_t e^{\chi_t}$, and $MC_t(i)/P_t = mc_t$. Normalizing (A.8) and (A.9) by $\Lambda_t Y_t^U$ and using the definitions of Θ_t and $\mathcal{M}_{t+1}$ in Section 2.3 gives (2.9) and (2.10).

Assume an optimal price plan exists, the value function is differentiable in the log reference and lagged posted prices, and the derivatives of per-period profits weighted by the household's marginal utility of income with respect to the log reference price are uniformly bounded across dates, admissible plans, and states. The envelope formula then gives the present-value representation

$$\Theta_t = \mathbb{E}_t \sum_{j=0}^{\infty} (\beta\rho)^j \frac{\Lambda_{t+j} Y_{t+j}^U}{\Lambda_t Y_t^U} \Gamma(1 - mc_{t+j}).$$

The assumed derivative bound also bounds $\partial \mathcal{V}_t / \partial \log R_t(i)$, since a log-reference perturbation shrinks by a factor ρ each period along a fixed price plan. The discounted continuation term therefore vanishes:

$$\lim_{J \to \infty} (\beta\rho)^J \mathbb{E}_t[\Lambda_{t+J} Y_{t+J}^U \Theta_{t+J}] = 0.$$

This condition excludes a nonzero discounted remainder without requiring Θ_t itself to converge to zero.

The firm's transversality conditions concern both inherited states. Under Assumption B.1, $\Lambda_t Y_t^U$ and $\Pi_t(i)$ are uniformly bounded, so the lagged-price envelope derivative is bounded, as is the reference-price derivative above. Holding aggregates fixed, let $\delta \log R_{t+J}(i)$ and $\delta \log P_{t+J-1}(i)$ denote uniformly bounded admissible variations in inherited log prices, equivalently in the corresponding relative log prices. Then

$$\lim_{J \to \infty} \beta^J \mathbb{E}_t \left[\frac{\partial \mathcal{V}_{t+J}}{\partial \log R_{t+J}(i)} \delta \log R_{t+J}(i) + \frac{\partial \mathcal{V}_{t+J}}{\partial \log P_{t+J-1}(i)} \delta \log P_{t+J-1}(i) \right] = 0,$$

because the absolute value of the discounted boundary term is bounded by a constant times β^J. Thus both terminal boundary terms converge to zero. This conclusion uses bounded marginal values; the continuation condition with discount factor $(\beta\rho)^J$ alone does not establish it.

Steady-state markup. At the deterministic zero-inflation steady state, $\bar{\Pi} = \bar{\mathcal{M}} = 1$, $\bar{\chi} = 0$, and $\bar{\Theta} = \Gamma(1 - \bar{mc})/(1 - \beta\rho)$. Substitution into the pricing condition gives

$$\bar{mc} = \frac{\varepsilon_{\text{eff}} - 1}{\varepsilon_{\text{eff}}} = \frac{1}{\bar{\mu}}, \qquad \varepsilon_{\text{eff}} = \varepsilon + \Gamma \frac{1 - \beta}{1 - \beta\rho}. \tag{A.10}$$

The future-reference benefit offsets part of the instantaneous demand elasticity. The subsidy $1 - \tau = 1/\bar{\mu}$ removes the steady-state monopoly distortion.

First-order pricing equation. Let $\widehat{mc}_t$ denote the log deviation of subsidized real marginal cost in posted-price units from its steady-state value, and let $\theta_t \equiv \Theta_t - \bar{\Theta}$. Under log utility, $\sigma = 1$, $c_t^U = y_t^U$ to first order, so $\mathcal{M}_{t+1}$ has no first-order variation. Linearizing (2.10) and (2.9) gives

$$\psi_p(\pi_t - \beta \mathbb{E}_t \pi_{t+1}) = \frac{\varepsilon + \Gamma}{\bar{\mu}} \widehat{mc}_t + \beta(1 - \rho)\mathbb{E}_t \theta_{t+1}, \tag{A.11}$$

$$\theta_t = -\frac{\Gamma}{\bar{\mu}} \widehat{mc}_t + \beta\rho \mathbb{E}_t \theta_{t+1}. \tag{A.12}$$

With $\rho = 0$, the costate equation is static. Substituting its one-period-ahead version into (A.11) gives

$$\psi_p(\pi_t - \beta \mathbb{E}_t \pi_{t+1}) = \frac{\varepsilon + \Gamma}{\bar{\mu}} \left(\widehat{mc}_t - \beta\gamma \mathbb{E}_t \widehat{mc}_{t+1} \right), \qquad \gamma = \frac{\Gamma}{\varepsilon + \Gamma}. \tag{A.13}$$

To express marginal cost in terms of physical output relative to its technology benchmark, use the log deviations defined in Section 2.2. Under log utility, labor supply (2.7) and marginal cost (2.8) imply $\widehat{mc}_t = c_t^U + \chi_t - a_t + (\varphi + \alpha) n_t$. Since $c_t^U = y_t^U$ to first order and (2.14) gives $\widetilde{y}_t = y_t - a_t = y_t^U + \chi_t - a_t = (1 - \alpha) n_t$, it follows that

$$\widehat{mc}_t = (1 + \varphi) n_t = \frac{1 + \varphi}{1 - \alpha} \widetilde{y}_t. \tag{A.14}$$

Thus, to first order, marginal cost is at its steady-state value when employment is at $\bar{N}$, or equivalently when physical output equals the technology benchmark $A_t \bar{N}^{1-\alpha}$. Substituting (A.14) into (A.13) and dividing by $\psi_p > 0$ proves Proposition 1.

A.3 The Welfare Approximation

Under log utility, period utility is

$$U_t = \log C_t^U - \frac{N_t^{1+\varphi}}{1 + \varphi}. \tag{A.15}$$

Expand around the zero-inflation steady state. Because the subsidy removes the steady-state labor wedge,

$$\bar{N}^{1+\varphi} = 1 - \alpha. \tag{A.16}$$

The second-order expansion of period utility is

$$U_t - \bar{U} = c_t^U - (1-\alpha)n_t - \frac{(1+\varphi)(1-\alpha)}{2} n_t^2 + O(3). \tag{A.17}$$

The resource constraint gives

$$c_t^U = y_t^U - \frac{\psi_p}{2}\pi_t^2 + O(3), \tag{A.18}$$

while the exact log production identity gives

$$n_t = \frac{y_t^U + \chi_t - a_t}{1-\alpha}. \tag{A.19}$$

Substituting these two relations into (A.17) yields

$$U_t - \bar{U} = -\chi_t - \frac{\psi_p}{2}\pi_t^2 - \frac{1+\varphi}{2(1-\alpha)}(y_t^U - a_t + \chi_t)^2 + a_t + O(3). \tag{A.20}$$

The term a_t depends only on the exogenous productivity path and is independent of policy to the relevant order. Using $\widetilde{y}_t = y_t - a_t = y_t^U + \chi_t - a_t$, changing signs, discounting, and collecting terms gives

$$\mathcal{L}_{t_0} = \frac{1}{2}\mathbb{E}_{t_0}\sum_{t=t_0}^{\infty}\beta^{t-t_0}\left[\frac{1+\varphi}{1-\alpha}\widetilde{y}_t^2 + 2\chi_t + \psi_p\pi_t^2\right] + \text{t.i.p.} + O(3). \tag{A.21}$$

For the experiment in Section 2.5, set $t_0 = 0$ and omit the expectation operator along the perfect-foresight transition. Under symmetric pricing and $\rho = 0$, the reference law implies $2\chi_t = 2\Gamma\pi_t/(\varepsilon - 1)$, giving Proposition 2.

A.4 Symmetric Equilibrium

This appendix defines the symmetric equilibrium described in Section 2.4. The definition covers the smooth model, $\Gamma_L = \Gamma_G = \Gamma$, and the kinked model, $\Gamma_L > \Gamma_G$; only the firm's necessary conditions in item 2 differ. Given exogenous sequences $\{A_t, \nu_t\}_{t=0}^{\infty}$ and symmetric initial conditions $B_0 = 0$, $R_0(i) = R_0$, and $P_{-1}(i) = P_{-1}$ for all i, a symmetric equilibrium is a sequence

$$\left\{\begin{array}{c} C_t^U, N_t, B_{t+1}, Y_t, Y_t^U, W_t, Q_t, I_t, P_t, P_t^{\mathrm{RD}}, D_t^F, T_t, \\ \{C_t(i), Y_t(i), N_t(i), P_t(i), R_t(i), D_t^F(i)\}_{i\in[0,1]} \end{array}\right\}_{t=0}^{\infty}$$

that satisfies the following conditions:

1) *Household optimality.* Given prices, wages, profits, and transfers, the representative household's variety demands solve the intratemporal expenditure-minimization problem, and $\{C_t^U, N_t, B_{t+1}\}_{t=0}^{\infty}$ solves the intertemporal problem in Section 2.2, including its no-Ponzi condition.

2) *Firm optimality.* Given aggregate prices and quantities, each firm satisfies its demand, production, reference-price, and price-corridor constraints and chooses its price sequence to maximize the present discounted value of its profits, with demand (B.1). In the smooth model, its optimal choices satisfy the necessary conditions (2.9) and (2.10). In the kinked model, these conditions hold with Γ replaced by the active branch's elasticity whenever $P_t(i) \neq R_t(i)$, and the one-sided conditions (B.11) hold whenever $P_t(i) = R_t(i)$.[34]

3) *Price aggregation, reference updating, and symmetry.* The conventional posted-price index is $P_t = [\int_0^1 P_t(i)^{1-\varepsilon}\,\mathrm{d}i]^{1/(1-\varepsilon)}$, the ideal price index P_t^{RD} and variety demands satisfy (2.2), and reference prices obey (2.3). Symmetry requires $P_t(i) = P_t$, $R_t(i) = R_t$, $P_{t-1}(i) = P_{t-1}$, $C_t(i) = e^{\chi_t} C_t^U$, $Y_t(i) = Y_t = e^{\chi_t} Y_t^U$, and $N_t(i) = N_t$ for all i. Aggregate physical output satisfies (2.14).

4) *Monetary policy.* The nominal interest rate satisfies the posted-price-inflation Taylor rule (2.16), with $I_t = Q_t^{-1}$.

5) *Fiscal policy.* The government finances the employment subsidy with lump-sum taxes, so net household transfers satisfy $T_t = -\tau W_t N_t$.

6) *Market clearing.* The nominal bond is in zero net supply, $B_{t+1} = 0$; labor and distributed profits satisfy $N_t = \int_0^1 N_t(i)\,\mathrm{d}i$ and $D_t^F = \int_0^1 D_t^F(i)\,\mathrm{d}i$; and the composite-good market satisfies the resource constraint (2.15).

[34] Global concavity is not imposed (Remark B.1), so these necessary conditions are not sufficient by themselves. For the quantitative model of Section 4, Appendix F.2 reports dynamic grid checks on the baseline cost-push paths at all reference-persistence values, with continuation choices re-optimized after each deviation. In a corridor of ±20 log points around the aggregate price, the equilibrium price is within 0.07 log point of the firm's best response at every date. These checks provide numerical support for firm optimality on the tested paths and grid.

B The Hard-Kink RD–CES Model

Section 2 derives closed-form pricing and welfare results under the smooth restriction $\Gamma_L = \Gamma_G$. This appendix restores the hard kink but maintains the RD–CES demand system and reference law. Throughout, I focus on the loss-aversion ordering $\Gamma_L > \Gamma_G \geq 0$. At the normalized steady state, posted and reference prices coincide and the reference wedge is zero, but the kink changes the firm's marginal incentive to move its price in either direction.

The extension has four economic consequences. First, the differentiable steady-state pricing condition becomes a pair of one-sided conditions, so an interval of employment subsidies can support $P = R$. Second, under the conditions stated below, the kink can generate an endogenous inaction region even without Rotemberg adjustment costs. Third, outside that region, firms satisfy branch-specific nonlinear pricing conditions whose first-order approximations differ across the loss and gain branches. Finally, inflationary and deflationary excursions can have asymmetric welfare effects. The results below provide the analytical counterpart to the rule-conditioned nonlinear transitions used in the quantitative analysis.

Customer-side behavioral models provide a natural precedent for this mechanism. Perceived unfairness and customer anger can restrain price increases and change the frequency of price adjustment (Rotemberg, 2005, 2011), while Rotemberg (2008) discusses the broader behavioral evidence and its monetary-policy implications. The RD–CES model gives these responses a specific demand-system representation based on posted prices relative to reference prices.

B.1 Steady-State Pricing at the Kink

At $P = R$, the firm's payoff is continuous but generally not differentiable. The usual equality first-order condition must therefore be replaced by a right derivative for a price increase and a left derivative for a price cut. This result does not require log utility or a one-period reference: take $\sigma \geq 1$ and $\rho \in [0, 1)$. Define $d_t(i) \equiv p_t(i) - r_t(i) = \log[P_t(i)/R_t(i)]$. The hard-kink demand schedule is

$$Y_t(i) = \begin{cases} \left[\dfrac{P_t(i)}{R_t(i)}\right]^{-\Gamma_L} \left[\dfrac{P_t(i)}{P_t^{\mathrm{RD}}}\right]^{-\varepsilon} Y_t^U, & d_t(i) \geq 0, \\ \left[\dfrac{P_t(i)}{R_t(i)}\right]^{-\Gamma_G} \left[\dfrac{P_t(i)}{P_t^{\mathrm{RD}}}\right]^{-\varepsilon} Y_t^U, & d_t(i) < 0, \end{cases} \qquad \Gamma_L > \Gamma_G \geq 0. \tag{B.1}$$

Here Γ_L and Γ_G are the reference-price elasticities above and below the reference price. With $R_{t+1}(i) = R_t(i)^\rho P_t(i)^{1-\rho}$, the firm's flow profit and Bellman equation are

$$\begin{aligned} D_t^F(i) &= P_t(i)Y_t(i) - (1-\tau)W_t N_t(i) - \frac{\psi_p}{2}\left[\frac{P_t(i)}{P_{t-1}(i)} - 1\right]^2 P_t^{\mathrm{RD}} Y_t^U, \\ Y_t(i) &= A_t N_t(i)^{1-\alpha}, \\ \mathcal{V}_t(R_t(i), P_{t-1}(i)) &= \max_{P_t(i)} \left\{ \Lambda_t \frac{D_t^F(i)}{P_t^{\mathrm{RD}}} + \beta \mathbb{E}_t \mathcal{V}_{t+1}(R_{t+1}(i), P_t(i)) \right\}. \end{aligned} \tag{B.2}$$

The firm chooses its posted price; (B.1) determines sales, and the technology in (B.2) determines the employment required to meet them.

The analysis maintains two assumptions on admissible prices, aggregate paths, and the firm's objective.

Assumption B.1 (Admissible prices and aggregate paths)**.** *Posted prices lie in the corridor $P_t(i)/P_t \in [1/K, K]$ of the firm's problem in Section 2.3. Aggregate log inflation is uniformly bounded in absolute value and initial relative prices are finite and positive, so inherited posted and reference prices, expressed relative to the aggregate price, remain in a compact set. The consumption composite C_t^U, composite output Y_t^U, productivity A_t, the posted real wage W_t/P_t, and the ratio P_t^{RD}/P_t are uniformly bounded and bounded away from zero across dates and aggregate states.*

Assumption B.2 (Branchwise single-peakedness)**.** *On the admissible domain, the maximand of (B.2), viewed as a function of $d_t(i) = \log[P_t(i)/R_t(i)]$ holding states and aggregates fixed, is strictly quasiconcave on each branch $\{d_t(i) \geq 0\}$ and $\{d_t(i) \leq 0\}$.*

Consider a unilateral deviation at date t from the deterministic zero-inflation allocation $\bar{P} = \bar{R} = \bar{P}_{-1}$, evaluated against the firm's own continuation policy. Suppose that after a sufficiently small price deviation the firm tracks its updated reference price in each subsequent period, $P_{t+k}(i) = R_{t+k}(i)$ for $k \geq 1$; Proposition B.1 maintains this tracking condition. A date-t log-price change h then raises all subsequent references and posted prices permanently: $\log R_{t+k}(i) = \log P_{t+k}(i) = \log \bar{P} + (1-\rho)h$ for every $k \geq 1$. Holding the aggregate path fixed, the deviation changes the posted price relative to its reference only at date t. In subsequent periods, posted and reference prices move together, so only the relative-price effect governed by ε remains.

The coefficient $\Phi(\rho)$ summarizes the discounted future effect of today's price change on the firm's relative price. Averaging the one-period branch penalty over the $1 + \Phi(\rho)$ discounted periods of exposure yields the directional effective elasticities

$$\Phi(\rho) \equiv \sum_{k=1}^{\infty} \beta^k (1-\rho) = \frac{\beta(1-\rho)}{1-\beta}, \qquad \varepsilon^{\uparrow} \equiv \varepsilon + \Gamma_L \frac{1-\beta}{1-\beta\rho}, \qquad \varepsilon^{\downarrow} \equiv \varepsilon + \Gamma_G \frac{1-\beta}{1-\beta\rho}, \tag{B.3}$$

using $1+\Phi(\rho)=(1-\beta\rho)/(1-\beta)$. These expressions follow from equation (2.11) by setting $\Gamma=\Gamma_L$ for upward deviations and $\Gamma=\Gamma_G$ for downward deviations. An upward deviation pays the loss-branch penalty Γ_L during its single transition period and the ordinary elasticity ε thereafter, and symmetrically for a downward deviation with Γ_G.

At the MRS–MPL allocation, $\mathrm{MRS}_{N,C^U} \equiv -U_N/U_{C^U} = (C^U)^\sigma N^\varphi = \mathrm{MPL}$. At this allocation, household labor supply implies that unsubsidized real marginal cost in posted-price units is $W/(\bar{P}\,\mathrm{MPL})=1$. The firm's subsidized real marginal cost is therefore $\bar{mc} = (1-\tau)W/(\bar{P}\,\mathrm{MPL}) = 1-\tau$.

Let $\mathscr{B}_t(h)$ denote the firm's Bellman return when $P_t(i)=\bar{P}e^h$, holding inherited states and the aggregate path fixed and optimizing continuation choices. Where tracking is locally optimal, the continuation term follows the policy above. Define

$$\mathcal{D}^+ \equiv \frac{1}{\bar{\Lambda}\bar{Y}^U}\lim_{h\downarrow 0}\frac{\mathscr{B}_t(h)-\mathscr{B}_t(0)}{h}, \quad \mathcal{D}^- \equiv \frac{1}{\bar{\Lambda}\bar{Y}^U}\lim_{h\downarrow 0}\frac{\mathscr{B}_t(-h)-\mathscr{B}_t(0)}{-h}.$$

The normalization $\bar{\Lambda}\bar{Y}^U$ is steady-state output in composite units valued at the marginal utility of consumption, which under symmetric pricing equals the utility value of revenue, so the derivatives are measured in current-revenue-equivalent units. They are derivatives with respect to the current log price, not derivatives with respect to a state variable. Direct computation of the deviation value gives

$$\mathcal{D}^+ = (1+\Phi(\rho))\big[1-\varepsilon^{\uparrow}(1-\bar{mc})\big], \qquad \mathcal{D}^- = (1+\Phi(\rho))\big[1-\varepsilon^{\downarrow}(1-\bar{mc})\big]. \tag{B.4}$$

No profitable price increase requires $\mathcal{D}^+ \le 0$, while no profitable price cut requires $\mathcal{D}^- \ge 0$.

Proposition B.1 (Subsidies supporting the reference price)**.** *Suppose the Bellman return is continuous at $P=R$, tracking is optimal following sufficiently small deviations, and $\varepsilon^{\uparrow},\varepsilon^{\downarrow}>1$. Then the one-sided first-order no-deviation conditions at $P=R=\bar{P}$ hold if and only if*

$$\frac{\varepsilon^{\downarrow}-1}{\varepsilon^{\downarrow}} \le 1-\tau \le \frac{\varepsilon^{\uparrow}-1}{\varepsilon^{\uparrow}} \qquad \Longleftrightarrow \qquad \frac{1}{\varepsilon^{\uparrow}} \le \tau \le \frac{1}{\varepsilon^{\downarrow}}. \tag{B.5}$$

If, in addition, Assumptions B.1–B.2 hold for the Bellman maximand with continuation choices optimized after every deviation, then $P=R=\bar{P}$ is the unique optimal price whenever both inequalities are strict.

Substituting the one-sided conditions $\mathcal{D}^+ \le 0 \le \mathcal{D}^-$ into (B.4) gives the interval. The conditions are necessary. Under the proposition's additional assumptions, they are also sufficient strictly inside the interval: small price changes in either direction lower firm value, and branchwise single-peakedness of the Bellman maximand rules out larger profitable deviations. Continuity on the compact domain ensures that a maximizer exists. At either endpoint, the first-order effect of a price change in one direction is zero. This does not rule out a profitable

finite price change in that direction, even under branchwise single-peakedness (Remark B.1). The quantitative implementation calibrates the subsidy to the midpoint of the interval and checks firm optimality numerically by re-solving the firm's dynamic pricing problem on a grid. The interval is nonempty because $\varepsilon^{\uparrow} - \varepsilon^{\downarrow} = (\Gamma_L - \Gamma_G)(1-\beta)/(1-\beta\rho) > 0$.

When $\Gamma_L = \Gamma_G = \Gamma$, the two directional effective elasticities coincide at $\varepsilon_{\text{eff}} = \varepsilon + \Gamma(1-\beta)/(1-\beta\rho)$ of Section 2.3. The subsidy interval therefore collapses to the smooth-model subsidy $\tau = 1/\varepsilon_{\text{eff}}$. Thus, the smooth model pins down a unique subsidy, whereas a range of subsidies can sustain the same allocation in the kinked model under Proposition B.1. Reference dependence affects firms' one-sided pricing incentives even though the reference wedge is zero at $P = R$. The kink in demand provides a source of nominal rigidity distinct from Rotemberg adjustment costs.

B.2 The Inaction Region

Under the conditions below, the hard kink can make the reference price optimal over an interval of pricing pressure. Inaction here means setting the posted price equal to the current reference price. With persistent references, this need not imply an unchanged posted price upon entry into the region. Define

$$z_t \equiv mc_t = \frac{(1-\tau)W_t}{P_t \text{MPL}_t}. \tag{B.6}$$

Section 4 adds a cost-push factor υ_t, so that $z_t = \upsilon_t mc_t$ there. Higher marginal cost relative to the posted price, z_t, increases the incentive to raise that price. Upward and downward deviations affect future references differently, shifting the two thresholds separately.

Related inaction mechanisms arise from asymmetric consumer search (Dupraz, 2017) and managers' reference-dependent mental accounting (Angelis, 2025). Here the asymmetry enters household demand, and the thresholds depend on the firm's continuation values.

Consider a date on a deterministic perfect-foresight path at which $P_t(i) = R_t(i)$. Let $\Pi_t = P_t/P_{t-1}$ and define the intertemporal scaling term and directional reference costates as

$$\mathcal{M}_{t+1} \equiv \left(\frac{C^U_{t+1}}{C^U_t}\right)^{-\sigma} \frac{Y^U_{t+1}}{Y^U_t}, \qquad \Theta^{R,\pm}_{t+1} \equiv \frac{R_{t+1}}{\Lambda_{t+1} Y^U_{t+1}} D^{\pm}_R \mathcal{V}_{t+1}. \tag{B.7}$$

Here $D^{\pm}_R \mathcal{V}_{t+1}$ is the one-sided partial derivative of optimized firm value with respect to its reference state, holding its inherited posted-price state fixed. The signs correspond to the reference change induced by a current price increase $(+)$ or cut $(-)$. Future pricing choices remain optimized. The adjustment-cost and continuation terms are

$$\begin{aligned} \mathcal{A}_t &\equiv -\psi_p(\Pi_t - 1)\Pi_t + \beta\psi_p \mathcal{M}_{t+1}(\Pi_{t+1} - 1)\Pi_{t+1}, \\ \mathcal{R}^{\pm}_t &\equiv \beta(1-\rho)\mathcal{M}_{t+1}\Theta^{R,\pm}_{t+1}. \end{aligned} \tag{B.8}$$

As in Appendix B.1, let $\mathcal{D}_t^+$ and $\mathcal{D}_t^-$ denote the right and left derivatives of the Bellman return with respect to the current log posted price, normalized by $\Lambda_t Y_t^U$. Holding inherited states and the aggregate path fixed, and evaluating continuation value under optimal future pricing, their values at the kink are

$$\begin{aligned} \mathcal{D}_t^+ &= 1 - (\varepsilon + \Gamma_L)(1 - z_t) + \mathcal{A}_t + \mathcal{R}_t^+, \\ \mathcal{D}_t^- &= 1 - (\varepsilon + \Gamma_G)(1 - z_t) + \mathcal{A}_t + \mathcal{R}_t^-. \end{aligned} \tag{B.9}$$

Holding $\mathcal{A}_t$ and $\mathcal{R}_t^{\pm}$ fixed, the roots of these derivatives define the lower and upper thresholds:

$$\begin{aligned} \underline{z}_t &\equiv 1 - \frac{1 + \mathcal{A}_t + \mathcal{R}_t^-}{\varepsilon + \Gamma_G}, \qquad && \mathcal{D}_t^-(\underline{z}_t) = 0, \\ \overline{z}_t &\equiv 1 - \frac{1 + \mathcal{A}_t + \mathcal{R}_t^+}{\varepsilon + \Gamma_L}, \qquad && \mathcal{D}_t^+(\overline{z}_t) = 0. \end{aligned} \tag{B.10}$$

Because $\mathcal{R}_t^+$ and $\mathcal{R}_t^-$ depend on future regimes, both thresholds are state-contingent equilibrium objects.

Proposition B.2 (Local threshold characterization)**.** *Suppose the Bellman return is continuous on the admissible domain and directionally differentiable at $P_t(i) = R_t(i)$, and that $R_t(i)$ lies in the interior of the corridor of Assumption B.1. Conditional on inherited states and the aggregate path, hold $\mathcal{A}_t$ and $\mathcal{R}_t^{\pm}$ fixed in the threshold calculation. The directional derivatives in (B.9) are then affine and strictly increasing in z_t. If $0 < \underline{z}_t \le \overline{z}_t$, then the reference price satisfies the one-sided first-order no-deviation conditions if and only if*

$$\underline{z}_t \le z_t \le \overline{z}_t \quad \Longleftrightarrow \quad \mathcal{D}_t^+ \le 0 \le \mathcal{D}_t^-. \tag{B.11}$$

Throughout the interval, the one-sided first-order no-deviation conditions hold. Under the proposition's conditions and Assumptions B.1–B.2, an optimal price $P_t^\star(i)$ exists and satisfies

$$\begin{cases} P_t^\star(i) < R_t(i), & z_t < \underline{z}_t, \\ P_t^\star(i) = R_t(i), & \underline{z}_t < z_t < \overline{z}_t, \\ P_t^\star(i) > R_t(i), & z_t > \overline{z}_t, \end{cases} \tag{B.12}$$

with a unique maximizer in each strict case. Strictly inside the threshold interval, small price changes in either direction lower firm value. Branchwise single-peakedness then makes the reference price the unique global maximizer. At either endpoint, one derivative is zero, so these conditions alone do not rule out a profitable finite deviation.

Remark B.1 (Role of the two assumptions). Assumption B.1 ensures that firm value is finite and continuous and that the Bellman maximand attains a maximum on the compact price domain. Assumption B.2 separately imposes branchwise single-peakedness: this property is not guaranteed by the regularity conditions and does not require global concavity. Ap-

pendix F.2 documents checks of single-peakedness and firm optimality on the numerical grid, with continuation choices re-optimized after each deviation.

The threshold ordering required by Proposition B.2 is equivalent to

$$\frac{1+\mathcal{A}_t+\mathcal{R}_t^-}{\varepsilon+\Gamma_G} \geq \frac{1+\mathcal{A}_t+\mathcal{R}_t^+}{\varepsilon+\Gamma_L}. \tag{B.13}$$

A sufficient condition is $1+\mathcal{A}_t+\mathcal{R}_t^+ > 0$ and $\mathcal{R}_t^- \geq \mathcal{R}_t^+$. When the continuation value at date $t+1$ is differentiable in its reference state, $\Theta_{t+1}^{R,+} = \Theta_{t+1}^{R,-}$, so $\mathcal{R}_t^+ = \mathcal{R}_t^- = \mathcal{R}_t$. Weak threshold ordering then requires $1+\mathcal{A}_t+\mathcal{R}_t \geq 0$; strict ordering requires a strict inequality, as at the policy-consistent zero-inflation continuation in Appendix B.1. Away from that continuation, the nonlinear equilibrium determines the thresholds and their ordering jointly with the active regime.

Corollary (Inaction without Rotemberg costs)**.** *Maintain the continuity and directional differentiability conditions of Proposition B.2 and Assumptions B.1–B.2. Set $\psi_p = 0$ and evaluate the policy-consistent zero-inflation continuation used in Appendix B.1. Then*

$$\frac{\varepsilon^{\downarrow}-1}{\varepsilon^{\downarrow}} \leq z_t \leq \frac{\varepsilon^{\uparrow}-1}{\varepsilon^{\uparrow}} \tag{B.14}$$

is an interval on whose interior the reference price is the unique optimal price, even though changing prices carries no quadratic cost; at its endpoints the one-sided conditions hold with equality. Its width is

$$\frac{\Gamma_L-\Gamma_G}{\varepsilon^{\uparrow}\varepsilon^{\downarrow}}\,\frac{1-\beta}{1-\beta\rho} > 0. \tag{B.15}$$

Inaction here comes from the different marginal returns to raising and cutting prices at the reference. It survives even when price changes consume no resources.

With log utility and no Rotemberg adjustment costs, let $\lambda_y \equiv (1+\varphi)/(1-\alpha)$ and recall from Section 2.3 that $\widetilde{y}_t \equiv y_t - a_t$ is the log deviation of physical output from its technology benchmark. The household and production conditions imply

$$z_t = (1-\tau)\exp\{\lambda_y \widetilde{y}_t\}. \tag{B.16}$$

These pricing thresholds therefore translate into thresholds for $\widetilde{y}_t$, the physical-output deviation from its technology benchmark, between which setting the price equal to the reference price remains optimal under the same conditions.

Proposition B.2 is conditional on inherited states and continuation values. It supplies the complementarity conditions used by the nonlinear equilibrium solution; it does not by itself establish aggregate existence, uniqueness, or regime selection.

Assumption B.1 also bounds the shadow values of the firm's state variables. Holding the price plan fixed, a change in the log reference price shrinks by a factor ρ each period, while the

lagged posted price affects only the current adjustment cost. Discounting yields a Lipschitz bound that is uniform across dates and admissible plans. Maximizing over these plans preserves the bound (Milgrom and Segal, 2002, Theorem 2). The directional costates defined above therefore satisfy

$$\lim_{J\to\infty} \beta^J \mathbb{E}_t\left[\left|\Lambda_{t+J} Y_{t+J}^U \Theta_{t+J}^{R,\pm}\right|\right] = 0.$$

Before normalization by $\Lambda_t Y_t^U$, the continuation multipliers are $\beta\rho$ on smooth branches and β under tracking. Together with the uniform bound on shadow values, this makes the discounted remainder converge to zero under either regime. This rules out non-fundamental solutions without requiring the costate itself to converge to zero.

B.3 Inflation Dynamics Away from the Kink

Outside the inaction region, pricing depends on the active branch's demand elasticity and the continuation value of a higher reference price. Maintain $\sigma = 1$, $\rho = 0$, symmetric firms, and deterministic perfect foresight. Then $d_t = \log(P_t/R_t) = \pi_t$: positive inflation selects the loss branch $b_t = L$, and negative inflation the gain branch $b_t = G$.

On a smooth branch, let $\Theta_t \equiv [R_t/(\Lambda_t Y_t^U)]\partial\mathcal{V}_t/\partial R_t$ denote the ordinary reference-state costate. For an interior optimum, where the price corridor does not bind, the exact pricing system is

$$\begin{aligned} \mathcal{P}_t &\equiv \psi_p[(\Pi_t - 1)\Pi_t - \beta\mathcal{M}_{t+1}(\Pi_{t+1} - 1)\Pi_{t+1}], \\ \Theta_t &= \Gamma_{b_t}(1 - z_t), \\ \mathcal{P}_t &= (\varepsilon + \Gamma_{b_t})z_t - (\varepsilon + \Gamma_{b_t} - 1) + \beta\mathcal{M}_{t+1}\Theta_{t+1}. \end{aligned} \tag{B.17}$$

Here $\Pi_t = P_t/P_{t-1}$ is gross inflation, $\mathcal{P}_t$ is the Rotemberg residual, and z_t and $\mathcal{M}_{t+1}$ are defined in Appendix B.2. If the economy switches branch tomorrow, Θ_{t+1} uses tomorrow's elasticity; if it reaches the kink, the appropriate directional costate $\Theta_{t+1}^{R,+}$ or $\Theta_{t+1}^{R,-}$ replaces the ordinary costate.

Let $\bar{mc} = 1 - \tau$ and recall $\widetilde{y}_t = y_t - a_t = y_t^U + \chi_t - a_t$ and $\lambda_y = (1 + \varphi)/(1 - \alpha)$.

Proposition B.3 (Off-kink branchwise Phillips curves)**.** *Conditional on interior pricing on the same branch $b \in \{L, G\}$ at dates t and $t + 1$, the directional first-order pricing equation is*

$$\psi_p(\pi_t - \beta\pi_{t+1}) = \Delta_b + \bar{mc}\lambda_y\left[(\varepsilon + \Gamma_b)\widetilde{y}_t - \beta\Gamma_b\widetilde{y}_{t+1}\right], \qquad \Delta_b \equiv 1 - [\varepsilon + (1 - \beta)\Gamma_b](1 - \bar{mc}). \tag{B.18}$$

Use $b = L$ when $\pi_t, \pi_{t+1} > 0$ and $b = G$ when $\pi_t, \pi_{t+1} < 0$.

Under log utility, $c_t^U = y_t^U$ to first order, so $\mathcal{M}_{t+1}$ has no first-order variation. Substituting $z_t = \bar{mc}(1 + \lambda_y\widetilde{y}_t) + O(2)$ into (B.17) and retaining first-order terms gives (B.18), as in Appendix A.2.

The branch elasticity enters both current demand and the continuation value of a higher reference price. The intercept also differs: setting $\Delta_b = 0$ requires $1 - \bar{m}c = 1/[\varepsilon + (1-\beta)\Gamma_b]$. One employment subsidy cannot set both intercepts to zero when $\Gamma_L \neq \Gamma_G$, even though it can support the kink allocation of Proposition B.1. Equal-sized inflationary and deflationary shocks therefore need not generate mirror-image responses.

Under Proposition B.2 and Assumptions B.1–B.2, a shock that raises z_t above $\overline{z}_t$ selects the loss branch; one that lowers it below $\underline{z}_t$ selects the gain branch. Equation (B.17) governs interior pricing away from the kink. On reentry at $d_t = 0$, the one-sided condition $\mathcal{D}_t^+ \leq 0 \leq \mathcal{D}_t^-$ of (B.11) replaces the pricing equality.

Equation (B.18) is a fixed-branch directional approximation. If $\Delta_b \neq 0$, that branch cannot approach $(\pi_t, \widetilde{y}_t) = (0,0)$ as an ordinary smooth perturbation. The nonlinear solution determines exit, persistence on a branch, and reentry into the inaction region.

B.4 Welfare with Kinked Reference Dependence and CRRA Utility

This subsection extends the welfare approximation in Appendix A.3 to unequal reference-price elasticities and CRRA curvature $\sigma \geq 1$, with log utility at $\sigma = 1$. Maintain symmetric pricing and allow any reference persistence $\rho \in [0,1)$. The benchmark is the zero-inflation MRS–MPL allocation, with $\bar{P} = \bar{R}$, $\bar{\chi} = 0$, and $\bar{C}^U = \bar{Y}^U = \bar{Y}$. Interpreting this benchmark as a steady-state equilibrium requires the firm-optimality conditions of Proposition B.1; membership in the subsidy interval alone does not establish optimality at its endpoints.

Lowercase real quantities denote log deviations from this benchmark. Recall $\widetilde{y}_t = y_t - a_t = y_t^U + \chi_t - a_t$ and $\lambda_y = (1+\varphi)/(1-\alpha)$. Let $\bar{\Lambda} = (\bar{C}^U)^{-\sigma}$ denote steady-state marginal utility. Normalize period utility losses by $\bar{\Lambda}\bar{C}^U = (\bar{C}^U)^{1-\sigma}$:

$$\ell_t \equiv \frac{\bar{u} - u_t}{(\bar{C}^U)^{1-\sigma}}. \tag{B.19}$$

Here u_t is period utility and $\bar{u}$ is its benchmark value.

Proposition B.4 (Welfare with kinked reference dependence and CRRA utility)**.** *Under the assumptions above, the normalized period utility loss has the second-order expansion*

$$\ell_t = \frac{\sigma - 1}{2}(y_t^U)^2 + \frac{\lambda_y}{2}\widetilde{y}_t^2 + \chi_t + \frac{\psi_p}{2}\pi_t^2 + \text{t.i.p.} + O(3), \tag{B.20}$$

where the policy-independent term is $-a_t$. Taking the expected discounted sum of this approximation defines the normalized lifetime loss $\mathcal{L}_{t_0}^{(2)}$. At $\sigma = 1$, the consumption-curvature term vanishes, yielding the log-utility welfare kernel underlying (A.21), with the kinked reference wedge.

Proof. The benchmark condition implies $\bar{N}^{1+\varphi} = (1-\alpha)\bar{\Lambda}\bar{C}^U$. After normalization, the expansion of period utility losses has the same linear and labor terms as the log-utility loss in

Appendix A.3, plus $(\sigma-1)(c_t^U)^2/2$. Substituting $c_t^U = y_t^U - \psi_p \pi_t^2/2 + O(3)$ from (A.18) and $n_t = \widetilde{y}_t/(1-\alpha)$ from (2.14) gives (B.20). These aggregation identities hold on both sides of the kink and at the reference price. □

The curvature term is quadratic in composite output y_t^U, while the activity term is quadratic in $\widetilde{y}_t$. The reference term is signed and first order in the realized price–reference gap $d_t = \log(P_t/R_t)$: $\chi_t = \Gamma_L d_t/(\varepsilon-1)$ above the reference and $\chi_t = \Gamma_G d_t/(\varepsilon-1)$ below it. Thus equal and opposite gaps generate larger first-order losses than gains when $\Gamma_L > \Gamma_G$. At the kink, $\chi_t = 0$, but $\widetilde{y}_t$ need not vanish. This remains possible without Rotemberg costs under the conditions in Appendix B.2.

When $\rho = 0$, $d_t = \pi_t$, so the reference term can be written directly in terms of current inflation. With persistent references, the sign of d_t selects the branch. The expansion is directional at the kink. Absolute losses relative to steady state must retain the productivity term $-a_t$; it vanishes in the cost-push exercises of Section 4, where $a_t = 0$.

Consumption-equivalent losses. The permanent consumption-equivalent loss λ reduces steady-state consumption to $(1-\lambda)\bar{C}^U$, holding labor at $\bar{N}$. For an exact discounted utility loss ΔW, define $x = (1-\beta)\Delta W/(\bar{\Lambda}\bar{C}^U)$. The exact conversion is $\lambda = 1 - \big(1-(1-\sigma)x\big)^{1/(1-\sigma)}$ for $\sigma > 1$, with log-utility limit $\lambda = 1 - e^{-x}$. Its second-order expansion is

$$\lambda = x - \frac{\sigma}{2}\,x^2 + O(x^3). \tag{B.21}$$

Using $x = (1-\beta)\mathcal{L}_0^{(2)}$ produces an approximation even when the exact conversion formula is applied. An order-consistent second-order conversion retains the square of only the full first-order part of x, including the productivity term when it is present.

C Survey Instrument

This appendix documents the survey instrument. The first subsection describes the structure of the survey in the order respondents encounter it—the sequence of blocks, the three randomizations, the piping of respondent-specific information, and the attention-check logic—as implemented in the Qualtrics program. The second reproduces the question screens.

C.1 Structure, Randomization, and Logic

Entry. Respondents recruited on Prolific first pass an entry block: the Prolific ID (auto-filled), a captcha, and a consent form. The consent form states the study's purpose and length (about ten minutes), notes that the survey contains two attention checks and that compensation requires passing both, and instructs respondents not to use AI tools. Declining consent ends the survey. A short introduction then previews the three parts of the survey: naming a product, reporting the price the respondent typically expects to pay for it, and answering hypothetical shopping scenarios. Page timers record time spent on the entry screens, and Qualtrics records total completion time.

Product setup and piping. The respondent selects a product category (a grocery staple, or a self-described alternative) and then enters three free-text fields: the specific product, the package or unit size they typically buy, and the store where they usually buy it (Figure C.1). These three fields are stored and piped into every subsequent question, so each later screen opens with a reminder of the respondent's own item, unit, and store and all prices are quoted per unit of the respondent's own package size. A first attention check follows (a six-point agreement scale with the instruction to select "Disagree").

Baseline elicitation and constructed prices. The respondent reports the price they typically expect to pay for one unit of their item at their usual store, R_0, as a free-entry number (Figure C.2); the entry is cleaned to a numeric value before reuse. From this single input the program computes every price used later in the survey, rounded to the cent: the scenario usual prices $(1 \pm s)R_0$ for $s \in \{0.12, 0.20, 0.28\}$, the updating price $1.2\,R_0$, and a trend-inflation price $1.02^5 R_0$ (described below). The respondent then reports the baseline quantity q_0: the number of units they typically purchase per month at a price of exactly R_0, with R_0 restated in the question text (Figure C.3).

Demand scenarios and their randomization. Two randomizers, both drawn once per respondent with even presentation, then govern the demand block. The first assigns the gap size $g \in \{12\%, 20\%, 28\%\}$; because it is drawn once, the same g applies to both scenarios. The second assigns the scenario order (loss first or gain first), and branch logic routes the respondent

through the two scenario blocks in the assigned order. Within each scenario block, display logic shows only the question pair matching the assigned g, always in the same order: first the quantity question, then the going-forward expected price. The loss-scenario text reads that the price is *usually* $(1-s)R_0$, will *temporarily increase* to R_0 next month only, and will return to $(1-s)R_0$ after next month (Figures C.4 and C.5); the gain scenario is the mirror image, with a usual price of $(1+s)R_0$ and next month's price described as a sale at R_0 (Figures C.6 and C.7). Each quantity question instructs the respondent to hold income and the prices of other goods fixed, to answer only for next month's use, and to ignore stockpiling or delaying purchases; zero is an allowed answer. A second attention check follows the scenario blocks (select "Agree"), and the survey terminates early for respondents who have failed both checks.

Descriptive questions. Three short screens follow: whether the item is perishable, how often the respondent shops for the item at their usual store (five categories, from about twice a week to less than once a month; Figure C.8), and how a posted price equal to R_0 would feel (too low, about right, or too high; Figure C.9).

Reference-price updating. A third evenly-presented randomizer assigns each respondent to the E arm or the N arm. Both arms read the same scenario—the respondent paid R_0 on previous trips and finds a price of $1.2\,R_0$ on the next trip—and differ only in the question: the E arm asks what price the respondent would *typically expect to pay* going forward (Figure C.10), while the N arm asks what price would *feel like the normal price* going forward, glossed as "a price you've gotten used to paying that feels neither too high nor too low" (Figure C.11). E-arm respondents receive one follow-up: after the higher price has persisted for a full month, does it feel too low, about right, or too high (Figure C.12). All respondents then answer a trend-inflation question: after five years in which all prices and the respondent's income rise steadily by 2% per year, so that the item's price reaches $1.02^5 R_0 \approx 1.104\,R_0$, does that price feel too low, about right, or too high. This question is not used in the analysis in this paper.

Closing blocks. The survey ends with the reference-price formation questions—how often the respondent has their typically expected price in mind when shopping (Figure C.14) and which factors shape it (select all that apply; Figure C.13)—followed by two multiple-selection questions on the emotional response to finding the item priced 20% above or 20% below R_0 with no explanation (Figures C.15 and C.16), a privacy note, and an optional free-text feedback box.

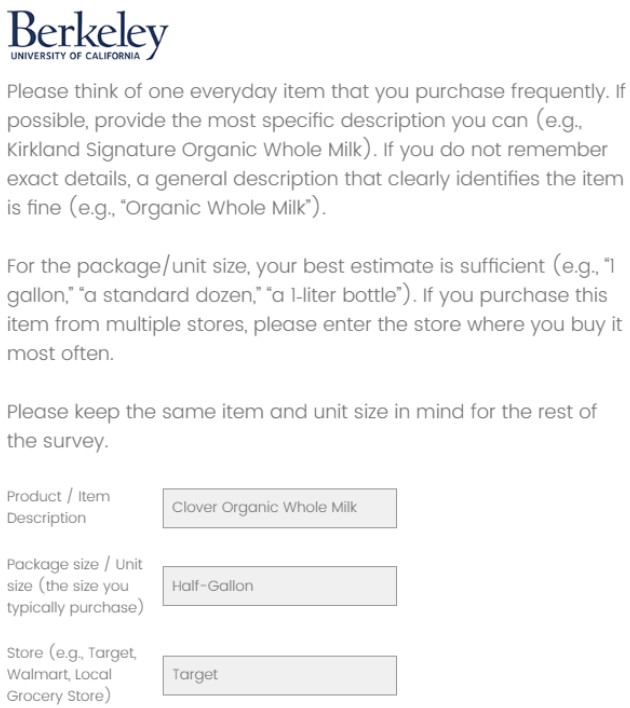
Recorded treatment fields. The randomizer draws and check outcomes are written to embedded data fields—the assigned gap (`ref_shift_price_change`), the scenario order (`ref_shift_order`), the updating arm (`ref_price_update_arm`), and the two attention-check

indicators—which are the treatment variables used in the analysis.

C.2 Question Screens

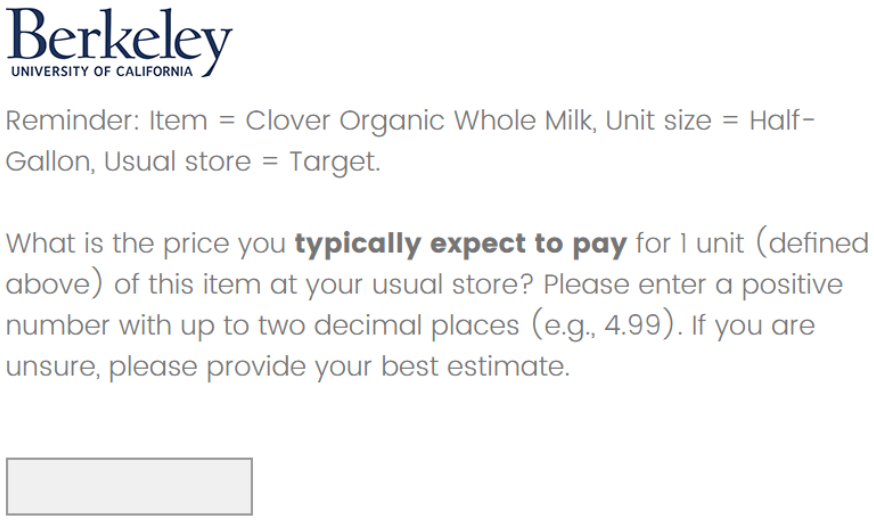
The screens below reproduce each analysis-relevant question in the order respondents encounter them. Throughout, the respondent's own product, package size, store, and elicited reference price are piped into the question text; the screens show these fields filled in for one example respondent (Clover Organic Whole Milk, half-gallon, at Target, with $R_0 = \$4.99$).

Figure C.1: Setup: Product, Package Size, and Usual Store



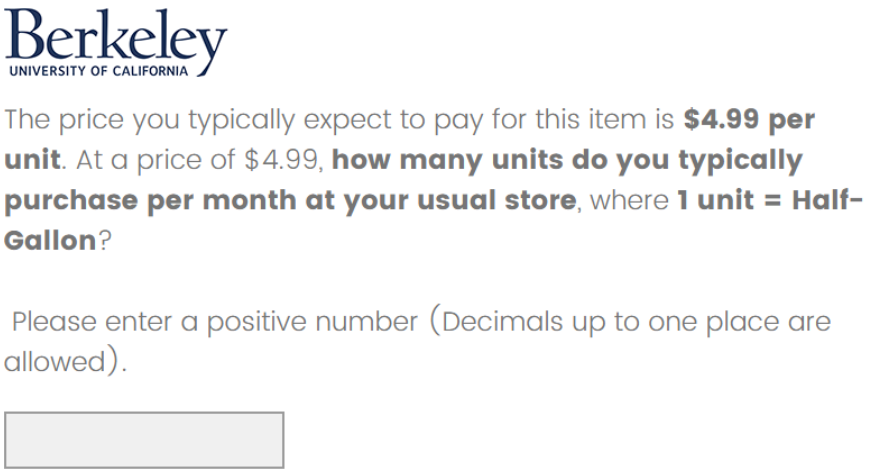
Notes: All later questions pipe the selected product, package/unit size, and store into the question text.

Figure C.2: Baseline Reference-Price Elicitation (R_0)



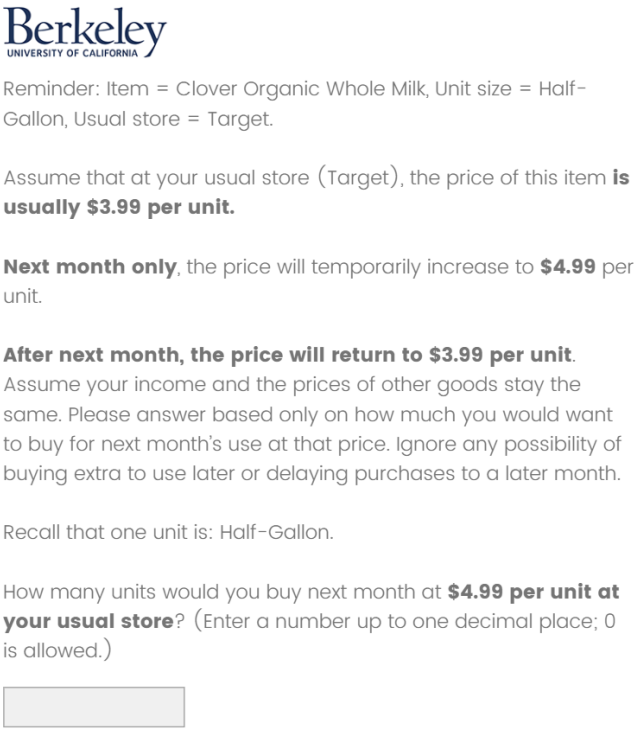
Notes: The per-unit price the respondent typically expects to pay for the item at the usual store.

Figure C.3: Baseline Quantity (q_0)



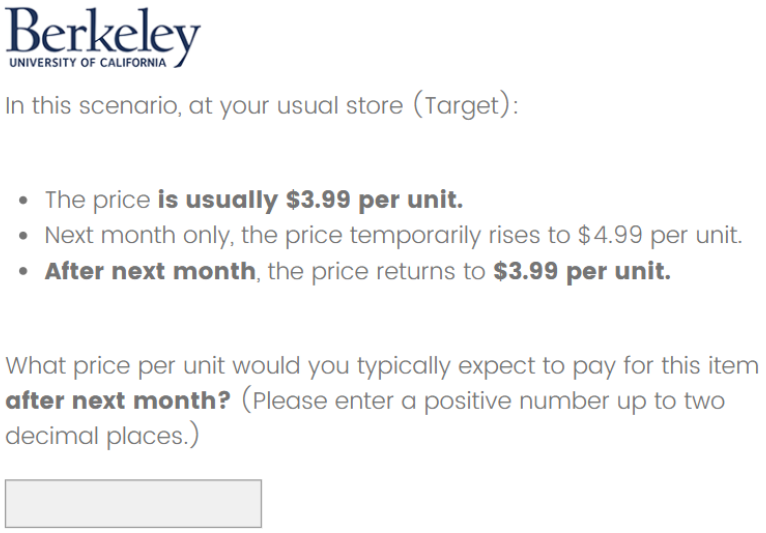
Notes: Units purchased per month at the baseline reference price R_0; q_0 is the offset in the estimating equations.

Figure C.4: Loss Scenario: Quantity (q_L)



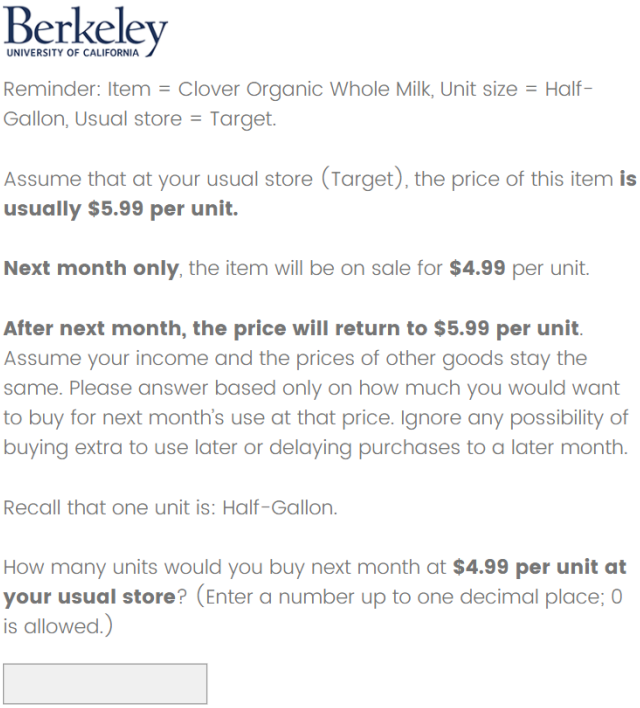
Notes: A usual price of $(1-s)R_0$, a temporary one-month increase to the posted price $P = R_0$, and the units the respondent would buy that month.

Figure C.5: Loss Scenario: Elicited Reference (R_L)



Notes: The per-unit price the respondent would expect to pay going forward.

Figure C.6: Gain Scenario: Quantity (q_G)



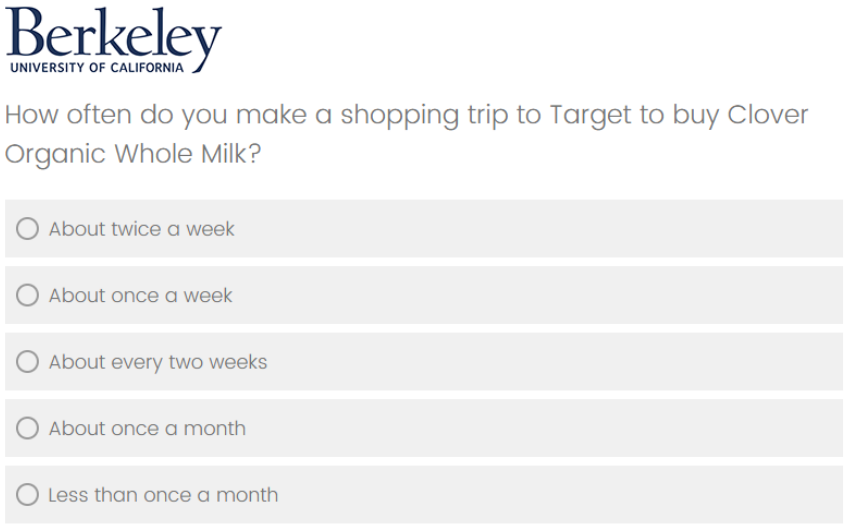
Notes: A usual price of $(1+s)R_0$, a temporary one-month sale at the posted price $P = R_0$, and the units the respondent would buy that month.

Figure C.7: Gain Scenario: Elicited Reference (R_G)



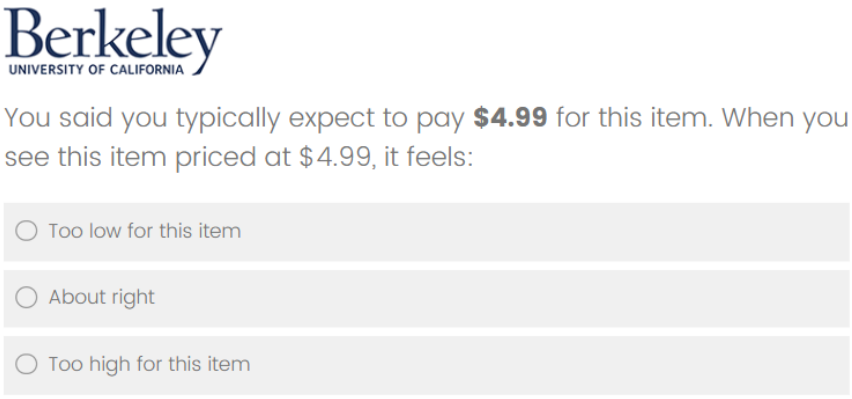
Notes: The per-unit price the respondent would expect to pay going forward.

Figure C.8: Shopping Frequency



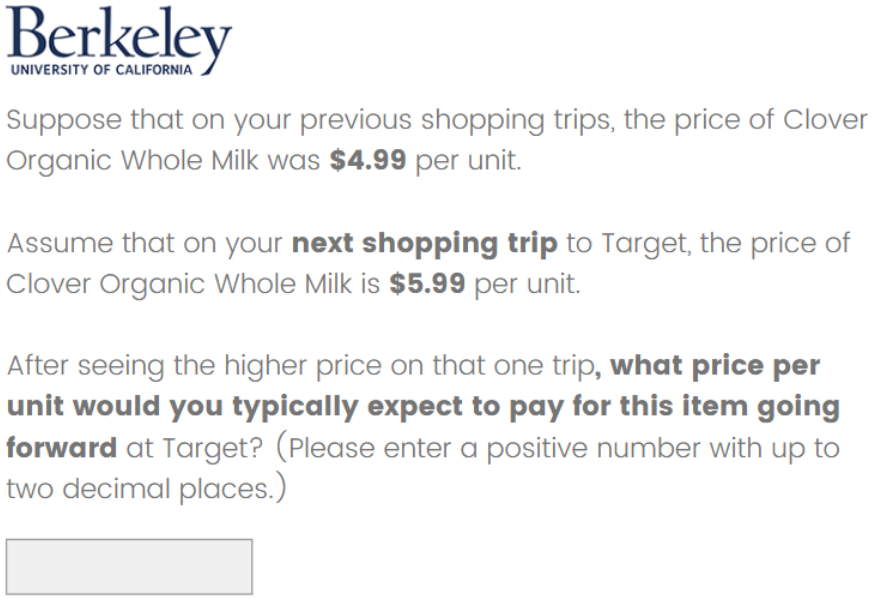
Notes: How often the respondent makes a trip to the usual store to buy the item. This response supplies the trips-per-quarter exponent n_i used to convert $\hat{\rho}_{\text{trip},i}$ to $\hat{\rho}_{\text{qtr},i}$.

Figure C.9: Baseline Reference-Price Feeling



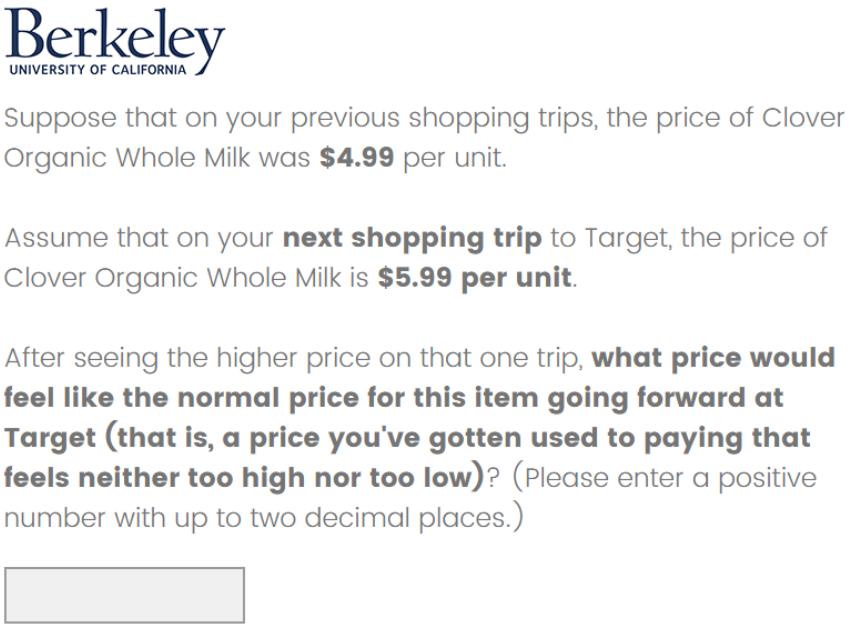
Notes: Whether a posted price equal to R_0 feels too low, about right, or too high for the item.

Figure C.10: Reference-Price Persistence Elicitation, E Arm



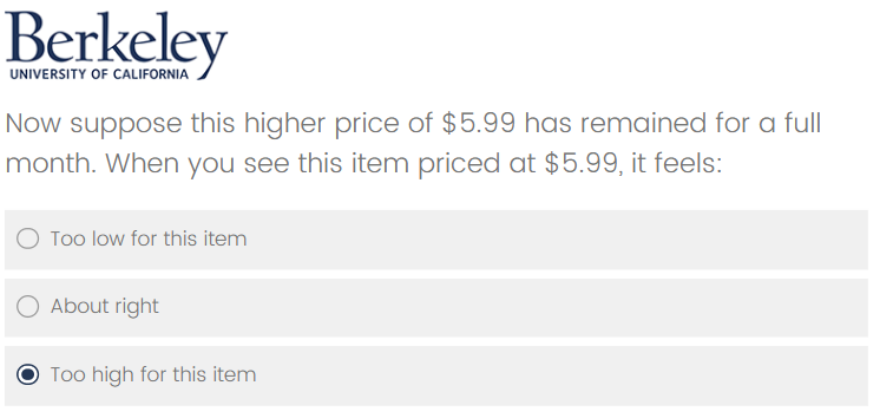
Notes: The price the respondent would *typically expect to pay* going forward after one trip at a price 20% above R_0.

Figure C.11: Reference-Price Persistence Elicitation, N Arm



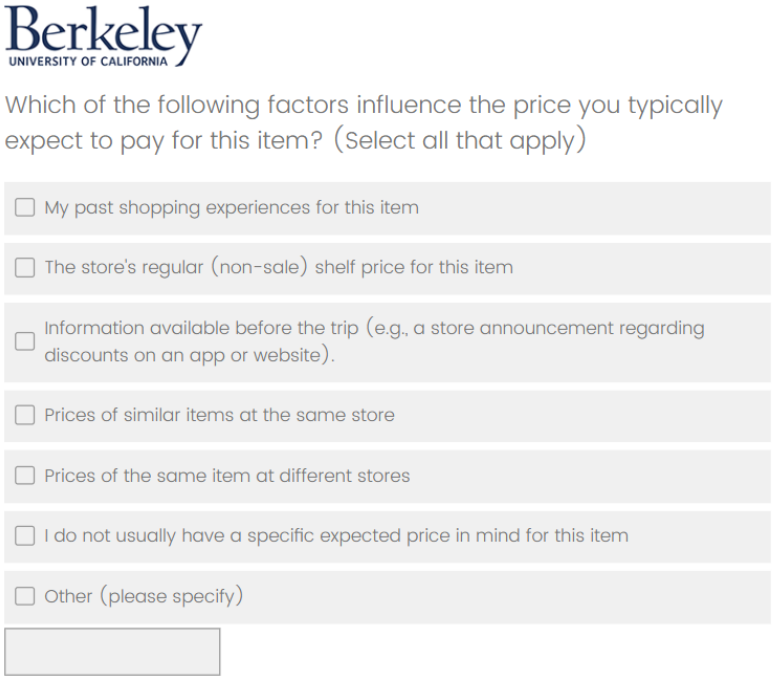
Notes: The price that would *feel like the normal price* going forward after one trip at a price 20% above R_0.

Figure C.12: Post-Shock Feeling Follow-Up (E Arm)



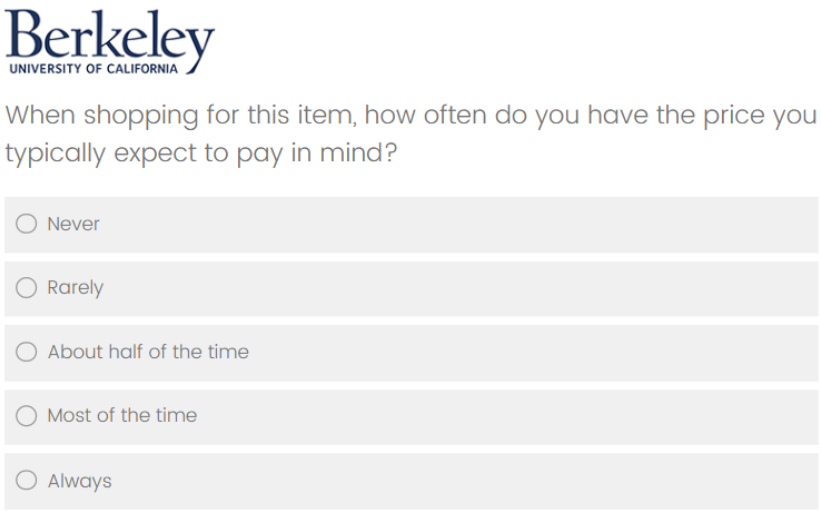
Notes: Whether, once the higher price has persisted for a full month, it feels too low, about right, or too high. This screen underlies the diagnostic separating price expectations from the internal benchmark.

Figure C.13: Reference-Price Formation



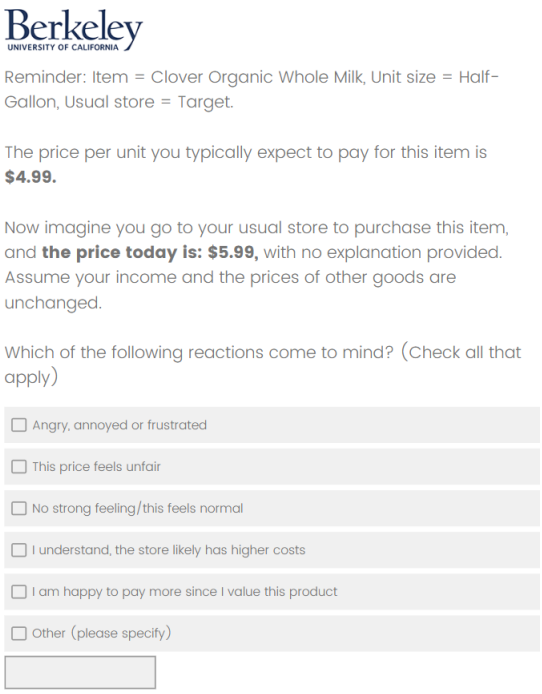
Notes: The factors the respondent reports as influencing the price they expect to pay (select all that apply). This question underlies Figure 2.

Figure C.14: Reference-Price Salience



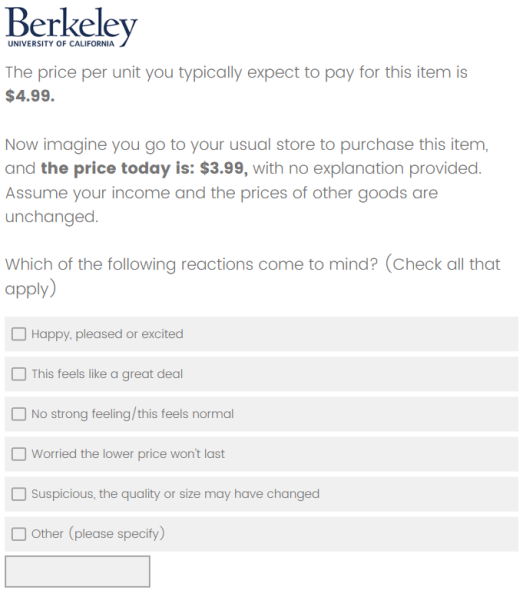
Notes: How often the respondent has the expected price in mind while shopping. This question underlies panel (b) of Figure 1.

Figure C.15: Emotional Response to a Surprise Price Increase



Notes: Reactions to paying 20% above R_0 with no explanation (check all that apply). Reproduced for completeness of the instrument; these responses are not analyzed in this paper.

Figure C.16: Emotional Response to a Surprise Price Decrease



Notes: Reactions to paying 20% below R_0 with no explanation (check all that apply). Reproduced for completeness of the instrument; these responses are not analyzed in this paper.

D Survey Identification and Additional Results

D.1 Derivation of Proposition 3

This appendix derives Proposition 3 from the choice problem (3.1), maintaining the assumptions stated in Section 3.2. The argument applies separately within each reference-price domain $b \in \{L, G\}$ with slope Γ_b; to keep the notation light I write Γ for the active domain's slope.

Setup. Write the focal product's baseline attractiveness as $A_i \equiv e^{\delta_{if}} P_i^{-\varepsilon}$, the exponential of the non-stochastic part of its baseline choice utility in (3.1); because P_i and δ_{if} are held fixed, it does not change across survey scenarios for the same respondent. Let $D_{i,-f} \equiv \sum_{j \neq f} e^{\delta_{ij}}$ denote the contribution of the outside option and all non-focal alternatives to the choice denominator; because all δ_{ij} are held fixed, it too is fixed across scenarios. At baseline the reference price equals the posted price, so the transaction-utility term in (3.1) is zero and the baseline denominator is $D_{i0} = D_{i,-f} + A_i$. In a survey scenario the posted price remains P_i while the reference moves to R_{ib}; define

$$x_{ib} \equiv \log\left(\frac{R_{ib}}{P_i}\right), \qquad s_{ib} \equiv e^{\Gamma x_{ib}}, \qquad \omega_i \equiv \frac{A_i}{D_{i0}}, \tag{D.1}$$

so that s_{ib} is the transaction-utility shifter, ω_i is the focal product's baseline share in the respondent's choice set, and the scenario denominator is $D_{i1} = D_{i,-f} + A_i s_{ib}$.

Step 1: choice probabilities. Under the choice model in Section 3.2, the taste shocks are i.i.d. standard Type I extreme-value draws, so each purchase occasion is a multinomial logit choice: the probability of selecting an alternative is its exponentiated non-stochastic utility divided by the sum over all alternatives. For the focal product,

$$\Pr_{i0}(f) = \frac{A_i}{D_{i0}} \quad \text{at baseline}, \qquad \Pr_{i1}(f) = \frac{A_i\, s_{ib}}{D_{i1}} \quad \text{in the scenario}, \tag{D.2}$$

because moving the reference multiplies the focal product's exponentiated utility by $s_{ib} = e^{\Gamma x_{ib}}$ and leaves every other term unchanged.

Step 2: expected quantities. The respondent makes M_i purchase occasions during the scenario month, and the draws are independent across occasions, so expected purchases are the number of occasions times the per-occasion probability:

$$\mathbb{E}[q_{0i}] = M_i \frac{A_i}{D_{i0}}, \qquad \mathbb{E}[q_{ib}] = M_i \frac{A_i\, s_{ib}}{D_{i1}}. \tag{D.3}$$

Step 3: the exact ratio. Because only the focal product's term changes, the scenario denominator decomposes around the baseline denominator:

$$D_{i1} = D_{i,-f} + A_i s_{ib} = D_{i0} + A_i\,(s_{ib} - 1) = D_{i0}\big[1 + \omega_i(s_{ib} - 1)\big]. \tag{D.4}$$

Dividing the two expressions in (D.3) and using (D.4), the occasion count M_i, the attractiveness A_i—and with it every fixed respondent–product taste δ_{if}—and the baseline denominator D_{i0} all cancel:

$$\frac{\mathbb{E}[q_{ib}]}{\mathbb{E}[q_{0i}]} = \frac{s_{ib}}{1 + \omega_i(s_{ib} - 1)} = \frac{e^{\Gamma x_{ib}}}{1 + \omega_i\,(e^{\Gamma x_{ib}} - 1)}. \tag{D.5}$$

This is the exact finite-share counterpart to the limit in Proposition 3.

Step 4: the small-share approximation. Taking logs of (D.5),

$$\log\!\left(\frac{\mathbb{E}[q_{ib}]}{\mathbb{E}[q_{0i}]}\right) = \Gamma x_{ib} - \log\!\Big[1 + \omega_i\left(e^{\Gamma x_{ib}} - 1\right)\Big]. \tag{D.6}$$

The scenario design bounds the elicited gap—the reference-price filter restricts R_{ib} to a corridor around P_i of width proportional to the announced shift—so $e^{\Gamma x_{ib}} - 1$ is bounded and the argument $u_{ib} \equiv \omega_i(e^{\Gamma x_{ib}} - 1)$ is of order ω_i. Using $\log(1 + u) = u + O(u^2)$,

$$\log\!\left(\frac{\mathbb{E}[q_{ib}]}{\mathbb{E}[q_{0i}]}\right) = \Gamma x_{ib} - \omega_i\left(e^{\Gamma x_{ib}} - 1\right) + O(\omega_i^2) = \Gamma x_{ib} + O(\omega_i). \tag{D.7}$$

The first-order correction $-\omega_i(e^{\Gamma x_{ib}} - 1)$ works against the slope in both domains: when the reference rises above the posted price the product becomes more attractive within the choice set and the denominator absorbs part of the demand increase, and symmetrically below. For a single brand-size grocery item in a household's full grocery basket, ω_i is tiny and the correction is negligible.

Step 5: the estimating equation. Rearranging (D.7), the conditional mean of the scenario quantity is

$$\mathbb{E}[q_{ib} \mid i, x_{ib}] = \mathbb{E}[q_{0i}]\; e^{\Gamma x_{ib}}\,\big[1 + O(\omega_i)\big], \tag{D.8}$$

so $\log \mathbb{E}[q_{ib} \mid i, x_{ib}] = \alpha_i + \Gamma x_{ib} + O(\omega_i)$ with respondent-level intercept $\alpha_i \equiv \log \mathbb{E}[q_{0i}]$. The survey measures the baseline mean directly: q_{0i} is the respondent's reported baseline quantity at $R_{ib} = P_i$, so $\log q_{0i}$ implements α_i as an offset. This is the specification estimated in Section 3.2. Letting $\omega_i \to 0$ in (D.7) gives (3.2) and completes the proof of Proposition 3. □

Pooling across products rests on one assumption: a common slope within each domain. Identification comes from within-respondent changes in x_{ib} rather than from level comparisons across products, so respondent-specific reference prices and fixed respondent–product tastes need

no modeling; they cancel in (D.5). If Γ instead varies across consumers or products, the pooled coefficient is a design-weighted average of their slopes rather than the representative-agent parameter of Section 2.

D.2 Sample Construction and Descriptive Supplements

The baseline filters referenced in the main text are five exclusion rules applied before any analysis. The instrument contains two attention checks, and I drop respondents who fail both. I further exclude respondents who report a baseline quantity above 50 units per month, report a baseline price outside $[\$0.50, \$100]$, report any scenario quantity above three times their baseline quantity, or enter a price or reference response at or below one cent. These rules remove 69 of the 1,501 collected responses and leave the descriptive sample of $N = 1{,}432$; Table D.1 gives the count behind each rule and the resulting analysis samples.

Estimation of the reference-price elasticities applies an additional scenario-level filter. Let $g_i \in \{0.12, 0.20, 0.28\}$ denote the assigned proportional gap between the scenario's usual price and the posted price $P_i = R_{0i}$. I retain loss-scenario observations with $R_{iL} \in [P_i(1 - 1.5g_i),\ 1.03P_i]$ and gain-scenario observations with $R_{iG} \in [0.97P_i,\ P_i(1 + 1.5g_i)]$. The outer bound permits the elicited reference price to overshoot the scenario's usual price by up to half the assigned gap; the inner bound permits it to lie up to 3% on the opposite side of the posted price.

For example, a respondent with a \$5 baseline reference price assigned a 20% loss-scenario gap is told that the item usually costs \$4, costs \$5 next month, and returns to \$4 thereafter. Elicited reference prices between \$4 and \$5 reflect partial updating. The filter retains values between \$3.50 and \$5.15, excluding an answer such as \$6, which exceeds every price specified in the scenario. In the corresponding gain scenario, the usual price is \$6 and next month's price is \$5; the filter retains values between \$4.85 and \$6.50, excluding an answer such as \$4. The filter leaves 1,238 loss-scenario respondents and 1,223 gain-scenario respondents, with 1,137 respondents (2,274 scenario observations) retained in both scenarios for the balanced sample.

Table D.2 summarizes the products respondents chose to answer about: everyday staples dominate, led by dairy milk (35%), eggs (16%), and soda, water, and energy drinks (9%), consistent with the design's request for a frequently purchased item whose price the respondent knows; 75% of the chosen products are perishable. Figure D.1 shows the distributions of the baseline reference price and the baseline monthly quantity in the descriptive sample: the median reference price is \$3.99 (mean \$5.15), the median quantity is three units per month (mean 4.8), and 2.7% of respondents report more than 20 units. The median respondent completed the survey in 470 seconds (mean 567, standard deviation 354). Table D.3 reports the raw demand responses by scenario and assigned gap that the main text summarizes.

Table D.4 collects the validation shares cited in the main text. At baseline, 83% of respondents say a posted price equal to their elicited R_0 feels "about right" for the item. The

post-shock follow-up responses, trip-level persistence estimates, and conversion to the quarterly frequency are discussed in Appendix D.5.

Table D.1: Sample Construction: Exclusion Counts and Analysis Samples

	N
Raw completed responses	1,501
Respondent-level exclusions (may overlap)	
Failed both attention checks	1
Baseline quantity $q_0 > 50$ units per month	8
Baseline price outside [\$0.50, \$100]	11
Any scenario quantity above $3q_0$	25
Any price or reference elicitation $\leq$ \$0.01	27
Full descriptive sample after baseline exclusions	**1,432**
Γ *estimation: row-level reference-price filter*	
Loss scenario, $R_L \in [P(1-1.5g),\ 1.03P]$	1,238
Gain scenario, $R_G \in [0.97P,\ P(1+1.5g)]$	1,223
Balanced (both scenarios retained)	1,137
ρ *estimation*	
After dropping post-shock reference below the original reference	1,330
E arm with $\hat{\rho}_{\text{trip}} \in [0,1]$	651
N arm with $\hat{\rho}_{\text{trip}} \in [0,1]$	629

Notes: Respondent-level exclusions are applied to every analysis; a respondent can appear in more than one exclusion row. The reference-price filter operates at the row level: it removes a respondent's loss (gain) scenario when the elicited going-forward reference lies outside the stated interval, where P is the temporary posted price and g the randomized shift size; the outer edge is 1.5 times the announced shift and the inner edge carries a 3% tolerance. The balanced sample retains respondents whose loss and gain scenarios both pass the window. The ρ sample additionally drops respondents whose post-shock reference report lies below the original reference price, and the in-range rows drop $\hat{\rho}_{\text{trip}} \notin [0,1]$ (28 E-arm and 22 N-arm respondents).

Table D.2: Main product categories in the survey analysis sample. Each of the $N = 1{,}432$ respondents passing all exclusion restrictions named one frequently purchased everyday product in a free-text field; responses are coded into categories by an ordered keyword rule list, with author review of ambiguous responses.

Category	N	Share (%)
Milk (dairy)	498	34.8
Eggs	230	16.1
Soda, water, and energy drinks	135	9.4
Fresh produce	80	5.6
Milk (plant-based)	68	4.7
Pantry, frozen, and prepared foods	68	4.7
Bread	67	4.7
Snacks and sweets	48	3.4
Coffee and coffee creamer	44	3.1
Other dairy (cheese, butter, ice cream)	42	2.9
Meat, poultry, and seafood	34	2.4
Juice, tea, and other beverages	33	2.3
Yogurt	25	1.7
Paper and paper products	16	1.1
Cereal	4	0.3
Other	40	2.8
Total	1,432	100.0

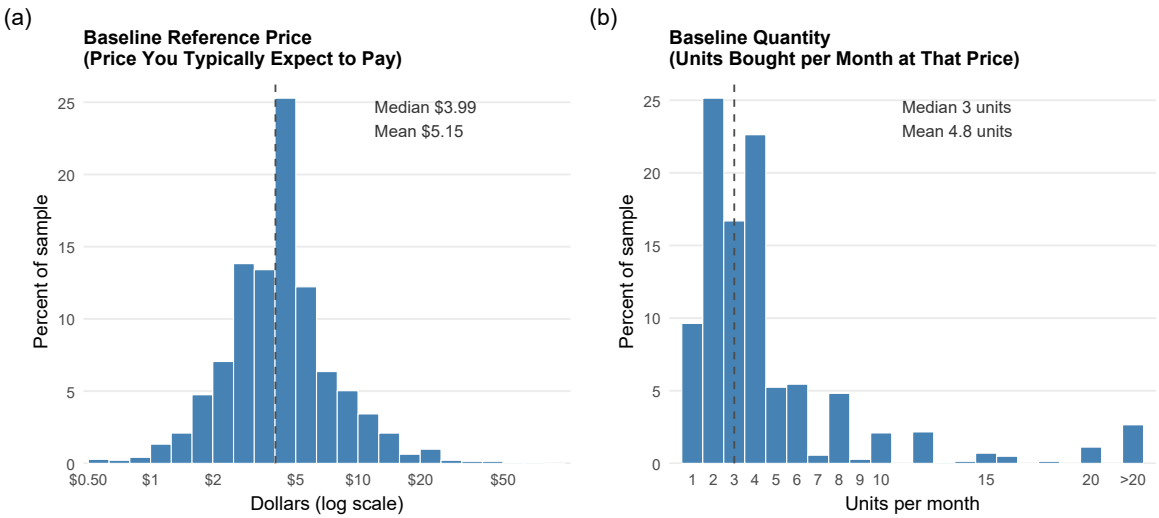
Notes: "Other" comprises non-paper household and personal-care items, alcohol and tobacco, pet food and medical items, and responses too vague to classify (e.g., a bare number or store-brand name). Energy drinks are grouped with soda and water; sweet bakery items (muffins, croissants, donuts) with snacks; granola and hot cereal with pantry; and half-and-half with other dairy.

Table D.3: Raw Demand Responses by Reference-Price Domain

Scenario	N	Mean %	Median %	% Decrease	% Unchanged	% Increase
Panel A: posted price above the reference (loss domain)						
Pooled	1,238	−12.1	0.0	30.8	64.1	5.1
12% temporary increase	412	−8.0	0.0	26.5	67.5	6.1
20% temporary increase	423	−13.3	0.0	31.2	64.3	4.5
28% temporary increase	403	−14.9	0.0	34.7	60.5	4.7
Panel B: posted price below the reference (gain domain)						
Pooled	1,223	11.7	0.0	11.1	60.7	28.2
12% temporary decrease	407	11.7	0.0	9.8	65.1	25.1
20% temporary decrease	413	11.0	0.0	10.9	62.2	26.9
28% temporary decrease	403	12.5	0.0	12.7	54.6	32.8

Notes: Each cell summarizes the percentage change in the stated scenario quantity relative to the baseline quantity, $q_b/q_0 - 1$. Rows split the sample by the randomized shift size g. The sample applies the baseline exclusions and the reference-price filter described in the text: on the loss side the elicited reference must satisfy $R_L \in [P(1 - 1.5g),\ 1.03P]$, on the gain side $R_G \in [0.97P,\ P(1 + 1.5g)]$, where P is the temporary posted price. The theory-consistent direction is a quantity decrease in Panel A and an increase in Panel B.

Figure D.1: Baseline Reference Prices and Monthly Quantities



Notes: Descriptive sample after baseline exclusions ($N = 1{,}432$). Panel (a): the price respondents typically expect to pay for their chosen product, R_0, on a log scale (median \$3.99, mean \$5.15). Panel (b): the units they would buy over the next month at that price, q_0, in unit bins, with quantities above 20 pooled in the last bar (median 3, mean 4.8). Dashed lines mark the medians.

Table D.4: Baseline Validation and Post-Shock Feeling (Percent of Respondents)

	Too low	About right	Too high	N
A. Baseline: a posted price equal to the elicited R_0 feels...				
All respondents	1.1	83.1	15.8	1,432
B. E arm, after a month at the 20% higher price, it feels...				
Baseline price felt about right	0.2	21.6	78.3	575
and expectation moved ≥95% of the way	0.3	23.3	76.4	301
Expectation moved ≥95%, any baseline feeling	0.3	20.1	79.7	359

Notes: Panel A uses the descriptive sample after baseline exclusions (question in Figure C.9). Panel B uses E-arm respondents in the persistence sample who answered the follow-up question (Figure C.12); "moved at least 95% of the way" means $\hat{\rho}_{\text{trip},i} \leq 0.05$ in equation (D.11). Rows sum to 100 up to rounding.

D.3 Control-Function Estimation and Asymmetry

First stage. For the assigned proportional gap $g_i \in \{0.12, 0.20, 0.28\}$, the instrument is $z_{iL} = \log(1 - g_i)$ in the loss scenario and $z_{iG} = \log(1 + g_i)$ in the gain scenario. Column (2) of Table 2 omits the order terms from both the first stage (3.4) and the control function (3.5); columns (3) and (4) use the full first stage and include $\hat{v}_{ib}$ and $O_{ib}\hat{v}_{ib}$. The reported first-stage F statistic tests the excluded assigned-gap terms, $\pi_1 = 0$ in column (2) and $\pi_1 = \pi_{zO} = 0$ in columns (3)–(4); the preferred unweighted specification gives $F = 375.6$ on the loss side and $F = 262.0$ on the gain side. First stages are re-estimated in every bootstrap replication, and in the ACS-weighted specifications the calibration weights are re-estimated within each replication as well (Appendix D.6.3).

Asymmetry test. Table D.5 tests the difference directly on the balanced sample that stacks each respondent's two scenarios. Column (1) estimates the plain Poisson specification

$$\mathbb{E}[q_{ib} \mid q_{0i}, x_{ib}, D_{L,ib}] = q_{0i} \exp(\Gamma_G x_{ib} + \Delta\Gamma \, D_{L,ib} \, x_{ib}) \,, \tag{D.9}$$

where $D_{L,ib}$ indicates the loss scenario, so $\Delta\Gamma = \Gamma_L - \Gamma_G$ and its standard error is a direct test of the asymmetry. Columns (2)–(4) add the control function. The first stage is re-estimated on the stacked sample with z_{ib}, $D_{L,ib}$, and $D_{L,ib} z_{ib}$; columns (3)–(4) additionally include O_{ib} and $O_{ib} z_{ib}$. Its residual enters the second stage with domain and order interactions,

$$\mathbb{E}[q_{ib} \mid q_{0i}, x_{ib}, D_{L,ib}, \hat{v}_{ib}, O_{ib}] = q_{0i} \exp(\Gamma_G x_{ib} + \Delta\Gamma \, D_{L,ib} \, x_{ib} + \delta \hat{v}_{ib} + \delta_L D_{L,ib} \hat{v}_{ib} + \delta_O O_{ib} \hat{v}_{ib}) \,. \tag{D.10}$$

The preferred column (3) gives $\hat{\Gamma}_G = 0.329$ (0.112) and $\Delta\hat{\Gamma} = 0.511$ (0.178), significant at the 1% level, with implied $\hat{\Gamma}_L = 0.840$ (0.096): loss-side reference dependence is roughly two and a half times gain-side reference dependence. In the language of Section 2, demand is kinked at the reference price, the case Appendix B analyzes, and $\Delta\hat{\Gamma}$ measures the kink. With ACS weights the asymmetry is larger, 0.590 (0.249), and the gain-side slope is smaller (0.305, standard error 0.144). The asymmetry does not depend on the Poisson form or its treatment of zeros: the log-linear two-stage least squares counterpart on the positive-quantity sample gives $\Delta\hat{\Gamma} = 0.407$ (0.122) and the perishable-product subsample gives 0.496 (0.152), both significant at the 1% level (estimates not tabulated).

Table D.5: Asymmetric Reference-Price Elasticities

	PPML (1)	CF (2)	CF + Order (3)	CF + Order (ACS wt.) (4)
$\hat{\Gamma}_G$	0.331***	0.325***	0.329***	0.305**
	(0.111)	(0.113)	(0.112)	(0.144)
$\Delta\hat{\Gamma}$	0.442***	0.511***	0.511***	0.590**
	(0.170)	(0.183)	(0.178)	(0.249)
$\hat{\Gamma}_L$	0.773***	0.836***	0.840***	0.895***
	(0.086)	(0.099)	(0.096)	(0.136)
Observations	2,274	2,274	2,274	2,178
Respondents	1,137	1,137	1,137	1,089
First-stage F		657.3	439.1	410.2

Notes: Poisson pseudo-maximum likelihood on the balanced sample that stacks each respondent's loss and gain scenarios, estimating $\log \mathbb{E}[q_{ib}] = \log q_{0,i} + \Gamma_G x_{ib} + \Delta\Gamma\, D_{L,ib}\, x_{ib}$ plus control-function terms, where $x_{ib} = \log(R_{ib}/P_i)$ and $D_{L,ib}$ indicates the loss scenario; $\hat{\Gamma}_L = \hat{\Gamma}_G + \Delta\hat{\Gamma}$ carries a delta-method standard error. Column (1) treats x_{ib} as exogenous; columns (2)–(3) add the control-function residual, its loss-scenario interaction, and, in column (3), its interaction with scenario order, none of which are reported. Column (4) applies the ACS calibration weights, excluding the 48 respondents outside the calibration universe. Samples, weighting, and standard errors otherwise follow Table 2. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

D.4 Robustness of the Reference-Price Elasticities

Tables D.6 and D.7 re-estimate Γ_L and Γ_G under one change at a time to the sample or the estimator, each row in the four columns of Table 2. The rows answer three questions: whether the reference price is measured correctly, which product the respondent had in mind, and whether the estimator matters. The estimates quoted below are from the preferred column (3), loss side first.

Measurement of the reference. Identification sets the posted price equal to the baseline reference price, so baseline transaction utility must be zero; if the elicited R_{0i} is not the respondent's reference, it is not. Restricting the sample to the 83% of respondents who say that a posted price equal to R_{0i} feels "about right" leaves the loss-side estimate almost unchanged at 0.86 and lowers the gain-side estimate to 0.26 (row B). Learning or fatigue could change how carefully respondents read the second scenario they see. Letting the slope differ by presentation order gives 0.72 when the loss scenario comes first and 1.03 when it comes second, and 0.43 and 0.20 on the gain side (rows C and D); neither difference is significant at the 5% level in the control-function columns.

The product in mind. Rows E and F check that the estimates do not depend on respondents with unusually large baseline quantities. Restricting the sample to baseline quantities of at most 20 or 10 units gives 0.79 and 0.69 on the loss side and 0.39 and 0.52 on the gain side: heavy buyers respond more to a loss and less to a gain. The quantity question asks for next month's use, and stockpiling is least feasible for perishable products; restricting to them gives 0.77 and 0.24 (row G). The product screen used dairy milk as its example, and a third of respondents chose dairy milk; without it the estimates are 1.01 and 0.46 (row H), so an example-driven product choice does not drive the result. Finally, the model's Γ is the elasticity for a single variety, and substitution away from a specific product is easier than away from a whole category: a respondent who named a brand can switch brands when the deal turns bad, while a respondent who named "milk" cannot. Specific answers give 0.92 and 0.66 (row I), 0.93 and 0.61 with fresh produce added (row J), and generic answers excluding produce give 0.81 and -0.08 (row K). The loss-side response is common to both groups; the gain-side response comes from the specific answers, which are the closer counterpart to the model's object.

The estimator. The stacked specification of Table D.5 implies 0.84 and 0.33 (row L), in line with the single-domain estimates. The zero intercept is the theoretical restriction that purchases equal baseline when the reference is unchanged; freeing it leaves the loss slope at 0.79 with a standard error roughly three times larger and estimated levels that are small and at most marginally significant (row M), so the restriction is not rejected, while on the gain side the intercept, slope, and control-function residual are nearly collinear and the unrestricted model is uninformative. A log-linear two-stage least squares estimator changes the functional form and drops the zero quantities that Poisson retains; it gives 0.73 and 0.30 (row N).

Across every row the loss-side elasticity, which prices the welfare cost of inflation, lies

between 0.69 and 1.03; the gain side is the fragile one.

Table D.6: Robustness of the Loss-Side Reference-Price Elasticity Γ_L

Sample or specification	PPML (1)	CF (2)	CF + order (3)	ACS wt. (4)	N
A. Baseline, loss sample (Table 2)	0.788***	0.873***	0.875***	0.933***	1,238
	(0.084)	(0.095)	(0.097)	(0.131)	
B. Baseline price feels "about right"	0.769***	0.860***	0.864***	0.972***	1,025
	(0.091)	(0.107)	(0.111)	(0.148)	
C. Baseline, loss scenario presented first	0.594***	0.705***	0.718***	0.830***	1,238
	(0.083)	(0.104)	(0.105)	(0.151)	
D. Baseline, loss scenario presented second	0.986***	1.039***	1.033***	1.024***	1,238
	(0.145)	(0.165)	(0.158)	(0.209)	
E. Baseline quantity at most 20 units	0.758***	0.790***	0.791***	0.811***	1,210
	(0.071)	(0.074)	(0.073)	(0.104)	
F. Baseline quantity at most 10 units	0.653***	0.687***	0.688***	0.647***	1,152
	(0.055)	(0.058)	(0.059)	(0.068)	
G. Perishable products only	0.712***	0.764***	0.766***	0.848***	934
	(0.071)	(0.077)	(0.073)	(0.130)	
H. Excluding dairy milk	0.908***	1.007***	1.008***	1.059***	804
	(0.121)	(0.138)	(0.141)	(0.190)	
I. Specific product answers only	0.816***	0.927***	0.922***	1.047***	638
	(0.110)	(0.133)	(0.137)	(0.212)	
J. Specific answers plus fresh produce	0.818***	0.936***	0.933***	1.040***	697
	(0.102)	(0.126)	(0.125)	(0.185)	
K. Generic answers only, excluding fresh produce	0.762***	0.798***	0.810***	0.792***	534
	(0.141)	(0.137)	(0.148)	(0.157)	
L. Implied by Table D.5, $\hat{\Gamma}_G + \Delta\hat{\Gamma}$	0.773***	0.836***	0.840***	0.895***	1,137
	(0.086)	(0.099)	(0.096)	(0.136)	
M. Free intercept	0.479***	0.790***	0.789***	0.937***	1,238
	(0.141)	(0.277)	(0.259)	(0.311)	
N. Log-linear outcome, positive quantities	OLS	2SLS	2SLS + order	ACS wt.	
	0.675***	0.731***	0.732***	0.705***	1,182
	(0.056)	(0.060)	(0.060)	(0.070)	

Notes: Each row re-estimates Γ_L from the loss-side specification of Table 2, Poisson pseudo-maximum likelihood with $\log q_0$ as offset, on a different sample or with a different estimator; standard errors below each estimate. Columns as in Table 2: (1) elicited gap treated as exogenous, respondent-clustered SEs; (2)–(3) control function with the randomized gap, in (3) interacted with scenario order, respondent-bootstrap SEs (1,000 draws); (4) column (3) with ACS calibration weights (Appendix D.6), re-estimated in each draw, on the 95–96% of each sample inside the calibration universe. N is respondents. Rows C and D come from one specification that replaces $\Gamma_L x_{iL}$ with $\Gamma_L^{\text{first}} O_{iL} x_{iL} + \Gamma_L^{\text{second}}(1 - O_{iL}) x_{iL}$, with $O_{iL} = 1$ when the loss scenario was presented first as in equation (3.5), keeping the control-function terms; the difference between the two slopes is not significant at the 5% level in the control-function columns ($p = 0.09$ in column (3)), though it is in column (1) ($p = 0.02$). Rows I and J: the product answer names a brand, product line or retailer; J also keeps fresh produce. Row K: the complement, answers naming no brand, product line or retailer, with fresh produce excluded. Row L: delta-method SEs. Row M frees the level the demand system sets to zero, and in columns (3)–(4) an order-specific level as well; the estimated levels are small and at most marginally significant. Row N: log outcome, 2SLS with the randomized gap as instrument, respondent-clustered SEs. First-stage $F > 140$ in every instrumented column. $^{***}p < 0.01$, $^{**}p < 0.05$.

Table D.7: Robustness of the Gain-Side Reference-Price Elasticity Γ_G

Sample or specification	PPML (1)	CF (2)	CF + order (3)	ACS wt. (4)	N
A. Baseline, gain sample (Table 2)	0.347***	0.324***	0.326***	0.310**	1,223
	(0.105)	(0.108)	(0.109)	(0.141)	
B. Baseline price feels "about right"	0.279**	0.259**	0.263**	0.196	1,022
	(0.110)	(0.111)	(0.112)	(0.154)	
C. Baseline, gain scenario presented first	0.455***	0.428**	0.434**	0.508***	1,223
	(0.173)	(0.174)	(0.173)	(0.196)	
D. Baseline, gain scenario presented second	0.225**	0.207*	0.204*	0.067	1,223
	(0.108)	(0.111)	(0.117)	(0.198)	
E. Baseline quantity at most 20 units	0.404***	0.385***	0.386***	0.384***	1,198
	(0.088)	(0.086)	(0.094)	(0.126)	
F. Baseline quantity at most 10 units	0.549***	0.515***	0.517***	0.528***	1,140
	(0.059)	(0.063)	(0.067)	(0.079)	
G. Perishable products only	0.254***	0.244**	0.241**	0.146	916
	(0.092)	(0.095)	(0.094)	(0.199)	
H. Excluding dairy milk	0.505***	0.465***	0.465***	0.549***	801
	(0.141)	(0.138)	(0.142)	(0.169)	
I. Specific product answers only	0.676***	0.662***	0.657***	0.588***	630
	(0.133)	(0.137)	(0.137)	(0.189)	
J. Specific answers plus fresh produce	0.643***	0.617***	0.612***	0.558***	686
	(0.125)	(0.130)	(0.130)	(0.196)	
K. Generic answers only, excluding fresh produce	−0.074	−0.087	−0.085	−0.037	530
	(0.166)	(0.166)	(0.159)	(0.175)	
L. Asymmetric specification (Table D.5)	0.331***	0.325***	0.329***	0.305**	1,137
	(0.111)	(0.113)	(0.112)	(0.144)	
M. Free intercept	0.210	−0.329	−0.318	−0.210	1,223
	(0.205)	(0.363)	(0.365)	(0.367)	
N. Log-linear outcome, positive quantities	OLS	2SLS	2SLS + order	ACS wt.	
	0.348***	0.302***	0.303***	0.365***	1,212
	(0.071)	(0.078)	(0.077)	(0.086)	

Notes: Gain-side counterpart of Table D.6: each row re-estimates Γ_G from the gain-side specification of Table 2, Poisson pseudo-maximum likelihood with $\log q_0$ as offset, on a different sample or with a different estimator; standard errors below each estimate. Columns, standard errors, weights and row definitions as in Table D.6. Rows C and D come from one specification with $\Gamma_G^{\text{first}} O_{iG} x_{iG} + \Gamma_G^{\text{second}} (1 - O_{iG}) x_{iG}$, $O_{iG} = 1$ when the gain scenario was presented first; the difference is not significant at the 5% level in any column ($p = 0.28$ in column (3)). Row L reports $\hat{\Gamma}_G$ from the stacked asymmetric specification. In row M the intercept, slope and control-function residual are nearly collinear on the gain side, so the point estimates are not separately identified. First-stage $F > 150$ in every instrumented column. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

D.5 Reference-Price Persistence: Estimation and Additional Results

Trip-level estimator. In the persistence scenario of Section 3.3, the respondent has paid R_{0i} on previous trips, the next-trip posted price is $P_{1i} = 1.2\,R_{0i}$, and the reported going-forward price is $\hat{R}_{1i}$ (the typically expected price in the E arm, the price that feels normal in the N arm). One step of the reference law (2.3), applied at the frequency of shopping trips rather than the model's quarters, delivers the respondent-level persistence estimate

$$\hat{\rho}_{\text{trip},i} = \frac{\log \hat{R}_{1i} - \log P_{1i}}{\log R_{0i} - \log P_{1i}}, \tag{D.11}$$

so that $\hat{\rho}_{\text{trip},i} = 0$ if the reference jumps fully to the new price and $\hat{\rho}_{\text{trip},i} = 1$ if it does not move. Because the reference law updates the reference toward the observed price, any $\rho \in [0, 1]$ maps the reported $\hat{R}_{1i}$ into the interval between the old reference R_{0i} and the new price P_{1i}, and reports outside that interval cannot be rationalized by the law. Starting from the baseline exclusions, I therefore drop respondents whose $\hat{R}_{1i}$ lies below R_{0i}—after a price increase, a going-forward price below the original reference updates in the wrong direction (102 respondents)—and respondents whose $\hat{R}_{1i}$ lies above P_{1i}, who extrapolate beyond the observed price change (28 E-arm and 22 N-arm respondents). The analysis below uses the resulting sample.

The arms differ sharply: the mean of $\hat{\rho}_{\text{trip},i}$ is 0.331 in the E arm ($N = 651$) and 0.743 in the N arm ($N = 629$), a difference significant at the 0.1% level in a Welch t-test. Table D.8 reports the means by shopping frequency. Figure 5 shows that both arms are approximately two-type mixtures—a mass of full updaters near zero and a mass of anchored respondents near one—and that the arms differ mainly in the mixing weights: wording the question as a forecast shifts respondents from the anchored type to the updating type.

Quarterly persistence. The model's period is a quarter, over which a consumer makes many shopping trips, so the trip-level estimates must be converted to the quarterly frequency. The four shopping-frequency bins (Figure C.8) imply $n_i \in \{26, 13, 6.5, 3\}$ trips per quarter. I compound each respondent's trip-level estimate over their own trips, $\hat{\rho}_{\text{qtr},i} = \hat{\rho}_{\text{trip},i}^{\,n_i}$, and then average across respondents; compounding before averaging preserves the dispersion in trip-level persistence, whereas averaging first and then compounding would understate quarterly persistence.[35] The less-than-once-per-month bin is excluded from the quarterly aggregation because it contains only five E-arm and one N-arm respondents. The result, $\bar{\hat{\rho}}^{E}_{\text{qtr}} = 0.217$ and $\bar{\hat{\rho}}^{N}_{\text{qtr}} = 0.531$, is Table 3 in the main text, where the estimates are stable across shopping-frequency bins.[36] The median-based calibration $\rho = 0.88$ used in Section 4 caps each respondent's trip-level estimate

[35] I cap trip-level estimates at 0.995 before compounding, which keeps quarterly persistence below one, as the reference law requires.

[36] The exception is the N arm's twice-per-week bin, whose quarterly mean of 0.608 sits above the 0.51–0.54 of the other bins. Its weight is about 10%: excluding it lowers $\bar{\hat{\rho}}^{N}_{\text{qtr}}$ from 0.531 to 0.522.

Table D.8: Trip-Level Reference-Price Persistence by Arm and Shopping Frequency

	E arm (expected price)			N arm (normal price)		
Shopping frequency	N	Mean	Median	N	Mean	Median
Twice per week	73	0.351	0.000	63	0.851	0.995
Once per week	281	0.349	0.000	289	0.755	0.995
Every two weeks	235	0.307	0.000	208	0.728	0.988
Once per month	57	0.296	0.000	68	0.634	0.890
Less than once per month	5	0.647	0.771	1	0.995	0.995
Overall (ACS-weighted)	622	0.320	0.000	608	0.756	0.995

Notes: Cell entries summarize $\hat{\rho}_{\text{trip},i}$ within shopping-frequency bins, restricted to $\hat{\rho}_{\text{trip},i} \in [0, 1]$. The less-than-once-per-month bin is reported for completeness but is excluded from the quarterly aggregation in the main text because of its sample size. The in-range sample contains $N = 1{,}280$ respondents. The ACS-weighted row reports the overall weighted mean and median with calibration weights matching the 2024 ACS population ages 18–65 on the joint age–sex distribution and the aggregate employment margin (Appendix D.6); its sample excludes respondents outside the calibration universe.

at 0.995 before compounding over the respondent's own trips and takes the N-arm median of the resulting quarterly persistence, 0.878 unweighted and ACS-weighted alike.

D.6 Population Weighting

This section describes the calibration weights behind the ACS-weighted column of each estimation table and the ACS-weighted rows of the persistence tables.

D.6.1 Target Population and Benchmarks

The weighting target is the population of U.S. residents ages 18–65. Benchmarks come from the 2024 American Community Survey one-year Public Use Microdata Sample person files, restricted to persons ages 18–65 living in housing units or noninstitutional group quarters and weighted by the ACS person weight; the target population comprises 207,880,832 people, computed from 1,981,226 person records. Two sets of population shares enter the calibration: the joint distribution of five age groups (18–24, 25–34, 35–44, 45–54, 55–65) by binary sex, ten cells, and the marginal distribution of employment status (employed, unemployed, not in the labor force). These are the dimensions on which the survey and the ACS can be compared and on which the sample differs from the population (Table 1). Ethnicity is excluded because Prolific's five-category simplified ethnicity field has no Hispanic-origin item and cannot be crosswalked to the ACS race and Hispanic-origin questions. Prolific's full-time and part-time categories map to employed, "unemployed (and job seeking)" to unemployed, and "not in paid work" to not in the labor force; respondents about to start a job, reporting "other," or with missing employment form an explicit unclassified category whose weight share is fixed at its sample value, with the three ACS employment shares scaled proportionally.

D.6.2 Weight Construction

Weights are defined for respondents ages 18–65 who report Female or Male sex, the universe on which the ACS targets are defined, so each weighted specification drops the small number of respondents outside it: 51 of 1,238 in the loss-scenario sample, 49 of 1,223 in the gain-scenario sample, 48 of 1,137 in the balanced sample, and analogous counts in the remaining samples. Weights are rebuilt within every estimation sample, because the correct weight depends on the observations actually present in the regression.

The weights solve the calibration problem of Deville and Särndal (1992): with x_i collecting an intercept, nine age-by-sex indicators and three employment indicators, and t_x the corresponding ACS totals, the weights w_i satisfy $\sum_i w_i x_i = t_x$ exactly under the bounded logit link of the generalized raking family (Deville et al., 1993), with bounds $[1/8, 8]$ and Newton–Raphson with step halving. Tighter bounds are infeasible because matching the ACS unemployment share requires down-weighting unemployed respondents, who are heavily over-represented, by a factor of about four and a half; realized weights stay well inside the bounds, with a maximum of at most 4.3 and a minimum of about 0.17 across the main estimation samples. Weights are normalized to mean one. Table D.9 reports the weight distributions, the achieved balance, and the Kish effective sample sizes (Kish, 1992): weighting by age-by-sex cells and employment margins retains roughly two-thirds of the nominal sample in effective terms.

Table D.9: Weight Diagnostics and Achieved Demographic Balance

Sample	Scheme	N	Max. weight	Kish N_{eff}	N_{eff} (%)	Mean age	Female (%)	Unemployed (%)	Not in LF (%)
Reference sample	Unweighted	1,375	1.00	1375.0	100.0	37.83	52.58	14.04	7.93
Reference sample	Age × sex	1,375	2.20	1142.2	83.1	40.66	50.28	14.47	8.34
Reference sample	Age × sex + employment	1,375	4.36	927.1	67.4	40.70	50.28	3.13	18.44
Balanced Γ sample	Unweighted	1,089	1.00	1089.0	100.0	38.46	52.98	12.67	8.91
Balanced Γ sample	Age × sex	1,089	2.00	924.9	84.9	40.65	50.28	13.47	9.09
Balanced Γ sample	Age × sex + employment	1,089	3.62	773.8	71.1	40.70	50.28	3.11	18.32
ρ estimation sample	Unweighted	1,230	1.00	1230.0	100.0	38.01	52.36	13.01	8.13
ρ estimation sample	Age × sex	1,230	2.12	1036.0	84.2	40.63	50.28	13.74	8.36
ρ estimation sample	Age × sex + employment	1,230	4.27	845.6	68.7	40.67	50.28	3.13	18.46
ACS target	Target	–	–	–	–	40.98	50.28	3.57	21.00

Notes: Every diagnostic is tied to the sample named in the first column. The reference sample contains 1,375 U.S. residents aged 18–65 with binary sex. The Γ and ρ rows rebuild weights after their estimation-sample restrictions. Age × sex matches the ACS joint distribution; the employment specification adds the employment margin under bounded calibration on $[1/8, 8]$. The raw ACS employment targets appear in the final row. Calibration preserves each sample's measured Unclassified share and applies the ACS proportions to the remaining share, so the full-sample unemployed and not-in-labor-force percentages in the calibrated rows are proportionally below the raw ACS entries.

D.6.3 Inference and Interpretation

Weighted estimators enter their objective functions with one weight per respondent, applied to each of the respondent's scenario rows. In the control-function specifications both stages are weighted, and standard errors come from the respondent-level bootstrap with 1,000 replications in which the calibration weights are re-estimated within every replication, so the sampling

variation of the weighting step is reflected in the standard errors; the naive Poisson, OLS and two-stage least-squares columns retain respondent-clustered analytic standard errors. The ACS-weighted persistence estimates are Hájek means with a linearized variance that treats the calibration weights as fixed, which is conservative for calibrated means.

Two interpretive notes. The ACS classifies a respondent as unemployed only if they searched for work within a four-week window and were available to work, whereas Prolific records self-described status, so the employment calibration assumes the measured survey–ACS employment gap reflects sample composition rather than this definitional difference. And calibration adjusts only for the observed dimensions: it cannot correct selection on characteristics unmeasured in the survey, such as education or income, or on unobservables. The stability of the estimates across the unweighted and weighted columns is evidence that the findings are not driven by the sample's age, sex, or employment composition, not a guarantee of full population representativeness.

E Reference-Dependent Demand in Retail Scanner Data

This appendix estimates the RD–CES demand system of Section 2 on the store-level NielsenIQ Retail Measurement Services (RMS) scanner data. Where the survey of Section 3 identifies Γ_L and Γ_G from responses to hypothetical scenarios, the scanner data identify them from transactions: 22 grocery categories, 62 NielsenIQ product modules, and 1.16 billion UPC×store×week observations. The reference price is an exponentially weighted moving average of a product's own past posted prices in that store, and identification comes from comparing a UPC's price with the prices of other UPCs in the same module, store and week, instrumented by the same product's price in other markets.

Reference dependence is present and economically large in every category. The expenditure-weighted estimate is $\Gamma = 0.943$ (0.014); splitting the reference gap at zero gives $\Gamma_G = 1.117$ (0.023) on gains and $\Gamma_L = 0.761$ (0.023) on losses. The loss-side estimate is close to the survey's $\hat{\Gamma}_L = 0.875$, and it is the loss side that governs the welfare cost of an inflationary episode. The gain side is where the two designs produce very different results: 1.117 against the survey's $\hat{\Gamma}_G = 0.326$, which reverses the sign of the kink. I attribute that gap to stockpiling. The scanner records purchases, which include stockpiling during sales, while the survey elicits consumption and asks respondents to ignore purchases made today for future consumption.

E.1 Data and sample

The NielsenIQ RMS data, distributed by the Kilts Center for Marketing at the University of Chicago Booth School of Business, are point-of-sale records from the checkout systems of participating grocery, drug, and mass-merchandise chains across every U.S. media market. For every UPC sold in a participating store, RMS reports weekly unit sales and a weekly price, together with feature-advertising and in-store display indicators. NielsenIQ assigns every UPC to one of roughly 1,000 product modules—ready-to-eat cereal, canned tuna, bath tissue—so a module collects a product's close substitutes. The sample runs from 2006 through 2020, the years over which the module hierarchy is stable; NielsenIQ redrew module boundaries in the releases that follow, and the identification below is entirely within-module. The estimation sample is the national-brand grocery subset described in the rest of this subsection.

Quantity, price, and the reference price. Quantity q_{ist} is unit sales of UPC i in store s during week t; NielsenIQ records no row for a store-week with zero sales, so the log outcome $y_{ist} = \log q_{ist}$ is always defined. The regression price is the unit price, NielsenIQ's volume-weighted weekly price divided by its multi-buy multiplier, net of loyalty-card discounts and gross of manufacturer coupons, so P_{ist} is the shelf price faced by a card-holding shopper, with $p_{ist} = \log P_{ist}$. The reference price for each (UPC, store) pair is an exponentially weighted

moving average of that product's own past posted prices in that store,

$$r_{ist} \;=\; \rho_w \, r_{is,t-1} \;+\; (1-\rho_w)\, p_{is,t-1}, \tag{E.1}$$

the reference law (2.3) with zero trend inflation, applied one step per observation of the pair. I write the updating parameter as ρ_w to distinguish it from the quarterly ρ of the main text. The recursion uses lagged prices only, so r_{ist} is predetermined with respect to the week-t price, and I drop each pair's first 52 observations, so every observation enters with a reference built from at least a year of the pair's own price history. I build the panel at two updating speeds, $\rho_w = 0.6$ and $\rho_w = 0.9$, half-lives of about 1.4 and 6.6 observations. All results below use $\rho_w = 0.9$, which fits marginally better; Appendix E.5 gives the evidence and the reasons for treating the margin as weak, and Table E.2 reports the aggregate elasticities under both speeds.

Controls. Two sets of observable controls discipline the two sides of the kink. Feature and display indicate whether a product was advertised in the store's weekly flyer or given special in-store placement; a discount typically arrives with advertising, which raises demand on its own, and without the controls that effect would be attributed to the gain-side reference gap. Three post-sale dummies mark observations one, two, and three-to-four weeks after a temporary price reduction ends, where a reduction is any price at least 5% below the pair's 13-week rolling median. A sale drags the reference price down, so post-sale weeks sit on the loss side of the kink just as households run down the inventory they bought during the discount; without the dummies that demand trough would be attributed to the reference gap and Γ_L overestimated. Before 2013 NielsenIQ audited feature and display only in a subset of store-weeks, so the sample is all of 2013–2020 plus the audited 2006–2012 store-weeks; audit selection is unrelated to product-level demand shocks, so retaining the early years adds price history without selecting on the outcome.

Categories and filters. The survey of Section 3 drives category selection: I estimate the scanner analogue for the categories accounting for the bulk of respondents' chosen products, so that the survey and scanner measures of Γ are defined over comparable goods. The 22 categories cover 1,180 of 1,432 survey responses, or 82.4%: bacon, bread, butter and margarine, canned tuna, cereal, cheese, coffee and creamer, crackers, eggs, frozen pizza, ice cream, juice, milk, paper products, pasta, peanut butter, salty snacks, snack bars, soda and water, soup and ramen, liquid tea, and yogurt, spanning 62 NielsenIQ modules.[37] Soda and water alone contributes 20.4% of the rows and 26.3% of the sample's expenditure, so it carries substantial weight in every aggregate below. The one selected category never estimated is packaged salad, excluded on data grounds.

[37] Seven categories (bacon, canned tuna, eggs, frozen pizza, ice cream, peanut butter, and liquid tea) consist of a single module, so for them the module×store×week fixed effects below collapse to store×week.

Restricting the sample to national brands is what rules out the fresh categories. Fresh produce, butcher-counter meat and in-store bakery are random-weight goods sold under PLU or store-assigned codes rather than manufacturer UPCs, so there is no fixed product whose price can be compared across markets, which the instrument constructed below requires; nor is there a meaningful reference price for "a pound of apples," whose unit price moves with grade, size, varietal and season rather than with a posted promotional decision. Private-label items are excluded for the related reason that their UPCs are retailer-masked and not comparable across chains. Every good estimated here is packaged, branded, UPC-coded, and carries a well-defined posted price. After the burn-in I drop free-with-purchase items and data errors, pairs averaging under one unit per week, pairs observed fewer than 104 weeks or with a gap of more than 13 weeks between observations, private label, UPCs sold in fewer than two markets, and observations lacking a reference price or instrument. The same filter applies to every specification, so specifications differ only in their regressors and fixed effects, never in their rows.

E.2 The estimating equation and identification

Consider a consumer allocating expenditure across the UPCs i within a product module. Under CES preferences with elasticity of substitution ε, demand for i in store s at week t is $q_{ist} \propto P_{ist}^{-\varepsilon}$, conditional on the module price index and total expenditure. Reference dependence enters as in Section 2: the consumer evaluates the posted price against the reference price R_{ist} of equation (E.1), and gain-loss utility shifts the marginal value of purchase by a term proportional to $\log R_{ist} - \log P_{ist}$. Taking logs of the demand schedule (B.1) under the symmetric restriction $\Gamma_L = \Gamma_G \equiv \Gamma$ gives an equation linear in the two prices,

$$y_{ist} \;=\; -(\varepsilon + \Gamma)\, p_{ist} \;+\; \Gamma\, r_{ist} \;+\; \gamma' X_{ist} \;+\; \alpha_{is} \;+\; \delta_{m(i)st} \;+\; u_{ist}, \tag{E.2}$$

where $y = \log q$, $p = \log P$ and $r = \log R$. The term α_{is} is a UPC×store effect and $\delta_{m(i)st}$ a module×store×week effect. The theory signs both price coefficients: demand slopes down in the posted price, and holding the posted price fixed, a higher reference price makes the same price look like a better deal and raises quantity, so the coefficient on r, which is Γ itself, is positive.

The symmetric specification constrains demand to respond identically whether the posted price sits above or below the reference, but the kinked welfare kernel of Proposition B.4 takes $\gamma_s = \Gamma_s/(\varepsilon - 1)$ separately on each side: an inflationary episode puts posted prices above their references, so Γ_L determines the reference wedge and with it the welfare cost of inflation, while Γ_G governs shocks that reduce inflation in the short run. Splitting the reference gap $g_{ist} \equiv r_{ist} - p_{ist}$ at zero, with $g^+ = \max(g, 0)$ on the gain side and $g^- = \min(g, 0)$ on the loss

side, the asymmetric specification is

$$y_{ist} \;=\; -\varepsilon\, p_{ist} \;+\; \Gamma_G\, g^{+}_{ist} \;+\; \Gamma_L\, g^{-}_{ist} \;+\; \gamma' X_{ist} \;+\; \alpha_{is} \;+\; \delta_{m(i)st} \;+\; u_{ist}, \tag{E.3}$$

so one estimation recovers both elasticities. The two equations separate two elasticities of substitution that standard CES demand holds equal. A price change that leaves the reference where it is, as a one-week promotion does, moves demand by the short-run elasticity $\varepsilon + \Gamma$; a price change that persists until the reference catches up moves it by the long-run elasticity ε, the same ε that appears in Section 2. In equation (E.2) the coefficient on p is the short-run elasticity; in equation (E.3), where the reference enters only through the gaps, it is the long-run elasticity, and the short-run elasticity is $\varepsilon + \Gamma_G$ below the reference and $\varepsilon + \Gamma_L$ above it. I summarize the distance between the two sides by $\Delta \equiv \Gamma_L - \Gamma_G$, and call a category *gain-dominant* when $\Delta < 0$.

Fixed effects, controls, and the instrument. Every specification carries two sets of fixed effects. UPC×store intercepts α_{is} absorb each product's permanent demand level in each store. Module×store×week effects $\delta_{m(i)st}$ absorb everything common to a module in a store in a week: traffic, local demand shocks, weather, holidays, and any module-wide pricing or promotional event. What remains is the price of one UPC relative to the other UPCs of its own module, in the same store and week. The controlled specifications add the feature and display indicators and the three post-sale dummies as X_{ist}; the no-controls specifications set $X = \emptyset$ but keep both sets of fixed effects.[38] I cluster standard errors two-way, by store and by week.

Retailers set prices while observing demand, so unobserved demand shocks that prompt a price response correlate with p_{ist} and, through the recursion, with r_{ist}. The bias is large: OLS puts Γ below its 2SLS counterpart in every category, with a median ratio of about 0.4. I instrument with the leave-own-market-out price: for UPC i in week t, z^{P}_{it} is the average log price of the same UPC in the same week across stores in all other Nielsen DMAs, a Hausman-style instrument (Hausman, 1996) whose identifying assumption is that national cost and wholesale-pricing shocks move a product's price in many markets at once while the demand shocks that threaten identification are local.[39] Each instrument is transformed exactly as the regressor it instruments: the same recursion at the same speed gives an exogenous reference z^{R}_{it} and gap $z^{g}_{it} = z^{R}_{it} - z^{P}_{it}$, split at zero into z^{g+}_{it} and z^{g-}_{it} for the asymmetric model. Both models are just-identified. Instrument strength, the minimum Sanderson–Windmeijer conditional F across the endogenous regressors, is uniformly large: at least 591 in every symmetric specification and 258 in every asymmetric one.

[38] Contemporaneous sale flags are deliberately not controls: a sale is a price action rather than a demand shifter, and a sale week is almost by definition a week on the gain side of the reference gap, so conditioning on it would absorb the very variation that identifies Γ.

[39] Nielsen DMAs (designated market areas) partition the United States into 210 media markets; metropolitan areas like Chicago or Atlanta are single markets.

E.3 Results

Table E.1 reports the 2SLS estimates at $\rho_w = 0.9$ with controls and both sets of fixed effects: the symmetric Γ from equation (E.2) and the gain- and loss-side elasticities from equation (E.3). The no-controls counterparts, and the OLS estimates that serve the reference-speed comparison in Appendix E.5, are in the replication package and summarized below.

Table E.1: **Reference dependence by category**, 2SLS, $\rho_w = 0.9$, with controls, UPC×store and module×store×week fixed effects. Γ is the symmetric estimate from equation (E.2); Γ_G and Γ_L from equation (E.3) are the responses to a price below the reference (a gain) and above it (a loss); $\Delta = \Gamma_L - \Gamma_G$, with p from the Wald test of $\Gamma_G = \Gamma_L$. Class: Gain = gain-dominant ($\Delta < 0$, $p < 0.05$), Loss = loss-dominant ($\Delta > 0$, $p < 0.05$), Sym. = not distinguishable. Standard errors two-way clustered by store and week. The minimum Sanderson–Windmeijer conditional F is 885 across the symmetric and 315 across the asymmetric regressions. † single-module category.

Category	Γ (s.e.)	Γ_G (s.e.)	Γ_L (s.e.)	Δ (s.e.)	p	Class
Bacon†	1.823 (0.052)	1.872 (0.084)	1.764 (0.101)	-0.108 (0.153)	0.481	Sym.
Bread	0.537 (0.040)	0.940 (0.072)	0.091 (0.043)	-0.848 (0.089)	0.000	Gain
Butter & margarine	1.438 (0.094)	1.954 (0.142)	0.351 (0.147)	-1.602 (0.236)	0.000	Gain
Canned tuna†	0.713 (0.076)	0.423 (0.226)	1.055 (0.227)	0.632 (0.427)	0.139	Sym.
Cereal	1.248 (0.037)	1.464 (0.095)	0.990 (0.091)	-0.474 (0.173)	0.006	Gain
Cheese	1.189 (0.077)	1.184 (0.125)	1.196 (0.099)	0.012 (0.169)	0.943	Sym.
Coffee & creamer	0.977 (0.033)	0.938 (0.073)	1.020 (0.062)	0.083 (0.118)	0.484	Sym.
Crackers	1.027 (0.048)	0.946 (0.095)	1.111 (0.088)	0.166 (0.156)	0.289	Sym.
Eggs†	0.622 (0.079)	0.953 (0.122)	0.168 (0.110)	-0.785 (0.177)	0.000	Gain
Frozen pizza†	1.662 (0.056)	1.582 (0.103)	1.813 (0.136)	0.231 (0.212)	0.277	Sym.
Ice cream†	2.016 (0.079)	2.947 (0.140)	0.924 (0.125)	-2.023 (0.222)	0.000	Gain
Juice	0.628 (0.039)	0.269 (0.090)	1.052 (0.081)	0.783 (0.154)	0.000	Loss
Milk	0.487 (0.059)	0.363 (0.112)	0.633 (0.119)	0.270 (0.198)	0.174	Sym.
Paper products	1.075 (0.056)	0.634 (0.108)	1.665 (0.100)	1.031 (0.178)	0.000	Loss
Pasta	1.143 (0.041)	1.177 (0.074)	1.105 (0.064)	-0.072 (0.113)	0.522	Sym.
Peanut butter†	1.257 (0.089)	1.426 (0.194)	1.033 (0.176)	-0.393 (0.332)	0.237	Sym.
Salty snacks	0.528 (0.044)	0.643 (0.083)	0.407 (0.071)	-0.236 (0.126)	0.062	Sym.
Snack bars	0.763 (0.042)	0.261 (0.087)	1.385 (0.081)	1.124 (0.148)	0.000	Loss
Soda & water	0.813 (0.042)	1.255 (0.056)	0.404 (0.065)	-0.851 (0.108)	0.000	Gain
Soup & ramen	0.868 (0.046)	1.647 (0.098)	-0.213 (0.079)	-1.859 (0.161)	0.000	Gain
Tea (liquid)†	0.649 (0.035)	0.828 (0.083)	0.478 (0.066)	-0.351 (0.132)	0.008	Gain
Yogurt	1.185 (0.061)	1.049 (0.111)	1.378 (0.115)	0.329 (0.191)	0.086	Sym.

Reference dependence is present everywhere. Γ is positive and precisely estimated in all 22 categories, ranging with controls from 0.49 in milk to 2.02 in ice cream, and the CES null $\Gamma = 0$ is rejected overwhelmingly everywhere. Adding the controls lowers Γ in 22 of 22 categories, with the expenditure-weighted mean falling from 1.109 (0.014) to 0.943 (0.014). That the two are close is itself informative: with module×store×week effects already in place, the observable promotion variables absorb only a small additional slice of price variation, so the estimates do not hinge on how well feature and display are measured.

The asymmetry is heterogeneous, and mostly gain-dominant or symmetric. With controls, 8 categories classify as gain-dominant, 11 as symmetric and 3 as loss-dominant; without controls the counts are 10, 9 and 3. Only three categories respond more to a price above the reference than to one below it: paper products, snack bars and juice. The strongest asymmetries in the other direction—soup & ramen, ice cream, butter & margarine, bread—are all categories in which a discount can be met by stockpiling.

Adding the controls compresses the gain side more than the loss side. Γ_G falls in 21 of 22 categories and Γ_L in 17; the expenditure-weighted gain response drops by roughly 0.23 and the loss response by about 0.08. Part of the measured gain response at coarser conditioning is therefore promotional co-movement, feature and display accompanying the discount, whereas the loss-side response is closer to a household behavioral response and survives the controls. The same mechanism shows up in the tests: the Wald test of $\Gamma_G = \Gamma_L$ rejects at the 5% level in 13 of 22 categories without controls but only 11 of 22 with them. Every rejection also survives at the 1% level, and only salty snacks ($p = 0.062$) and yogurt ($p = 0.086$) sit near the boundary.

Weighted aggregates. Table E.2 averages the category estimates with expenditure weights, $w_c = R_c / \sum_k R_k$ with R_c the category's revenue over the estimation sample, so that categories count in proportion to their economic size. The choice of weight turns out not to matter: weighting by row count or by estimation precision instead moves every entry by at most 0.08, even though moving from row counts to expenditure reweights eggs up by a factor of 4 and pasta down by a factor of 2.4, so reference dependence is not systematically stronger in either large or small categories. The revenue weights use total revenue over the whole sample period, since expenditure itself responds to the promotions being studied; they are shares within the estimation sample, not of the overall consumer basket; and the standard errors on the averages ignore cross-category covariance and may understate uncertainty.

Table E.2: Expenditure-weighted reference dependence, 22 categories, 2SLS, at both reference speeds. Γ is the symmetric estimate; Γ_G and Γ_L the gain- and loss-side estimates. Soup & ramen enters at its estimated Γ_L, a precise small negative value at $\rho_w = 0.9$.

Specification	Parameter	$\rho_w = 0.9$	$\rho_w = 0.6$
No controls	Γ	1.109 (0.014)	0.820 (0.012)
	Γ_G	1.350 (0.024)	1.126 (0.024)
	Γ_L	0.840 (0.027)	0.512 (0.026)
With controls	Γ	0.943 (0.014)	0.767 (0.015)
	Γ_G	1.117 (0.023)	0.920 (0.023)
	Γ_L	0.761 (0.023)	0.599 (0.028)

E.4 Comparison with the survey

The survey of Section 3 gives $\hat{\Gamma}_L = 0.875$ and $\hat{\Gamma}_G = 0.326$ (Table 2). The welfare calculation of Section 4 rests on the loss side: an inflationary episode puts posted prices above their references, so Γ_L governs the reference wedge. The scanner puts the loss-side response at 0.761 (0.023), against the survey's 0.875: the two estimates are reasonably close on the parameter that matters most. Exact agreement should not be expected, because the two designs define the reference price differently. The scanner imposes one definition: the moving average of the product's own price history in a single store, while the survey elicits the household's own reference directly. If consumers form reference prices for their products based on their shopping experiences in multiple stores and from similar competing products, then the store-level reference price series and the consumer's reference price are different objects, and the two designs estimate responses to different reference prices. Instrumenting does not close this gap: the instruments shift the store-level series itself.

On the gain side, the two estimates differ by a wide margin: 1.117 (0.023) in the scanner against 0.326 in the survey. Because the scanner gain response also exceeds its own loss response, the kink reverses sign; the scanner categories are gain-dominant on average, while the survey responses imply loss aversion.

The divergence can be explained by storability. The survey asks how much of a good the respondent would *consume* over the coming month, holding the shopping occasion fixed; the scanner records what shoppers *buy* in a week. A discount can be exploited by buying ahead, so a purchase response includes inventory accumulation that a consumption response does not. Buying ahead is worthwhile only when the price is temporarily low; when the price rises there is nothing to be gained from adjusting the timing of purchases. Stockpiling therefore predicts divergence on the gain side and agreement on the loss side. That is the pattern in the data. The categories with the strongest gain-dominance are exactly the storable ones—soup and ramen, ice cream, butter and margarine, bread—and the loss-side estimates line up across the two designs. Sample composition pushes the same way: the scanner categories cover about 80% of the consumption categories respondents chose in the survey, and the excluded remainder—fresh produce most prominently—consists of perishables that cannot be stockpiled, so the scanner Γ_G is estimated on a more storable basket than the survey's.

The calibration in Section 4 uses estimates from the survey of Section 3, and the scanner estimates provide independent evidence on the magnitude of Γ_L, which, along with ρ, is the main parameter governing the welfare cost of an inflationary episode. I prefer the survey estimates because in the model of Section 2, reference dependence operates on intratemporal consumption choices, and the model has no mechanism for building inventories of individual varieties. The survey elicits exactly this object: its Γ_L is identified from a question about *consumption* over a fixed horizon, and it is therefore the closer empirical counterpart of the demand response the model requires. The scanner estimates, which embed the purchase-timing

margin the model abstracts from, serve as corroborating evidence rather than as the calibration input.

E.5 Evidence on the reference-updating speed

The choice of $\rho_w = 0.9$ rests on fit. For each category I compare the within-R^2 of the four OLS specifications—symmetric and asymmetric, each without and with controls—at $\rho_w = 0.6$ and $\rho_w = 0.9$. Because the two references are computed on identical rows with identical fixed effects and the same number of regressors, a higher within-R^2 means a lower sum of squared residuals on the same data; I use the OLS specifications because 2SLS does not minimize that sum and so cannot adjudicate fit. The slow reference wins 65 of the 88 comparisons, and wins all four specifications in 10 of 22 categories; eggs prefers the fast reference in all four, and bread and yogurt in three of four. The margins are tiny: the median difference in within-R^2 is 0.00055 on levels running from 0.05 to 0.41, a median relative improvement of 0.17%, with the largest single gain in either direction under 1.5%. At samples of tens to hundreds of millions of rows these differences lie far outside sampling error, but they are economically negligible. I take $\rho_w = 0.9$ as the baseline because it fits better in most categories, not because the data sharply distinguish the two speeds, and Table E.2 reports the aggregates at both.

The scanner reference updates at the level of a pair's successive observations, which need not fall in consecutive weeks, so there is no straightforward mapping from ρ_w into the quarterly ρ of Section 3, and I do not attempt one. The survey elicits reference updating over the respondent's own shopping trips and records how often they shop, so its trip-level parameter converts to a quarterly ρ respondent by respondent; the scanner's irregular observation spacing permits no such conversion. This is a further reason the calibration uses the survey's estimate of ρ. The fit evidence here supports a statement at the observation level only: the data favor the slower reference.

F Quantitative Model, Numerical Solution, and Sensitivity

This appendix states the equilibrium conditions used in Section 4, documents the nonlinear solution and its verification, and reports sensitivity to the monetary-policy rule, demand elasticity, and size of the inflation episode.

F.1 Equilibrium conditions and calibration

Physical output is Y_t, composite output is Y_t^U, and composite consumption is C_t^U. Let $d_t = \log(P_t/R_t)$, $\Pi_t = P_t/P_{t-1}$, $\pi_t = \log \Pi_t$, and $\Lambda_t = (C_t^U)^{-\sigma}$. The reference wedge and reference-price law are

$$\chi_t = \begin{cases} \Gamma_L d_t/(\varepsilon-1), & d_t > 0, \\ \Gamma_G d_t/(\varepsilon-1), & d_t < 0, \\ 0, & d_t = 0, \end{cases} \qquad R_{t+1} = R_t^{\rho} P_t^{1-\rho}, \qquad P_t^{\mathrm{RD}} = P_t e^{\chi_t}. \tag{F.1}$$

The cost-push factor satisfies $\log \upsilon_t = \rho_u \log \upsilon_{t-1} + \varepsilon_t^u$, with a single initial innovation. Productivity remains at $A_t = 1$ throughout these experiments. Under perfect foresight, household optimality and market clearing imply

$$1 = \beta I_t \left(\frac{C_t^U}{C_{t+1}^U} \right)^{\sigma} \frac{e^{\chi_t - \chi_{t+1}}}{\Pi_{t+1}}, \qquad \frac{W_t}{P_t} = e^{\chi_t} (C_t^U)^{\sigma} N_t^{\varphi}, \tag{F.2}$$

$$Y_t = A_t N_t^{1-\alpha} = e^{\chi_t} Y_t^U, \qquad C_t^U = Y_t^U \left[1 - \frac{\psi_p}{2} (\Pi_t - 1)^2 \right], \tag{F.3}$$

$$mc_t = \frac{(1-\tau) W_t}{P_t (1-\alpha) A_t N_t^{-\alpha}}, \qquad z_t = \upsilon_t mc_t. \tag{F.4}$$

Here mc_t is subsidized real marginal cost in posted-price units, and z_t additionally includes the cost-push factor. The gross nominal interest rate I_t follows

$$I_t = I_{t-1}^{\rho_i} \left[\beta^{-1} \Pi_t^{\phi_\pi} \left(\frac{Y_t}{\bar{Y}} \right)^{\phi_y} \right]^{1-\rho_i}. \tag{F.5}$$

Thus policy responds to physical output relative to steady state. There is no monetary-policy innovation in the cost-push experiments. In logs, with $i_t \equiv \log(I_t/\bar{I})$ and $\bar{I} = 1/\beta$, the rule is $i_t = \rho_i i_{t-1} + (1-\rho_i)[\phi_\pi \pi_t + \phi_y \log(Y_t/\bar{Y})]$, which reduces to (2.16) without the policy shock when $\rho_i = 0$.

The factor υ_t is implemented as a tax on the subsidy-adjusted wage bill. Firm profits and

the government's net lump-sum transfer are

$$D_t^F(i) = P_t(i)Y_t(i) - \upsilon_t(1-\tau)W_tN_t(i) - \frac{\psi_p}{2}\left(\frac{P_t(i)}{P_{t-1}(i)} - 1\right)^2 P_t^{\mathrm{RD}}Y_t^U, \tag{F.6}$$

$$T_t = \left[(\upsilon_t - 1)(1-\tau) - \tau\right]W_tN_t. \tag{F.7}$$

Fiscal flows cancel when household income and firm profits are aggregated. The cost-push factor therefore changes pricing incentives without adding a resource cost to equation (F.3) or a separate term to household welfare.

Pricing and the costate. Let Γ_{b_t} equal Γ_L on the loss branch and Γ_G on the gain branch. Let η_t^+ and η_t^- denote the reference-price elasticities on the branches reached by a current price increase or cut, respectively. Both equal Γ_{b_t} off the kink; at $P_t = R_t$ they equal Γ_L and Γ_G, respectively. Define

$$\mathcal{M}_{t+1} = \frac{\Lambda_{t+1}Y_{t+1}^U}{\Lambda_tY_t^U}, \qquad \mathcal{A}_t = -\psi_p(\Pi_t - 1)\Pi_t + \beta\psi_p\mathcal{M}_{t+1}(\Pi_{t+1} - 1)\Pi_{t+1}. \tag{F.8}$$

Holding inherited states and the aggregate path fixed, the right and left derivatives of the firm's Bellman return with respect to its current log posted price, normalized by $\Lambda_tY_t^U$, are

$$\mathcal{D}_t^{\pm} = 1 - (\varepsilon + \eta_t^{\pm})(1 - z_t) + \mathcal{A}_t + \beta(1-\rho)\mathcal{M}_{t+1}\Theta_{t+1}. \tag{F.9}$$

Here Θ_t is the partial derivative of optimized firm value with respect to its log reference price, holding the inherited posted-price state fixed and normalizing by $\Lambda_tY_t^U$. The continuation derivative is common when the date-$t+1$ value function is differentiable in its reference state. Otherwise, the corresponding one-sided derivative $\Theta_{t+1}^{R,\pm}$ replaces Θ_{t+1}, as in Appendix B.2. Continuation choices are optimized after each deviation. The first-order pricing conditions are

$$\begin{aligned} P_t > R_t: &\quad \mathcal{D}_t^+ = 0, \\ P_t < R_t: &\quad \mathcal{D}_t^- = 0, \\ P_t = R_t: &\quad \mathcal{D}_t^+ \le 0 \le \mathcal{D}_t^-. \end{aligned} \tag{F.10}$$

Where firm value is differentiable, the costate recursion on a smooth branch and under the locally optimal tracking policy is

$$\begin{aligned} P_t \neq R_t: &\quad \Theta_t = \Gamma_{b_t}(1 - z_t) + \beta\rho\mathcal{M}_{t+1}\Theta_{t+1}, \\ P_t = R_t: &\quad \Theta_t = 1 - \varepsilon(1 - z_t) - \psi_p(\Pi_t - 1)\Pi_t \\ &\qquad + \beta\mathcal{M}_{t+1}\left[\Theta_{t+1} + \psi_p(\Pi_{t+1} - 1)\Pi_{t+1}\right]. \end{aligned} \tag{F.11}$$

At the kink, the second recursion applies in the interior of the tracking region, where the optimal posted price follows the reference, $P_t = R_t$. Raising both prices raises next period's reference one for one, so the normalized continuation multiplier is $\beta\mathcal{M}_{t+1}$ rather than $\beta\rho\mathcal{M}_{t+1}$. The adjustment-cost terms account for the change in the posted price. The cost-push factor enters both pricing and costate conditions through z_t.

Steady state. In RD-RANK, I set the employment subsidy to the midpoint of the interval in Proposition B.1, $\tau = (1/\varepsilon^{\uparrow} + 1/\varepsilon^{\downarrow})/2$, and $\bar{mc} = 1 - \tau$, so the steady-state one-sided pricing inequalities are strict. Firm optimality is checked numerically as described below. The terminal costate is $\bar{\Theta} = [1 - \varepsilon(1 - \bar{mc})]/(1 - \beta)$. RANK sets both reference elasticities to zero and uses $\tau = 1/\varepsilon$. The Rotemberg coefficient is calibrated in RANK to a three-quarter Calvo-equivalent price duration and held fixed across the two models.

F.2 Solution and numerical verification

I solve the nonlinear perfect-foresight system over 200 quarters, starting from the deterministic steady state and imposing its real allocation, zero inflation, $P_t = R_t$, and the steady-state costate at the terminal date. For a conjectured regime sequence, I solve the equality system using $\mathcal{D}_t^+ = 0$, $\mathcal{D}_t^- = 0$, or $d_t = 0$ at each date, together with the corresponding costate recursion. The active-set algorithm updates inconsistent regime assignments until all conditions in (F.10) hold. Dates with a numerically zero gap are assigned to the kink when its inequalities hold; continuation in shock size and warm starts reduce sensitivity to initialization. The solution is a locally valid equilibrium path, without a claim of global uniqueness.

The stopping tolerance for pricing and regime violations is 10^{-8}. Separate checks use a 2×10^{-6} tolerance for equilibrium identities and terminal convergence, and verify feasibility, peak matching, and the welfare identities. For the baseline, policy-grid, and demand-elasticity experiments, the largest equilibrium residual is below 3.4×10^{-11} and the largest peak-matching error is below 1.9×10^{-6} percentage point. The $\rho_u = 0.70$ and larger-episode runs also pass these checks.

Firm optimality. Satisfying the pricing conditions alone does not establish global firm optimality. To assess Assumption B.2, I re-solve a single firm's dynamic problem conditional on the aggregate baseline cost-push paths at all reference-persistence values. The state grid covers the inherited reference and lagged price; the price-choice grid covers ± 20 log points around the aggregate price in steps of 0.1 log point. Value-function iteration supplies the stationary terminal value, and backward induction re-optimizes continuation choices after each deviation. The objective is single-peaked on each side of the kink at all tested states and dates, and any price at least one log point from the optimum lowers firm value by at least 0.43% of a quarter's steady-state revenue. The equilibrium price is optimal to within a small tolerance: at every

date, the firm's best response, refined between grid points, lies within 0.07 log point of the equilibrium price, and moving to it raises firm value by at most 2.2×10^{-5} of a quarter's revenue. Expanding the price corridor to ± 50 log points, on a 0.25-log-point grid, gives the same result. These checks provide numerical support for firm optimality on the tested paths and grid.

Welfare computation. Exact discounted utility losses ΔW are evaluated from C_t^U and N_t along the nonlinear path, with the steady-state continuation contributing zero additional loss. Appendix B.4 derives their conversion to permanent consumption-equivalent losses λ, holding labor at its steady-state level. For the cost-push experiment, the order-consistent approximation used in Table 5 is

$$\lambda^{(2)} = (1-\beta)\mathcal{L}^{(2)} - \frac{\sigma}{2}(1-\beta)^2 \left(\sum_{t=0}^{T-1} \beta^t \chi_t \right)^2 . \tag{F.12}$$

Component shares use the four terms in $\mathcal{L}^{(2)}$ before this conversion correction.

F.3 Sensitivity of welfare results

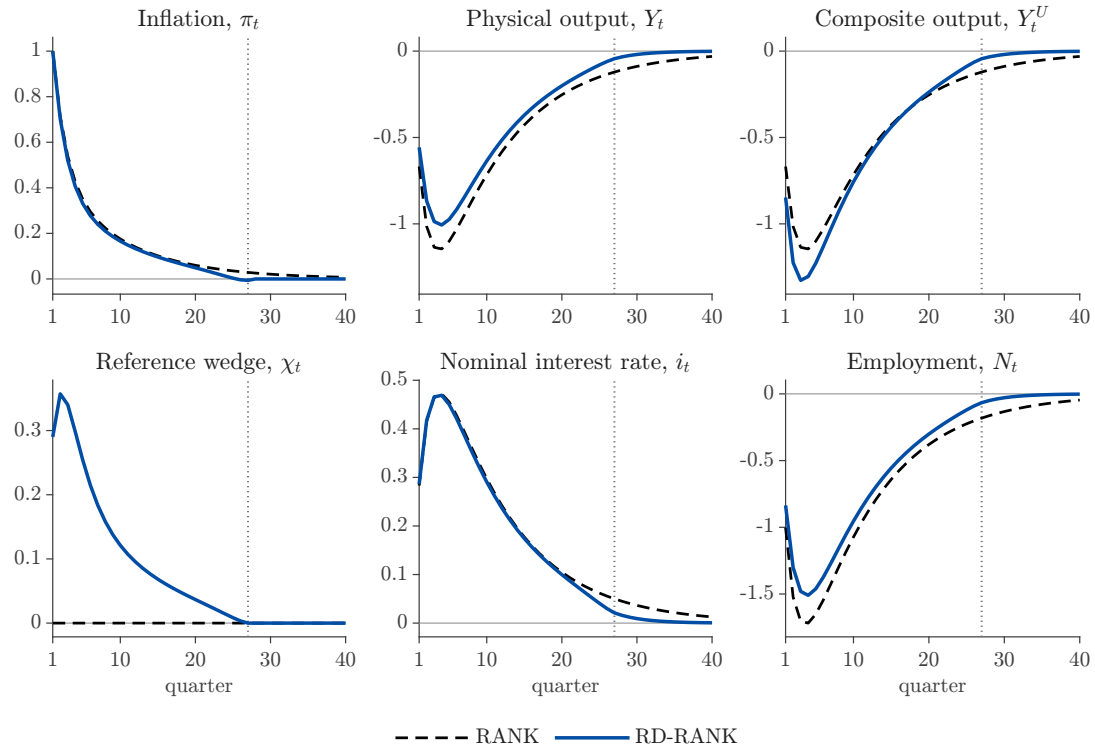
Baseline impulse responses. Figure F.1 shows the responses to the baseline shock of Table 5, with $\rho_u = 0.90$ and $\rho = 0.53$. RD-RANK stays above its reference price until quarter 26 and returns to $P_t = R_t$ in quarter 27.

Reference persistence. Figure F.2 plots the reference wedge along the baseline cost-push path at the three values of ρ in Table 5. All three paths start from the same wedge, because impact inflation is matched. With slower updating, the wedge decays more slowly; at $\rho = 0.88$ it rises until quarter 6.

Monetary-policy rule. Table F.1 compares the baseline rule with rules that remove interest-rate smoothing, output feedback, or both. Each model's cost-push innovation is rematched to a 1% peak in quarterly inflation. Reference dependence raises welfare losses in every case. The complete grid of 12 rules, $\rho_i \in \{0.7, 0.8, 0.9\}$ crossed with $\phi_y \in \{0.075, 0.125, 0.25\}$ plus $(\rho_i, \phi_y) = (0, 0)$, $(0, 0.125)$, and $(0.8, 0)$, gives RD-RANK/RANK ratios of 2.35–3.31 at $\rho = 0.22$ and 3.25–4.71 at $\rho = 0.53$.

Demand elasticity and episode size. Table F.2 varies the calibrated CES elasticity and the target peak inflation rate. For each elasticity, I recalibrate ψ_p to preserve the three-quarter Calvo-equivalent duration in RANK and use the same coefficient in RD-RANK. Higher ε reduces the welfare ratio in these experiments; it also directly reduces the wedge $\Gamma_{b_t} d_t/(\varepsilon - 1)$ for a given price-reference gap. Smaller episodes give the first-order wedge greater weight relative to the quadratic terms in the welfare decomposition. At a 4% peak, its share of the second-order RD-RANK loss falls to 50.6% at $\rho = 0.53$ and 38.5% at $\rho = 0.22$.

Figure F.1: Matched-peak cost-push shock at baseline persistence: $\rho_u = 0.90$



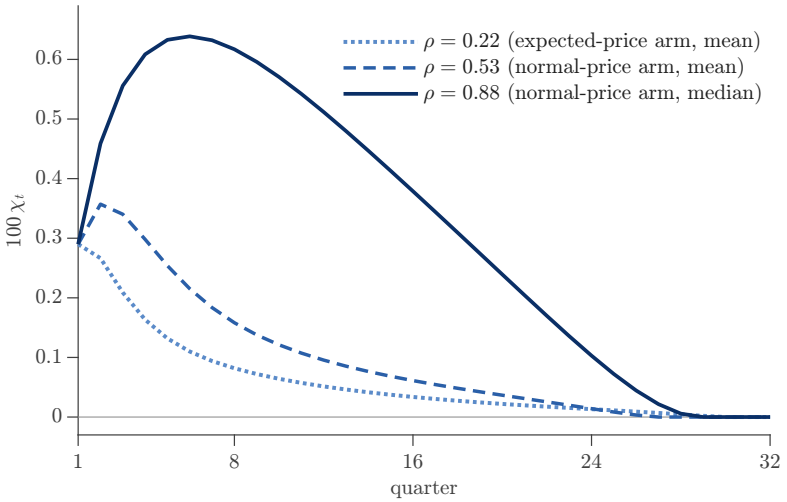
Notes: Responses to a cost-push shock with persistence $\rho_u = 0.90$, scaled separately in each model to produce a 1% peak in quarterly inflation (innovations ε_0^u of 8.36 log points in RANK and 7.89 in RD-RANK). Reference-price persistence is $\rho = 0.53$; all other parameters are those of Table 4. Quarter 1 is the quarter of the shock, $t = 0$. Physical output Y_t, composite output Y_t^U, employment N_t, and the gross nominal interest rate are expressed as log deviations from steady state multiplied by 100; inflation is $100\,\pi_t$ and the reference wedge is $100\,\chi_t$. The two output panels share a vertical scale. The vertical dotted line marks quarter 27, when the RD-RANK model returns to $P_t = R_t$; inflation is exactly zero from quarter 28 onward.

Table F.1: Welfare sensitivity to the monetary-policy rule

			$\rho = 0.22$		$\rho = 0.53$	
ρ_i	ϕ_y	RANK loss	RD-RANK loss	Ratio	RD-RANK loss	Ratio
0.80	0.125	0.0087	0.0266	3.052	0.0376	4.310
0.00	0.125	0.0163	0.0495	3.039	0.0722	4.430
0.80	0.000	0.0111	0.0262	2.355	0.0362	3.248
0.00	0.000	0.0209	0.0537	2.572	0.0771	3.691

Notes: Losses are exact permanent consumption equivalents in percent, 100λ. The first row is the baseline. Every row uses $\sigma = 2$, $\rho_u = 0.90$, $\phi_\pi = 1.50$, $\varepsilon = 4$, and $\psi_p = 52.94$, with innovations chosen separately to match a 1% inflation peak. Ratios use unrounded losses.

Figure F.2: The reference wedge by reference-price persistence



Notes: Reference wedge $100\,\chi_t$ in RD-RANK after a cost-push shock with persistence $\rho_u = 0.90$, scaled to a 1% peak in quarterly inflation, at each value of reference-price persistence ρ; all other parameters are those of Table 4. Quarter 1 is the quarter of the shock, $t = 0$. Multiplying $\sum_t \beta^t \chi_t$ by $1 - \beta$ gives the wedge's contribution to the approximate loss $100 \times \lambda^{(2)}$ in Table 5: 0.018%, 0.029%, and 0.094% at $\rho = 0.22$, 0.53, and 0.88.

Table F.2: Welfare sensitivity to demand elasticity and episode size

				$\rho = 0.22$		$\rho = 0.53$	
ε	ψ_p	Peak (%)	RANK loss	RD-RANK loss	Ratio	RD-RANK loss	Ratio
3	29.41	0.5	0.0014	0.0125	8.800	0.0166	11.733
3	29.41	1.0	0.0056	0.0300	5.375	0.0432	7.740
4	52.94	0.5	0.0022	0.0110	4.948	0.0158	7.100
4	52.94	1.0	0.0087	0.0266	3.052	0.0376	4.310
5	82.35	0.5	0.0033	0.0104	3.169	0.0146	4.471
5	82.35	1.0	0.0127	0.0267	2.100	0.0357	2.802
4	52.94	4.0	0.1255	0.1875	1.495	0.2312	1.843

Notes: Losses are exact permanent consumption equivalents in percent, 100λ, with $\sigma = 2$, $\rho_u = 0.90$, $\rho_i = 0.80$, $\phi_\pi = 1.50$, and $\phi_y = 0.125$. Innovations are rematched separately in each model to the target peak in quarterly inflation. Ratios use unrounded losses. The reference shares at the 4% peak discussed in the text refer to the second-order decomposition, not the exact consumption-equivalent loss.

G The Full Model and Equation System

This appendix states the model of Section 5.1 in the real, recursive form I compute, and lists the full equilibrium system.

G.1 The household block

Euler condition. The decision representation in (5.3), subject to (5.4), implies

$$\frac{c_{it}^{-\sigma}}{P_{z_{it}}^{C}(q_t)} \;\geq\; \frac{\beta_{it}}{\zeta_t}\,\mathbb{E}_{it}\left[R_{t+1}^{b}\,\frac{c_{i,t+1}^{-\sigma}}{P_{z_{i,t+1}}^{C}(q_{t+1})}\right], \tag{G.1}$$

with equality when $b_{it} > 0$. Here $\mathbb{E}_{it}$ conditions on household i's information at date t, including its current individual state. This condition incorporates the aggregate discount-factor shifter ζ_t; the baseline welfare recursions below fix that shifter at one.

Dividend income. Household i's dividend weight in (5.4) is $\xi_{z_{it}}$, where the state-dependent schedule is

$$\xi_z = \frac{e_z^{\lambda}}{\sum_{z'} \pi_z(z')\, e_{z'}^{\lambda}}, \qquad \lambda = 3.054. \tag{G.2}$$

Here $\pi_z(z)$ is the stationary population share in earnings state z. The weights satisfy $\sum_z \pi_z(z)\xi_z = 1$; the share of aggregate dividends received by earnings group z is therefore $\pi_z(z)\xi_z$.

The discount-factor chain. Following Auclert et al. (2025), household i has $\beta_{i,t+1} = \beta_{it}$ with probability $1 - p_\beta$ each quarter. With probability p_β, it draws $\beta_{i,t+1}$ anew, receiving β_H with probability $\pi_\beta(\beta_H)$ and β_L otherwise, independently of its previous discount factor and the earnings process. The two-point discount-factor process therefore has transition matrix

$$\Pi^{\beta} = \begin{pmatrix} 1 - p_\beta\,\pi_\beta(\beta_H) & p_\beta\,\pi_\beta(\beta_H) \\ p_\beta(1 - \pi_\beta(\beta_H)) & 1 - p_\beta(1 - \pi_\beta(\beta_H)) \end{pmatrix}, \qquad p_\beta = 0.01, \quad \pi_\beta(\beta_H) = 0.2295, \tag{G.3}$$

ordered (β_L, β_H). The patient redraw probability $\pi_\beta(\beta_H)$ also equals the stationary patient share. The joint state $s_{it} = (\beta_{it}, z_{it})$ follows a fourteen-state Markov chain with transition matrix $\Pi^s = \Pi^\beta \otimes \Pi_z$.

Sectoral demands. The CES composite behind (5.2) is

$$c_{it} = \left[\omega_R(z_{it})^{\frac{1}{\nu}} \left(c_{it}^{R}\right)^{\frac{\nu-1}{\nu}} + \left(1 - \omega_R(z_{it})\right)^{\frac{1}{\nu}} \left(c_{it}^{S}\right)^{\frac{\nu-1}{\nu}}\right]^{\frac{\nu}{\nu-1}}, \tag{G.4}$$

with demands

$$\begin{aligned} c_{it}^R &= \omega_R(z_{it})\Big(\frac{q_t}{P_{z_{it}}^C(q_t)}\Big)^{-\nu} c_{it}, \\ c_{it}^S &= (1-\omega_R(z_{it}))P_{z_{it}}^C(q_t)^{\nu} c_{it}, \\ q_t c_{it}^R + c_{it}^S &= P_{z_{it}}^C(q_t)\, c_{it}. \end{aligned} \tag{G.5}$$

The block's aggregate outputs are asset demand $A_t = \int_0^1 b_{it}\,\mathrm{d}i$, sectoral consumption $C_{j,t} = \int_0^1 c_{it}^j\,\mathrm{d}i$ for $j \in \{R, S\}$, and the wage-equation statistic

$$\ell_t^L = \int_0^1 e_{z_{it}} \frac{c_{it}^{-\sigma}}{P_{z_{it}}^C(q_t)}\,\mathrm{d}i. \tag{G.6}$$

G.2 Firms

Reference-dependent demand. In sector R, let $\mathcal{R}_{R,t}$ denote the inherited reference price and $d_{R,t} = \log(P_{R,t}/\mathcal{R}_{R,t})$ the price-reference gap. Reference updating implies

$$d_{R,t} = \rho_R d_{R,t-1} + \log \Pi_{R,t}. \tag{G.7}$$

The reference wedge and effective relative price are

$$\begin{aligned} X_{R,t} &= \exp(\chi_{R,t}), \qquad q_t = X_{R,t}\tilde{q}_t, \\ \chi_{R,t} &= \begin{cases} \dfrac{\Gamma_G}{\epsilon_R - 1} d_{R,t}, & d_{R,t} < 0, \\ 0, & d_{R,t} = 0, \\ \dfrac{\Gamma_L}{\epsilon_R - 1} d_{R,t}, & d_{R,t} > 0. \end{cases} \end{aligned} \tag{G.8}$$

Here $X_{R,t}$ abbreviates $e^{\chi_{R,t}}$ in the equations below. When posted prices exceed references, $X_{R,t} > 1$: a given physical consumption bundle delivers less utility, and the expenditure needed to attain a given c rises. This welfare interpretation maintains the reference-dependent utility specification of Section 2, including its treatment of transaction utility as welfare relevant.

Price-setting conditions. Sector-S pricing is given by (5.6). The sector-R conditions below adapt the pricing and reference-costate equations (2.10) and (2.9); Appendix B derives their one-sided counterparts. Let $\upsilon_{j,t}$ denote the cost-push factor, $mc_{j,t}$ the subsidy-adjusted real marginal cost, and $\delta_t = (1+i_t)^{-1}$ the nominal discount factor along a conditional perfect-foresight path. Write $\eta_t = \Gamma_G$ when $d_{R,t} < 0$ and $\eta_t = \Gamma_L$ when $d_{R,t} > 0$. At the kink, $\eta_t \in [\Gamma_G, \Gamma_L]$ is selected by the one-sided optimality conditions. The reference costate $\Theta_{R,t}$ summarizes the continuation effect of changing the reference. The pricing condition and off-kink

costate recursion are

$$\psi_R(\Pi_{R,t}-1)\Pi_{R,t} = 1-(\epsilon_R+\eta_t)(1-\upsilon_{R,t}\,mc_{R,t}) + \delta_t\,\mathbb{E}_t\Big[\frac{X_{R,t+1}Y_{R,t+1}}{X_{R,t}Y_{R,t}}\Big((1-\rho_R)\,\Theta_{R,t+1}\,\Pi_{R,t+1}+\psi_R(\Pi_{R,t+1}-1)\,\Pi_{R,t+1}^2\Big)\Big], \tag{G.9}$$

$$\Theta_{R,t} = \eta_t(1-\upsilon_{R,t}\,mc_{R,t})+\rho_R\,\delta_t\,\mathbb{E}_t\left[\frac{X_{R,t+1}Y_{R,t+1}}{X_{R,t}Y_{R,t}}\,\Pi_{R,t+1}\,\Theta_{R,t+1}\right]. \tag{G.10}$$

At a reference-tracking date, the equivalent costate condition is

$$\Theta_{R,t}\Big|_{d_{R,t}=0} = 1-\epsilon_R(1-\upsilon_{R,t}\,mc_{R,t})-\psi_R(\Pi_{R,t}-1)\Pi_{R,t} + \delta_t\,\mathbb{E}_t\left[\frac{X_{R,t+1}Y_{R,t+1}}{X_{R,t}Y_{R,t}}\,\Pi_{R,t+1}\Big(\Theta_{R,t+1}+\psi_R(\Pi_{R,t+1}-1)\,\Pi_{R,t+1}\Big)\right]. \tag{G.11}$$

The branch and tracking forms agree at a solution with a common selected slope, as shown in (H.5). Appendix H.4 states the complementarity condition and the numerical directional checks. Reference tracking need not imply an unchanged nominal price when the inherited reference moves.

Production, marginal costs, and dividends. Physical production is linear in labor:

$$X_{R,t}Y_{R,t} = Z_tZ_{R,t}N_{R,t}, \qquad Y_{S,t} = Z_tZ_{S,t}N_{S,t}, \qquad N_t = N_{R,t}+N_{S,t}. \tag{G.12}$$

Real marginal costs, inclusive of the constant production subsidies $\bar{mc}_j$, are

$$mc_{R,t} = \bar{mc}_R\,\frac{w_t}{Z_tZ_{R,t}\,\tilde{q}_t}, \qquad mc_{S,t} = \bar{mc}_S\,\frac{w_t}{Z_tZ_{S,t}}, \tag{G.13}$$

where $mc_{R,t}$ measures nominal cost per physical unit relative to the posted price. Sector-R nominal revenue is $P_{R,t}X_{R,t}Y_{R,t}$ and its operating margin is $1-\upsilon_{R,t}mc_{R,t}$. The cost-push factors $\upsilon_{R,t}$, $\upsilon_{S,t}$, and $\upsilon_{w,t}$ equal one in steady state. The sector-R disturbance is implemented as a structural tax with receipts rebated through aggregate dividends. Dividends include these receipts and sectoral revenues net of wage bills, the structural tax, and price-adjustment costs. The rebate therefore reaches household i through $\xi_{z_{it}}\mathcal{D}_t$, with incidence weights (G.2), and leaves the government budget (G.16) unchanged.

Steady-state objects of the reference-dependent block. The zero-inflation steady state has $d_{R,\mathrm{ss}}=0$. I select the midpoint slope $\bar{\Gamma}=(\Gamma_G+\Gamma_L)/2=0.60$ and the associated production

subsidy. The sector-R counterpart of the steady-state calculation in Section 2 gives

$$\Omega = \frac{\delta_{\text{ss}}(1-\rho_R)}{1-\delta_{\text{ss}}\rho_R}, \qquad \epsilon^{\text{eff}} = \epsilon_R + \bar{\Gamma}(1-\Omega), \qquad \bar{mc}_R = \frac{\epsilon^{\text{eff}}-1}{\epsilon^{\text{eff}}}, \qquad \Theta_{R,\text{ss}} = \frac{\bar{\Gamma}(1-\bar{mc}_R)}{1-\rho_R\delta_{\text{ss}}}. \tag{G.14}$$

At $r_{\text{ss}} = 0.005$, these expressions give $\Omega = 0.989$, $\epsilon^{\text{eff}} = 4.0063$, $\bar{mc}_R = 0.7504$, and $\Theta_{R,\text{ss}} = 0.317$; in sector S, $\bar{mc}_S = (\epsilon_S - 1)/\epsilon_S = 6/7$. Applying Proposition B.1 with δ_{ss} and ρ_R in place of β and ρ gives the admissible subsidy range $\bar{mc}_R \in [0.7502, 0.7506]$. The midpoint selection lies within this range. This selection is a calibration choice: the steady-state conditions alone do not select a slope and subsidy pair.

G.3 Policy details

Wages. Following Auclert et al. (2021), each union represents all households and sets the nominal wage for a differentiated labor service. Union k incurs the utility cost

$$\frac{\psi_w}{2}\left[\log\left(\frac{W_{k,t}}{W_{k,t-1}}\right)\right]^2, \qquad \psi_w = \frac{\epsilon_w}{\kappa_w}, \qquad \epsilon_w = \frac{\mu_w}{\mu_w - 1},$$

where ϵ_w is the elasticity of substitution across labor services. In the symmetric equilibrium, all unions set the same wage and households work the same hours. The resulting wage Phillips curve is (5.7), with the household statistic ℓ_t^L defined in (G.6). The calibration sets $\kappa_w = 0.085$ and $\mu_w = 1.2$. Wage adjustment costs enter no budget constraint or market-clearing condition; Section 5.3 includes them in household welfare. Matching the local Calvo slope through $(1-\theta_w)(1-\beta_w\theta_w)/\theta_w = \kappa_w\varphi_N$, where θ_w is the probability of not resetting a wage, gives a mean wage spell of approximately 4.1 quarters. This comparison matches the local Phillips-curve slope and does not imply equivalence of household welfare. The real wage evolves according to

$$w_t = w_{t-1}\frac{\Pi_{w,t}}{\Pi_{S,t}}. \tag{G.15}$$

Union discounting. The union discounts average household utility and wage adjustment costs at $\beta_w = (1-\pi_\beta(\beta_H))\beta_L + \pi_\beta(\beta_H)\beta_H \simeq 0.966$, the stationary population mean of household discount factors. This is an institutional assumption rather than an exact aggregation of heterogeneous lifetime preferences. The union discount factor remains fixed when ζ_t changes; aggregate discount-factor shocks affect wage setting through household marginal utilities and equilibrium prices and quantities.

Monetary policy. The Taylor rule (5.8) uses physical output, $Y_{R,t}^{\text{phys}} = X_{R,t}Y_{R,t}$ and $Y_{S,t}^{\text{phys}} = Y_{S,t}$, with fixed sectoral CPI weights ϑ_j^{Π} and steady-state GDP shares ϑ_j^{Y}. The calibration sets $(\rho_i, \phi_\pi, \phi_y) = (0.80, 1.50, 0.125)$, with the output coefficient of Taylor (1993) expressed in

quarterly units, and $(\vartheta_R^{\Pi}, \vartheta_R^{Y}) = (0.237, 0.262)$. The monetary-policy shock m_t follows an AR(1) process with persistence $\rho_m = 0.62$ (Appendix H.3). The model's steady state has zero inflation and a quarterly real rate of 0.5%. The rule's fixed-weight inflation aggregate differs from the measured CPI defined in (G.17).

Government. Real government debt is a state variable,

$$B_t = R_t^b\, B_{t-1} + G_t + Subs_t + Tr_t^{\text{exog}} - Tax_t^L - Tax_t^{\text{tail}}, \qquad Tax_t^{\text{tail}} = \mathbf{1}\{t \geq T_{tail}\}\, \phi_B (B_{t-1} - B_{\text{ss}}), \tag{G.16}$$

with $(T_{tail}, \phi_B) = (80,\ 0.05)$. Purchases G_t and the transfer instrument Tr_t^{exog} are the two exogenous fiscal inputs; labor-tax revenue $Tax_t^L = \tau_L \mathcal{T}^w w_t N_t$ moves with the wage bill; $Subs_t$ is the bill for the constant production and wage subsidies; and households receive $Tr_t^{\text{hh}} = Tr_t^{\text{exog}} - Tax_t^{\text{tail}}$. I choose the delayed tail tax so that no part of the government's revaluation gains is rebated to households during the 2021–2023 episode: it is identically zero inside the window and only later retires the debt gap, which allows the economy to eventually reach the steady state in the long run. Because the welfare measure follows the post-window path, the tail tax also enters welfare, from 2041.

The measured CPI aggregates posted prices with the CES weight ϑ_R^{Π}:

$$\frac{P_t^{\Pi}}{P_{S,t}} = \left[\vartheta_R^{\Pi}\, \tilde{q}_t^{\,1-\nu} + 1 - \vartheta_R^{\Pi}\right]^{\frac{1}{1-\nu}}, \qquad \Pi_t^{\Pi} = \Pi_{S,t}\, \frac{(P^{\Pi}/P_S)_t}{(P^{\Pi}/P_S)_{t-1}}. \tag{G.17}$$

The subsidy bill in (G.16) is

$$Subs_t = (1 - \bar{mc}_R)\, w_t N_{R,t} + (1 - \bar{mc}_S)\, w_t N_{S,t} + (\mathcal{T}^w - 1)\, w_t N_t, \tag{G.18}$$

government purchases fall on the S good ($G_{S,t} = G_t$, $G_{R,t} = 0$), and both fiscal instruments are i.i.d. level processes with zero steady state, $Tr_t^{\text{exog}} = \varepsilon_t^{Tr}$ and $G_t = \varepsilon_t^G$. The steady state runs a primary surplus $(R_{\text{ss}}^b - 1)B_{\text{ss}}$ financed by the labor tax, whose rate $\tau_L = 0.332$ balances the budget. The structural cost-push tax appears nowhere in (G.16) or (G.18): its receipts never cross the government's books.

G.4 The equilibrium system

Equilibrium. Following the block representation of Auclert et al. (2021, Appendix C.2), let Z collect the conditional paths of aggregate shocks and let

$$U = \left\{\left(\Pi_{w,t},\ \Pi_{S,t},\ \Pi_{R,t},\ \Theta_{R,t},\ Y_{R,t},\ Y_{S,t}\right)\right\}_{t \geq 0} \tag{G.19}$$

denote the six aggregate sequences. Given the initial household distribution and inherited aggregate states, an equilibrium consists of U, household policies and distributions, and the

remaining endogenous variables such that:

(i) Each model block is internally consistent. Household policies solve (5.3) subject to (5.4); the distribution Ψ_t evolves under these policies and the joint exogenous transition matrix Π^s; and household aggregates are obtained by integrating individual outcomes over this distribution. Production, reference updating, firm optimization, union wage setting, and fiscal and monetary policy satisfy the conditions stated above.

(ii) The equilibrium residuals vanish at every date, $H(U, Z) = 0$, with the sector-R branch admissibility and one-sided optimality conditions imposed as specified below.

Market clearing. The goods, labor, and bond market-clearing conditions are

$$Y_{R,t} = C_{R,t} + \frac{\psi_R}{2}(\Pi_{R,t} - 1)^2 Y_{R,t}, \tag{G.20}$$

$$Y_{S,t} = C_{S,t} + \frac{\psi_S}{2}(\Pi_{S,t} - 1)^2 Y_{S,t} + G_t, \tag{G.21}$$

$$N_{R,t} + N_{S,t} = N_t, \tag{G.22}$$

$$A_t = \int_0^1 b_{it} \, \mathrm{d}i = B_t. \tag{G.23}$$

Sector-R quantities in (G.20) are in utility units; multiplying both sides by $X_{R,t}$ gives the physical resource constraint. The labor condition uses the normalization of aggregate labor efficiency to one. Sector-S goods clearing follows from Walras' law and is verified ex post rather than imposed as an independent residual.

Equilibrium residuals and block outputs. The six residual equations in $H(U, Z) = 0$ are the wage Phillips curve (5.7), sector-S pricing (5.6), sector-R pricing (G.9), the reference-costate recursion (G.10) (equivalently (G.11) at a tracking date), sector-R goods clearing (G.20), and bond market clearing (G.23). Given U, the shock paths, and an admissible sector-R slope selection, the remaining aggregate variables follow from the production, reference, wage, and policy blocks. These determine the household inputs $(w_t, R_t^b, q_t, \zeta_t, \mathcal{D}_t, Tr_t^{\mathrm{hh}}, N_t)$, from which the household block generates policies, distributions, and aggregate outcomes.

Reference-price kink. The slope satisfies $\eta_t = \Gamma_G$ when $d_{R,t} < 0$ and $\eta_t = \Gamma_L$ when $d_{R,t} > 0$. At $d_{R,t} = 0$, a common $\eta_t \in [\Gamma_G, \Gamma_L]$ enters the pricing and costate equations, and the firm's objective must have a nonnegative left derivative and a nonpositive right derivative with respect to its log price. The notation $H(U, Z) = 0$ is understood subject to this admissibility requirement. The exact numerical system augments the six residual equations with the slope selector and complementarity condition (H.4). Firm profit maximization remains a requirement

of equilibrium; the directional conditions alone do not establish global optimality. Appendix H.4 gives the complementarity formulation and the additional optimality checks.

Information structure. The equilibrium definition applies to each conditional perfect-foresight transition. Agents observe current aggregate innovations and anticipate no further innovations, while household idiosyncratic risk remains present. The historical exercise retains each transition's current-quarter allocation and updates the inherited states and household distribution before the next innovations arrive (Appendix H.2).

Aggregate shocks. The model carries ten innovations: aggregate and two sectoral productivities ($\varepsilon^{a}, \varepsilon^{aR}, \varepsilon^{aS}$, persistence 0.90), three cost-push disturbances—the structural tax ε^{uR} and the wedges $\varepsilon^{uS}, \varepsilon^{uw}$, all persistence 0.89—the discount-factor innovation (ε^{ζ}, persistence 0.78), the monetary-policy innovation (ε^{i}, which drives the state m_t in eq. (5.8), persistence 0.62), with these three persistences estimated as described in Appendix H.3, and the two fiscal instruments ($\varepsilon^{Tr}, \varepsilon^{G}$, i.i.d.). The estimation of Appendix H.3 activates eight: the sectoral productivity innovations stay at zero because, without sectoral quantity observables, the within-sector split between productivity and markups is not identified.

H Solution and Estimation

The solution combines a sequence-space approximation for initialization with nonlinear pricing and household decisions. Each quarter's innovations are recovered jointly with the conditional equilibrium transition, imposing the reference-price kink exactly at convergence.

H.1 Sequence-space initialization

The household block maps seven aggregate input paths into asset demand, sectoral consumption, and the wage-setting statistic. Its steady-state Jacobian is

$$\mathcal{J}^{o,m}_{t,\tau} = \frac{\partial Y^{HA}_{o,t}}{\partial X^{HA}_{m,\tau}}, \qquad o \in \{A, C_R, C_S, \ell^L\}, \quad m \in \{w, R^b, q, \zeta, \mathcal{D}, Tr^{\text{hh}}, N\}. \tag{H.1}$$

I compute this Jacobian on the joint earnings and discount-factor chain using the fake-news algorithm of Auclert et al. (2021). Let H_U and H_Z denote the Jacobians of $H(U, Z)$ with respect to the endogenous sequences and aggregate shocks. Composing the block derivatives gives the initial linear equilibrium response

$$\mathrm{d}U = -H_U^{-1} H_Z \, \mathrm{d}Z. \tag{H.2}$$

The linearization uses the steady-state midpoint slope $\bar{\Gamma}$ and supplies starting values for the nonlinear solve. The horizon is $T = 300$ quarters. The household grid contains 1560 asset points from zero to 1640 times steady-state quarterly output and fourteen exogenous states. Backward policy iteration uses the endogenous grid method (Carroll, 2006); forward distribution iteration uses the lottery method of Young (2010).

H.2 The sequential solution

At date t, the economy inherits the shock levels, reference gap, lagged policy rate, relative price, real wage, government debt, and household distribution Ψ_t. Conditional on the current innovations, I solve the equilibrium transition with future innovations set to zero and existing shocks following their autoregressive laws. Reference updating and future pricing regimes are determined within this transition. Only the date-t allocation is retained, and the distribution advances under that date's household policies. I repeat the procedure for the twelve quarters 2021Q1–2023Q4. The transition anticipated in 2023Q4 supplies the subsequent continuation used in welfare calculations.

H.3 Data and identification

Observables. Table H.1 lists the eight series. Inflation rates and the policy rate are quarterly rates throughout the estimation, annualized only for display. Each series is measured against an anchor: wage inflation against a trend of 2.68% per year (a 2.0% price trend plus the 0.70% mean growth of the real employment cost index over 2015–2019); the policy rate against its 2015–2019 mean of 1.10% annual; the fiscal ratios against their 2015–2019 means (10.73% of GDP for transfers, 1.42% for the primary deficit); output and hours against 2015–2019 log-linear trends, with a -3.87% level break in hours anchored on 2023 alone, which keeps 2024 out of every anchor and hence out of the out-of-sample check.

Estimator. At each date, the eight innovations solve

$$y_{\text{model}}(\varepsilon_t \,|\, \text{history}_t) = y_{\text{data},t}, \tag{H.3}$$

using quasi-Newton iteration. Each trial evaluates the nonlinear conditional transition from the inherited states and distribution. The innovation Jacobian is initialized from the steady-state impact responses and updated by secant steps. The maximum fitted-observable residual across the four economies is 10^{-9} in model units.

Identification. The eight innovations are identified jointly within each quarter. Output and hours distinguish aggregate productivity from the discount-factor shock; transfers and the primary deficit jointly discipline the two fiscal instruments. Sectoral productivity innovations

Table H.1: The eight observables, data counterparts, and associated innovations

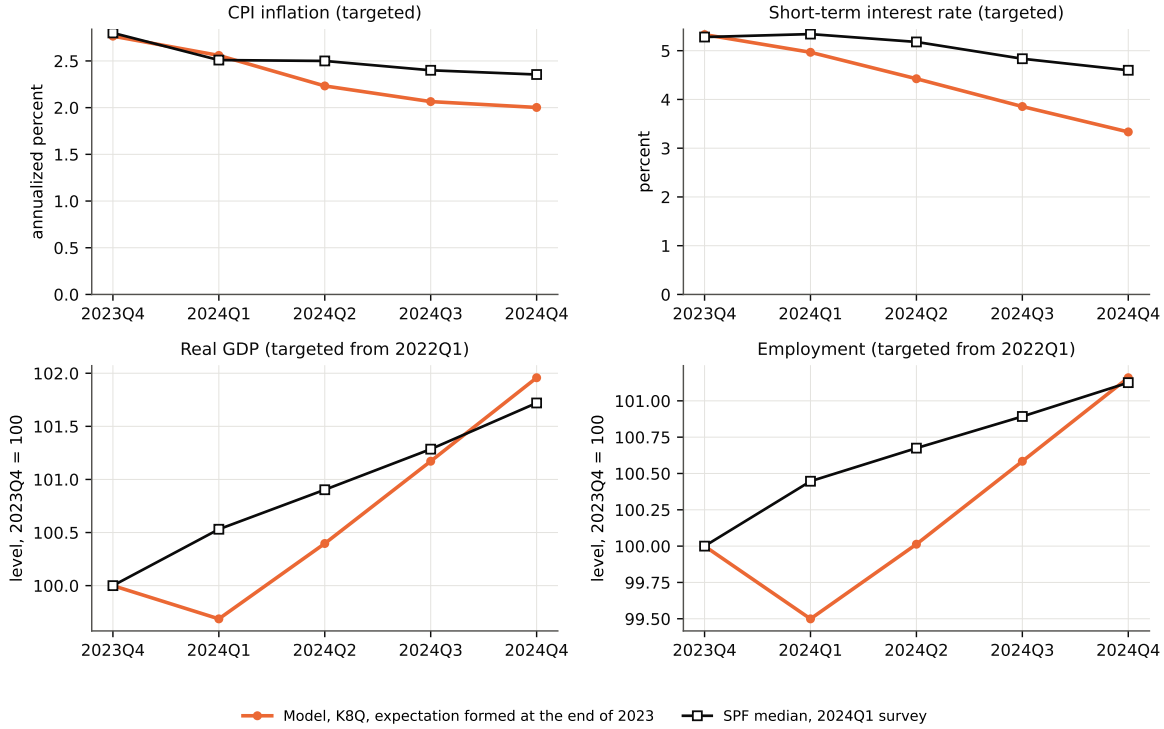
Model observable	Data counterpart	Innovations (joint)
Sector-R inflation, $\Pi_{R,t}$	BLS CPI items assigned to sector R	ε_t^{uR}
Sector-S inflation, $\Pi_{S,t}$	BLS CPI items assigned to sector S	ε_t^{uS}
Wage inflation, $\Pi_{w,t}$	BLS Employment Cost Index: private-industry wages and salaries	ε_t^{uw}
Policy rate, i_t	Effective federal funds rate, quarterly average	ε_t^{i}
Output, Y_t^{phys}	BEA real GDP	$\varepsilon_t^{a},\ \varepsilon_t^{\zeta}$
Hours, N_t	BLS nonfarm business hours	$\varepsilon_t^{a},\ \varepsilon_t^{\zeta}$
Transfers / GDP	NIPA federal social benefits to persons / nominal GDP	$\varepsilon_t^{Tr},\ \varepsilon_t^{G}$
Primary deficit / GDP	NIPA federal outlays less interest and receipts / nominal GDP	$\varepsilon_t^{Tr},\ \varepsilon_t^{G}$

Notes: Twelve quarters × eight series = 96 targets against 96 free innovations. All eight innovations are identified jointly each quarter. The last column indicates the associated shock groups: sectoral cost-push shocks, a wage-markup shock, a monetary-policy shock, aggregate productivity and discount-factor shocks, and transfer and government-purchases shocks. Output and hours jointly distinguish productivity from the discount-factor shock; the two fiscal observables jointly discipline transfers and purchases. Sectoral CPI indices use fixed December 2020 relative-importance weights and a 2% annual inflation benchmark. The paragraph above describes the remaining transformations. Transfers are NIPA table 3.2 federal social benefits to persons. Model output is physical output, $Y_t^{\text{phys}} = e^{\chi_{R,t}} Y_{R,t} + Y_{S,t}$, as in (G.12).

are held at zero because sectoral quantities are not observed. The exercise is exactly identified, so in-sample fit does not independently validate the shock attribution. An innovation dated t represents news arriving at t under the information structure in Appendix H.2.

Shock persistence. The in-sample fit alone does not identify shock persistence, because the innovations are recovered to match the eight observed series. I therefore estimate shock persistence by matching the model's expectations of CPI inflation, the short-term interest rate, real GDP growth, and employment growth to the corresponding forecasts from the Survey of Professional Forecasters. For each candidate vector of persistence parameters, the innovations are re-fitted to the 2021–2023 data. The comparison uses the model's expectations one to four quarters ahead, formed at the end of each quarter of 2021–2023, and the medians from the following survey. CPI inflation and the three-month Treasury bill rate are targeted over the whole window; real GDP growth and employment growth are targeted from the 2022Q1 survey on, after the one-off fiscal transfers of 2021. The criterion is evaluated with first-order sequence-space maps of the economy without reference dependence, and the estimates are verified in the nonlinear sequential solution. The estimated quarterly persistences are $\rho_u = 0.89$, common to the cost-push and wage-markup disturbances, $\rho_\zeta = 0.78$ and $\rho_m = 0.62$; productivity persistence is held at 0.90. Actual 2024 outcomes provide an out-of-sample check. Relative to a cost-push persistence of 0.5 with an i.i.d. monetary disturbance, the 2024 forecast errors fall from 3.4 to 0.9 annualized percentage points for sector-R inflation and from 2.3 to 1.0 for

Figure H.1: Expectations for 2024: the model at the end of 2023 and professional forecasters



Notes: Each panel compares the path the model's agents expected at the end of 2023—the continuation of the 2023Q4 sequential solution with no further innovations—with the median of the Survey of Professional Forecasters from the 2024Q1 survey, for 2024Q1–2024Q4; the 2023Q4 point is the model's fitted value and the survey's estimate of the previous quarter. The persistences of the cost-push, wage-markup, discount-factor and monetary-policy disturbances are estimated so that the model's expectations of CPI inflation and the three-month Treasury bill rate match the survey's over 2021–2023 (top row) and its real GDP and employment growth expectations match the survey's from 2022 on (bottom row). CPI inflation is the consumption-weighted aggregate of the two sectoral inflation rates, annualized; the short-term rate is the policy rate for the model and the three-month Treasury bill rate for the survey; real GDP and employment are indices with 2023Q4 = 100, cumulated from the expected quarterly growth rates, with the model's detrended series restored to their 2015–2019 trend growth. The model has no extensive margin; its employment counterpart is total hours.

the policy rate. The model's one-year-ahead CPI expectations also track the survey medians, which they undershot by about 1.7 percentage points at that persistence. Figure H.1 compares the expectations the model's agents held at the end of 2023 with the Survey of Professional Forecasters medians from the 2024Q1 survey for the four targeted series.

H.4 Nonlinear pricing at the reference-price kink

At each date, let $\alpha_t \in [0, 1]$ select the slope $\eta_t = \Gamma_G + (\Gamma_L - \Gamma_G)\alpha_t$. The pricing and reference-costate equations use this same slope, subject to

$$\alpha_t = \text{clip}\left(\alpha_t + d_{R,t}/d_{\text{scale}},\, 0,\, 1\right), \qquad d_{\text{scale}} = 0.01. \tag{H.4}$$

This condition selects the loss branch when $d_{R,t} > 0$, the gain branch when $d_{R,t} < 0$, and allows an interior selector when $d_{R,t} = 0$. The scaling parameter affects numerical conditioning but not the solution set. With a positive operating margin and differentiable continuation value, an interior selector implements the one-sided conditions in Appendix B: the firm's left price derivative is nonnegative and its right derivative is nonpositive.

Let $\mathcal{H}_t^P$ be the right-hand side of (G.9) minus its left-hand side, and let $\mathcal{H}_t^B$ and $\mathcal{H}_t^K$ be the residuals of the branch and tracking costate equations, respectively, with $\Theta_{R,t}$ on the left. Then

$$\mathcal{H}_t^K - \mathcal{H}_t^B = -\mathcal{H}_t^P \tag{H.5}$$

so both costate forms agree when the pricing residual is zero. These are necessary optimality conditions; sufficiency requires concavity of the full Bellman objective and the tail condition. Concavity is not guaranteed by the model's primitives, so residual and directional checks alone do not establish global optimality.

Global optimality. The firm-optimality check of Appendix F, applied to the fitted aggregate path of each window date in both baskets, finds the equilibrium price to be the global maximizer of the firm's objective at every quarter. On the ± 20 log-point price corridor the objective is single-peaked on each side of the kink at every grid state, the refined best response lies within 0.01 log point of the equilibrium price and raises firm value by at most 3×10^{-7} of a quarter's revenue, and a price one log point away lowers it by at least 0.2 percent.

I solve the augmented equilibrium system by damped semismooth Newton iteration, using shock-scale continuation and progressively tighter approximations to the clip function before imposing its exact form. Pricing regimes are endogenous. When using the tracking costate equation, the regime classification is held fixed within each Newton solve and updated between solves until it stabilizes. The residual identity above ensures that the two costate representations share the same roots. Section 4 uses an active-set implementation of the same one-sided conditions; Appendix F describes that algorithm.

The innovation estimation and nonlinear household correction are iterated with this pricing system. Table H.2 reports residuals for the four fitted economies; directional-check violations are zero in the forty-eight quarter-solves. The fitted baseline has eleven loss quarters, with 2023Q2 on the gain branch, and the broad basket has twelve.

H.5 Nonlinear household decisions

To account for nonlinear responses and the evolution of the wealth distribution, I solve household policies backward and the distribution forward from each quarter's inherited distribution. The

Table H.2: Numerical residuals for the four fitted economies

Economy	Obs fit	Equilibrium	Walras	Loss qtrs
Baseline basket	1×10^{-9}	9×10^{-10}	5.0×10^{-8}	11
Baseline, no reference dependence	1×10^{-9}	1×10^{-9}	9.7×10^{-6}	—
Broad basket	8×10^{-10}	8×10^{-10}	7.3×10^{-8}	12
Broad, no reference dependence	9×10^{-10}	1×10^{-9}	1.6×10^{-5}	—

Notes: Every entry is a worst case over the economy's twelve quarter-solves. "Obs fit" is the sup-norm residual of the eight observable targets in their own units; "equilibrium" the sup-norm residual of the exact equilibrium system; "Walras" the sector-S goods-market residual under the fully nonlinear household. The economies without reference dependence reuse the household steady state, policy functions, and Jacobians of the corresponding reference-dependent economy rather than re-solving them at their own firm-side steady state; this small mismatch leaves a residual of order 10^{-5} in sector-S goods clearing that is constant across quarters. Directional-check violations are zero in all forty-eight quarter-solves. "Loss quarters" counts window quarters with a positive reference gap; 2023Q2 is the baseline basket's one gain quarter.

correction to the sequence-space approximation is

$$\Delta h_t = h_{NL}(X_t) - h_{lin}(X_t), \qquad h \in \{A, C_R, C_S, \ell^L\}. \tag{H.6}$$

I feed this correction into the aggregate system and iterate jointly with the pricing and innovation solves until convergence. Welfare uses the resulting nonlinear household policies and distributions. The maximum asset-market residual is 4×10^{-7}; Table H.2 reports the goods-market residuals, including the offsets in the economies without reference dependence.

H.6 Welfare computation

The welfare recursions and consumption equivalent. Household i starts in state $(\beta_{i0}, z_{i0}, b_{i,-1})$, distributed according to Ψ_{ss}, where $t = 0$ denotes 2021Q1. Both scenarios $x \in \{\text{ss}, tr\}$ use the same idiosyncratic processes and include consumption utility, labor disutility, and wage adjustment costs. The baseline evaluation holds $\zeta = 1$, retaining the household's idiosyncratic discount-factor process β_{it}.

Let $V_{it}^{c,x}$, $V_{it}^{\ell,x}$, and $V_{it}^{w,x}$ denote household i's continuation values of consumption utility, labor disutility, and wage-adjustment costs. They satisfy

$$V_{it}^{c,x} = \frac{(c_{it}^x)^{1-\sigma}}{1-\sigma} + \beta_{it}\, \mathbb{E}_{it}^x \left[V_{i,t+1}^{c,x} \right], \tag{H.7}$$

$$V_{it}^{\ell,x} = \varphi_N \frac{(N_t^x)^{1+\varphi}}{1+\varphi} + \beta_{it}\, \mathbb{E}_{it}^x \left[V_{i,t+1}^{\ell,x} \right], \tag{H.8}$$

$$V_{it}^{w,x} = \frac{\psi_w}{2} \left(\log \Pi_{w,t}^x \right)^2 + \beta_{it}\, \mathbb{E}_{it}^x \left[V_{i,t+1}^{w,x} \right]. \tag{H.9}$$

Here $\mathbb{E}_{it}^x$ integrates over future idiosyncratic transitions conditional on household i's current state and the aggregate history in scenario x. The next individual state is $(\beta_{i,t+1}, z_{i,t+1}, b_{it}^x)$. In

the realized scenario the recursions run over $t = 0, \ldots, 299$ and end with the steady-state values at $t = 300$, $V^{a,tr}_{i,300} = V^{a,\mathrm{ss}}(\beta_{i,300}, z_{i,300}, b^{tr}_{i,299})$ for $a \in \{c, \ell\}$ and $V^{w,tr}_{i,300} = 0$; the steady-state scenario's values are the steady-state values themselves. Write $V^{a,x}_i \equiv V^{a,x}_{i0}$ for $a \in \{c, \ell, w\}$. Labor disutility and wage adjustment costs are common across households and are discounted using each household's own β_{it} process. The wage cost is $\frac{\psi_w}{2}(\log \Pi_{w,t})^2$, with $\psi_w = \epsilon_w/\kappa_w = 71$, and vanishes in steady state. Price adjustment costs enter resource constraints and therefore affect welfare through consumption.

The consumption equivalent reduces steady-state consumption in every state of the four quarters 2021Q1–2021Q4, and leaves it unchanged afterwards, until the household is indifferent, as in (5.10). Let $\underline{V}^{c,\mathrm{ss}}_i$ be the part of $V^{c,\mathrm{ss}}_i$ that accrues in those four quarters and $\overline{V}^{c,\mathrm{ss}}_i = V^{c,\mathrm{ss}}_i - \underline{V}^{c,\mathrm{ss}}_i$. With $V^{w,\mathrm{ss}}_i = 0$,

$$1 - \mathrm{CE}_i = \left(\frac{V^{c,tr}_i - V^{\ell,tr}_i - V^{w,tr}_i + V^{\ell,\mathrm{ss}}_i - \overline{V}^{c,\mathrm{ss}}_i}{\underline{V}^{c,\mathrm{ss}}_i} \right)^{\frac{1}{1-\sigma}}. \tag{H.10}$$

The cost of the episode is the population mean $\int_0^1 \mathrm{CE}_i \, \mathrm{d}i$, with initial states distributed according to Ψ_{ss}. Through 2023Q4, welfare is evaluated along the realized history using the policies chosen under each quarter's information. From 2024Q1, it follows the equilibrium path anticipated at the end of 2023, with no further aggregate innovations, including the subsequent fiscal adjustment. All reported costs are percentages of 2021 steady-state consumption, $100 \int_0^1 \mathrm{CE}_i \, \mathrm{d}i$.

Continuation values and sensitivity. Replacing the steady-state terminal values in 2096Q1 with zero changes welfare costs by less than 0.005 percentage points of 2021 steady-state consumption. Table H.3 instead varies the date at which household positions are valued using steady-state value functions, net of the first-order value of future taxes retiring the remaining public debt gap. The first row values positions at the end of 2023; later rows retain more of the model's anticipated continuation.

Later switching dates raise the cost: the estimated persistence of the cost-push, wage-markup and monetary disturbances keeps sector-R inflation, hours and the real return away from their steady-state values for several years after 2023, and each year of the model's continuation that is valued adds to the loss, from 3.78% of 2021 consumption at a 2024Q1 switch to 5.49% by 2031Q1 and 5.80% under the full continuation. The contribution of reference dependence rises from 0.61 to 0.91 percentage points over the same horizons as debt accumulates and the reference gap reopens before the delayed fiscal adjustment. This sensitivity reflects the assumed continuation and fiscal rule.

Realized preferences and a re-optimized benchmark. Two variants value both allocations with the realized weights β_{it}/ζ_t in place of β_{it}. The first keeps the steady-state-policy benchmark. The second replaces it with households that face steady-state prices, wages, bond

returns, hours, dividends, and transfers but re-optimize consumption and saving against each quarter's ζ_t as it is revealed, with the beliefs of the fitted economy, and follow their own continuation thereafter. Markets need not clear in this partial-equilibrium benchmark. Their within-period consumption bundles are chosen at steady-state relative prices.

Table H.4 crosses these two variants and the baseline welfare convention with six treatments of post-2023 values. Besides full continuation and zero value after 2023Q4, the table values household positions at the end of 2023 with steady-state value functions, either without a charge for the remaining public debt or net of the first-order value of lump-sum taxes retiring that debt from 2041Q1 or 2024Q1. These terminal adjustments change welfare valuation without re-solving the fitted equilibrium. The final treatment replaces the baseline's steady-state terminal value in 2096Q1 with zero as a numerical check.

With full continuation, re-optimizing the benchmark reduces total welfare costs by at most 0.30 percentage points of 2021 steady-state consumption relative to following steady-state policies under realized discounting; part of this gap reflects that the percentage is of the re-optimized benchmark's own 2021 consumption, which lies below steady state. The zero-terminal-value comparison gives much larger changes: costs fall from 4.58 to 1.88% in the baseline basket and from 4.65 to 2.07% in the broader basket. That calculation omits the post-2023 consequences of the benchmark households' changed saving decisions. All panels retain the household's idiosyncratic discount-factor process and include wage adjustment costs.

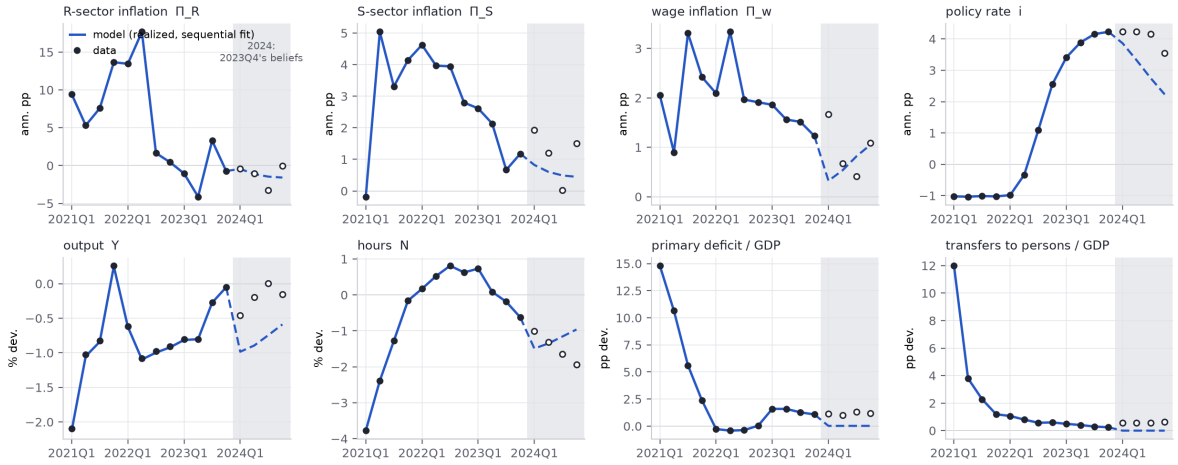
H.7 The fitted episode and estimated innovations

Figures H.2 and H.3 report the observed and fitted series and the recovered innovations. Monetary-policy surprises keep the rate below the Taylor rule through 2022Q2 and above it thereafter; transfer innovations are largest in early 2021, and the sector-R cost-push innovation is largest in 2021Q1, with further large positive surprises in 2021Q4 and 2022Q2. The baseline reference gap peaks at 0.075 in 2022Q2, raising the effective sector-R price approximately 2.2% above its posted price.

Innovation correlations across the economies with and without reference dependence exceed 0.994 in every channel; the sector-R cost-push peak is 0.12 in both economies. The shocks are nevertheless re-estimated, and exact fit does not independently validate their attribution.

The anticipated continuation tracks sector-R inflation in 2024 but reduces interest rates faster than the data. Mean absolute errors are 0.9 annualized percentage points for sector-R inflation and 1.0 for the policy rate. Welfare therefore includes the continuation households anticipated at the end of 2023, rather than the subsequent realized history; Appendix H.6 reports sensitivity to this valuation.

Figure H.2: The eight fitted series and the 2024 out-of-sample check



Notes: Filled markers are data; the solid line is the realized path of the sequential fit over the twelve estimation quarters 2021Q1–2023Q4. The dashed line in the shaded region is the model's continuation from 2024Q1 with no further innovations—under sequential information this is 2023Q4's own belief path, the forecast the model's agents held as the window closed—and open markers are the 2024 data, which enter neither the estimation nor the detrending anchors. Inflation rates and the policy rate are annualized percentage points in deviation from their anchors; output and hours are percent deviations from a 2015–2019 log-linear trend; the fiscal series are percentage points of GDP in deviation from 2015–2019 means. Appendix H.3 details the transformations.

Table H.3: The welfare cost by the date at which steady-state prices resume

	Baseline basket			Broad basket		
Steady-state prices from	Total	No RD	RD	Total	No RD	RD
2024Q1	3.78	3.17	0.61	3.95	3.19	0.76
2025Q1	4.00	3.39	0.61	4.19	3.43	0.76
2027Q1	4.79	4.15	0.64	5.14	4.22	0.92
2031Q1	5.49	4.64	0.85	6.00	4.73	1.27
2040Q4	5.15	4.16	0.99	5.82	4.34	1.48
2043Q3	5.37	4.43	0.94	6.05	4.64	1.41
2051Q1	5.78	4.88	0.90	6.38	5.04	1.34
2096Q1	5.80	4.89	0.91	6.40	5.05	1.35
Full model continuation (baseline)	5.80	4.89	0.91	6.40	5.05	1.35

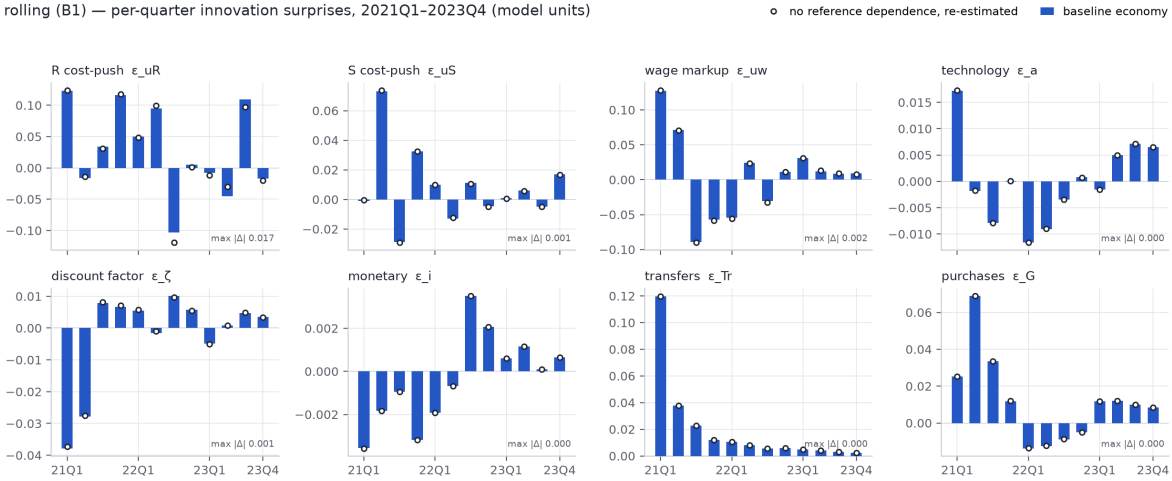
Notes: Percent of 2021 steady-state consumption (the constant percentage of consumption in each quarter of 2021 that a steady-state household would give up to be indifferent to the episode), using pre-episode aggregate discounting and the steady-state-policy benchmark. "Total" and "No RD" are population-mean welfare costs with and without reference dependence; "RD" is their difference. Entries are rounded independently. The table reports selected switching dates. Each dated row follows the model's continuation until the stated quarter and then values each household's position using steady-state value functions, less the first-order value of its lump-sum share of the taxes retiring the remaining public-debt gap. Repayment begins in 2041Q1 or at the switch date, whichever is later. The full-model baseline includes the fiscal adjustment in the equilibrium path and adds no separate debt charge at its 2096Q1 terminal date. Table H.4 reports the zero-continuation-value comparison and the alternative preference conventions.

Table H.4: Robustness of welfare costs to valuation and benchmark choices

	Baseline basket			Broad basket		
Post-2023 valuation	Total	No RD	RD	Total	No RD	RD
A. Pre-episode aggregate discounting; steady-state policies						
Full model continuation	5.80	4.89	0.91	6.40	5.05	1.35
Zero terminal value in 2024	4.37	3.77	0.60	4.43	3.68	0.75
Steady state; no debt charge	3.64	3.03	0.61	3.90	3.14	0.76
Steady state; repayment from 2041	3.78	3.17	0.61	3.95	3.19	0.76
Steady state; repayment from 2024	4.15	3.55	0.60	4.10	3.34	0.76
Full continuation; zero value in 2096	5.80	4.89	0.91	6.40	5.05	1.35
B. Realized aggregate discounting; steady-state policies						
Full model continuation	6.08	5.10	0.98	6.70	5.25	1.45
Zero terminal value in 2024	4.58	3.93	0.65	4.65	3.83	0.82
Steady state; no debt charge	3.75	3.10	0.66	4.04	3.21	0.83
Steady state; repayment from 2041	3.90	3.24	0.66	4.09	3.26	0.83
Steady state; repayment from 2024	4.31	3.66	0.65	4.25	3.42	0.82
Full continuation; zero value in 2096	6.08	5.10	0.98	6.70	5.25	1.45
C. Realized aggregate discounting; re-optimized benchmark						
Full model continuation	5.82	4.88	0.94	6.40	5.01	1.39
Zero terminal value in 2024	1.88	1.34	0.54	2.07	1.37	0.69
Steady state; no debt charge	3.55	2.91	0.64	3.80	3.00	0.80
Steady state; repayment from 2041	3.69	3.05	0.64	3.85	3.05	0.80
Steady state; repayment from 2024	4.08	3.44	0.63	4.01	3.21	0.80
Full continuation; zero value in 2096	5.82	4.88	0.94	6.41	5.01	1.40

Notes: Percent of 2021 steady-state consumption: the constant percentage of consumption in each quarter of 2021 that a steady-state household would give up to be indifferent to the episode. "Total" and "No RD" are population-mean welfare costs with and without reference dependence; "RD" is their difference. Entries are rounded independently. Panel A values both scenarios using β_{it}; panels B and C use β_{it}/ζ_t. In panel C, comparison households re-optimize consumption and saving against sequentially revealed ζ_t, with all other aggregate inputs at steady state; the percentage is of that benchmark's own 2021 consumption. The fitted equilibrium paths are unchanged across panels. Full continuation follows the fitted episode and the path anticipated in 2023Q4, ending with steady-state values in 2096Q1; the last row instead sets those terminal values to zero. Zero value in 2024 counts only 2021–2023 utility. The three steady-state rows value households' end-2023 positions with steady-state value functions, either without a debt charge or net of the first-order value of lump-sum taxes retiring the outstanding debt gap from the indicated date. Wage adjustment costs are included. See Appendix H.6 for the recursions.

Figure H.3: The eight estimated innovation surprises, 2021Q1–2023Q4



Notes: Innovations are in model units—each entry is the news arriving that quarter. Bars are the baseline economy's sequential estimates; open markers are the no-reference-dependence economy's, re-estimated against the same eight series. The annotation in each panel is the largest absolute gap between the two economies: the histories are similar (correlation above 0.994 in every channel), although innovations are re-estimated separately. Panels: sectoral cost-push innovations ε^{uR} and ε^{uS}, wage markup ε^{uw}, aggregate productivity ε^{a}, discount-factor innovation ε^{ζ} (demand), monetary-policy innovation ε^{i}, transfers ε^{Tr}, and government purchases ε^{G}.

I Calibration Details

Table I.1 collects the model's parameters and steady state. The rest of this appendix documents the two survey calibrations behind the household block: the sectoral expenditure weights $\omega_R(z)$ of (5.2), from the 2020 Consumer Expenditure Survey, and the dividend-incidence exponent λ of (G.2), from the 2019 Survey of Consumer Finances.

Untargeted wealth moments. Table I.2 compares the model's steady-state wealth distribution with the 2019 SCF. The model closely matches the wealth share of the bottom half of households but understates concentration at the top.

I.1 Sectoral expenditure shares across earnings states

I calibrate $\omega_R(z)$ from the 2020 Consumer Expenditure Survey (CEX). The baseline basket B1 assigns 200 consumption categories to sector R: food, energy, and frequently purchased nondurables. Basket B2 adds 18 recurring services, including cash rent and communications. Sector S is the complement within a fixed universe of 656 current-consumption categories. Basket membership is common across households; expenditure patterns generate heterogeneity in the weights.

Expenditure measurement. For each Universal Classification Code (UCC), I use the Interview or Diary survey designated by the BLS to avoid double counting. Interview and Diary expenditures are annualized by four and 52, respectively. For an Interview recall window with m_i months in 2020, expenditure and survey weight are adjusted by $3/m_i$ and $m_i/3$, so calendar overlap enters once. The consumption denominator excludes saving, pension contributions, debt principal, cash gifts, and income records.

Let $v \in \{I, D\}$ denote the survey, w_i^v its sampling weight, and M_{ip}^v household i's fractional membership in percentile cell p. Annual expenditure on UCC j in cell p is

$$\bar{x}_{pj}^v = a_v \frac{\sum_{i \in v} w_i^v M_{ip}^v x_{ij}^v}{\sum_{i \in v} w_i^v M_{ip}^v}, \qquad a_I = 4, \quad a_D = 52, \tag{I.1}$$

where x_{ij}^v incorporates the calendar adjustment. Summing over the designated survey for each UCC gives reference-sector expenditure $\bar{X}_{pk}^R$ for basket k and total consumption $\bar{X}_p^C$.

Income states and aggregation. The sample includes reference persons aged 25–60 and retains zero and negative earners. Labor income combines household wages and members' self-employment income, averaged across five implicates. I construct 100 weighted income-percentile cells from the Interview survey and apply that income-to-percentile map to the Diary sample, splitting tied groups fractionally at boundaries.

Table I.1: Calibration and steady state

Parameter	Description	Value	Source or target
Households			
σ	Relative risk aversion	2	Standard
φ	Inverse Frisch elasticity	2	Standard
β_H	Patient discount factor	0.99	Assigned
β_L	Impatient discount factor	0.9593	Calibrated (MPC, debt)
$\pi_\beta(\beta_H)$	Patient redraw mass (= ergodic share)	0.2295	Calibrated (MPC, debt)
p_β	Quarterly β-redraw probability	0.01	Auclert et al. (2025)
λ	Dividend exponent, $\xi_z \propto e_z^\lambda$	3.054	SCF 2019 top-7/64 equity
ν	CES elasticity between R and S	0.40	Bobasu et al. (2025)
$\underline{b}$	Borrowing limit	0	Standard
ρ_z	Persistence of log earnings	0.966	Auclert et al. (2021)
σ_z	Unconditional s.d. of log earnings	0.50	Auclert et al. (2021)
n_z	Discretization	7, Rouwenhorst	Standard
Firms			
ϵ_R	Demand elasticity, sector R	4	Scanner-data substitution
ϵ_S	Demand elasticity, sector S	7	Standard
ψ_R, ψ_S	Rotemberg adjustment cost	34.6, 89.7	Price-change freq. 10.6, 8.8%/mo
ρ_a	Productivity persistence	0.90	Assigned
ρ_u	Cost-push persistence (υ_R structural tax; υ_S, υ_w wedges)	0.89	Estimated (Appendix H.3)
ρ_ζ	Discount-factor shock persistence	0.78	Estimated (Appendix H.3)
ρ_m	Monetary-policy shock persistence	0.62	Estimated (Appendix H.3)
Reference dependence			
Γ_G	Gain-side loading	0.33	Survey, Section 3
Γ_L	Loss-side loading	0.87	Survey, Section 3
ρ_R	Reference adaptation	0.53	Survey, Section 3
Wage setting			
ϵ_w	Wage-demand elasticity	6	Erceg et al. (2000)
$\mu_w = \mathcal{T}^w$	Wage markup; offsetting subsidy	1.20	Undistorted steady state
κ_w	Wage Phillips-curve slope	0.085	$\psi_w = 71$; annual wage adjustment
β_w	Union discount factor	0.966	Population mean of β_{it}
φ_N	Labor-disutility scale	1.0035	Steady state
Policy			
ρ_i	Interest-rate smoothing	0.80	Standard
ϕ_π	Taylor coefficient on measured inflation	1.50	Standard
ϑ_R^Π	Sector-R weight, measured CPI	0.237	CPI weight
ϕ_y	Taylor coefficient on output	0.125	Standard
ϑ_R^Y	Gap weight: R share of SS nominal GDP	0.262	Steady-state value
T_{tail}	Tail-tax delay (quarters)	80	Assigned
ϕ_B	Tail-tax coefficient on the debt gap	0.05	Terminal condition
ρ_{Tr}, ρ_G	Fiscal-shock persistence	0 (iid)	Assigned
Basket and exposure (2020 CEX)			
$\omega_R(z_1..z_7)$	Sector-R expenditure weights	0.274, 0.287, 0.290, 0.271, 0.263, 0.228, 0.229	
ξ_{z_1}, ξ_{z_7}	Dividend weights, bottom and top bands	0.008, 14.06	SCF 2019
Steady state (targeted moments first)			
	Aggregate quarterly MPC	0.250	Targeted
	Debt / annual GDP	1.400	Targeted
	Sector-R share of output and hours	0.262	Untargeted
	Labor-income tax rate, τ_L	0.332	Balanced govt. budget, steady state
	Share at exactly zero bonds	0.186	Untargeted
	Real rate r_{ss} (quarterly)	0.005	Standard

Notes: "Standard" and "Assigned" rows are both imposed rather than calibrated, the former at conventional values; β_L and $\pi_\beta(\beta_H)$ are the only calibrated parameters, solved jointly so that the aggregate quarterly MPC equals 0.25 and government debt equals 1.40 times annual GDP (bottom panel). All rates are quarterly. $\pi_\beta(\beta_H)$ is both the probability that a discount-factor redraw lands on the patient state and the ergodic patient share. $\omega_R(z)$ is the sector-R expenditure weight of earnings state z and ξ_z the per-household dividend weight in that state, normalized to have population mean one; ϑ_R^Π is the sector-R weight in the measured CPI, which aggregates posted prices. T_{tail} is in quarters and ϕ_B is a per-quarter coefficient on the debt gap $B_{t-1} - B_{\text{ss}}$.

Table I.2: Untargeted moments of the wealth distribution

	Model		2019 SCF
Moment	Baseline basket	Broader basket	Net worth
Wealth Gini	0.765	0.766	0.852
Bottom-50% wealth share (%)	1.7	1.7	1.5
Top-10% wealth share (%)	59.3	59.5	76.5
Top-1% wealth share (%)	13.9	14.0	37.2

Notes: None of these moments is targeted in the calibration. Model columns report the steady-state distribution of bond holdings in the reference-dependent economies. SCF moments use net worth for all families, with survey-weighted statistics calculated separately in each of the five implicates and then averaged. Households are ranked by wealth within each distribution. Wealth shares are percentages; the Gini is on a zero-to-one scale. SCF net worth includes assets and liabilities absent from the single-asset model.

The seven earnings states have stationary masses $\{1, 6, 15, 20, 15, 6, 1\}/64$. If A_{pg} is the overlap in percentile points between cell p and state g, the sectoral weights are

$$\omega_{gk}^{R} = \frac{\sum_{p=1}^{100} A_{pg} \bar{X}_{pk}^{R}}{\sum_{p=1}^{100} A_{pg} \bar{X}_{p}^{C}}, \qquad \omega_{gk}^{S} = 1 - \omega_{gk}^{R}. \tag{I.2}$$

These are ratios of expenditure totals, rather than averages of household budget shares. At the normalized steady-state relative price of one, they equal the CES distribution weights. Table I.3 reports the resulting schedules and sampling uncertainty.

Baseline weights vary modestly across earnings states. The broader basket places more weight on reference-dependent goods among lower and middle earners, reflecting spending on rent and recurring services.

I.2 Dividend incidence from the 2019 SCF

I use the public 2019 Survey of Consumer Finances (SCF), whose income questions refer to 2018, to calibrate dividend incidence across earnings states. The sample includes families with a reference person aged 25–60, retains zero and negative labor income and zero dividends, and does not trim or winsorize the dividend tail. Labor income comprises wages and salaries plus net sole-proprietorship and farm income; it is a proxy for current earnings, since business income can also contain capital returns.

Within each of the five implicates, I rank families by labor income using survey weights and form seven bands with the model's stationary masses π_j. Boundary observations and tied groups are split fractionally to match these masses. Each band's share is its weighted dividend total divided by the sample total, averaged across implicates.

The top two bands, comprising 7/64 of families, receive 59.84% of dividend income. SCF replicate weights and between-implicate variation give a standard error of 4.96 percentage

Table I.3: Calibrated sectoral expenditure weights, 2020

Income state	Population mass	Basket B1		Basket B2	
		ω^R	ω^S	ω^R	ω^S
1	0.015625	0.2738	0.7262	0.4643	0.5357
		(0.0159)		(0.0343)	
2	0.093750	0.2871	0.7129	0.5163	0.4837
		(0.0113)		(0.0163)	
3	0.234375	0.2903	0.7097	0.5386	0.4614
		(0.0063)		(0.0074)	
4	0.312500	0.2708	0.7292	0.4457	0.5543
		(0.0046)		(0.0074)	
5	0.234375	0.2629	0.7371	0.3710	0.6290
		(0.0063)		(0.0067)	
6	0.093750	0.2278	0.7722	0.3064	0.6936
		(0.0136)		(0.0136)	
7	0.015625	0.2290	0.7710	0.2989	0.7011
		(0.0177)		(0.0189)	
Stationary mean	1.000000	0.2704	0.7296	0.4415	0.5585

Notes: Author's calculations from the 2020 CEX. The table reports the primary calibration, which ranks consumer units by labor income and restricts the reference person to ages 25–60 while retaining zero earners. Standard errors, shown in parentheses under the R weights, use the 44 CEX balanced-repeated-replication weights. The age restriction and full-sample income-rank map are held fixed across replicates, so the standard errors measure expenditure-sampling uncertainty conditional on sample membership and rank. Since $\omega^S = 1 - \omega^R$, the S weight has the same standard error. The stationary mean weights each state by the population mass in column 2; the seven state-specific rows are the model inputs.

points and a 95% confidence interval of 50.12–69.55%.

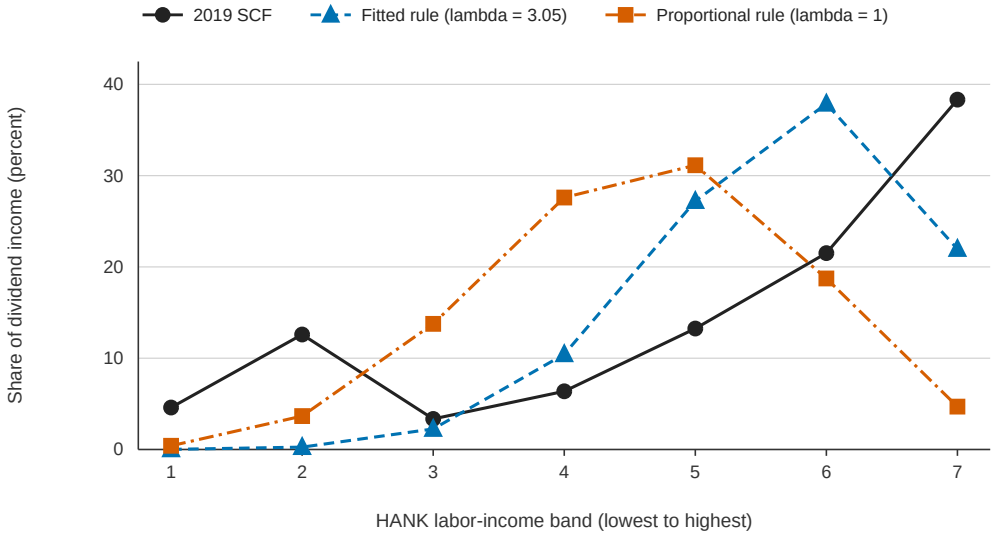
Mapping the survey moment to λ. Using the normalized dividend weights in (G.2), I choose λ to match the combined dividend share of the top two states:

$$\frac{\pi_6 e_6^\lambda + \pi_7 e_7^\lambda}{\sum_{j=1}^7 \pi_j e_j^\lambda} = 0.5984.$$

Here e_j is productivity in earnings state j. The fitted value is $\widehat{\lambda} = 3.054$, with confidence interval $[2.512, 3.654]$ obtained by inverting the empirical moment's confidence limits. The proportional benchmark $\lambda = 1$ assigns 23.43% of dividends to these two states.

Figure I.1 compares both rules with the data. The fitted rule matches only the combined upper-tail share: it assigns 37.87 and 21.96% to bands 6 and 7, compared with 21.51 and 38.33% in the SCF, and 0.27% to the bottom two bands, compared with 17.19%. Low current earners can own substantial portfolios, and current labor income need not equal permanent earnings. The full schedule therefore provides a diagnostic beyond the targeted upper-tail moment.

Figure I.1: Dividend incidence by labor-income band, 2019 SCF



Notes: The black line reports five-implicate average dividend shares for seven survey-weighted labor-income bands whose population masses equal the model's stationary masses. Labor income equals wages and salaries plus net sole-proprietorship and farm income; the sample includes SCF families with a reference person age 25–60. The fitted $\lambda = 3.05$ rule matches the combined dividend share of bands 6 and 7. The $\lambda = 1$ rule is the proportional-productivity benchmark. Income refers to calendar year 2018.